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United States Patent	12410255
Kind Code	B2
Date of Patent	September 09, 2025
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Natriuretic peptide receptor 1 antibodies and methods of use

Abstract

This disclosure relates to anti-Natriuretic Peptide Receptor 1 (NPR1) antibodies including agonist antibodies which are able to activate the NPR1 receptor, pharmaceutical compositions comprising the same, and methods of treatment comprising the same.

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Appl. No.: 18/193581

Filed: March 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230365674 A1	Nov. 16, 2023

Related U.S. Application Data

division parent-doc US 17348949 20210616 ABANDONED child-doc US 18193581
division parent-doc US 16897935 20200610 US 11066469 20210720 child-doc US 17348949
us-provisional-application US 62860508 20190612

Publication Classification

Int. Cl.: **C07K16/28** (20060101); **C12N15/63** (20060101); A61K39/00 (20060101); A61K39/395 (20060101); A61K45/06 (20060101); A61P9/00 (20060101); A61P13/12 (20060101); C07K14/58 (20060101); C07K14/72 (20060101)

U.S. Cl.:

CPC **C07K16/2869** (20130101); **C12N15/63** (20130101); A61K39/3955 (20130101); A61K2039/505 (20130101); A61K45/06 (20130101); A61P9/00 (20180101); A61P13/12 (20180101); C07K14/58 (20130101); C07K14/72 (20130101); C07K2317/50 (20130101); C07K2317/54 (20130101); C07K2317/55 (20130101); C07K2317/56 (20130101); C07K2317/565 (20130101); C07K2317/622 (20130101); C07K2317/75 (20130101)

Field of Classification Search

CPC: C07K (16/2869); A61K (39/3955); C12N (15/63)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9090695	12/2014	Waterman et al.	N/A	N/A
9987330	12/2017	Hoon et al.	N/A	N/A
10184942	12/2018	Mohapatra et al.	N/A	N/A
2004/0253242	12/2003	Bowdish et al.	N/A	N/A
2008/0181903	12/2007	Bhaskar et al.	N/A	N/A
2008/0214437	12/2007	Mohapatra et al.	N/A	N/A
2012/0270923	12/2011	Mohapatra et al.	N/A	N/A
2014/0343120	12/2013	Mohapatra et al.	N/A	N/A
2016/0168251	12/2015	Waterman et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104059154	12/2013	CN	N/A
9100292	12/1990	WO	N/A
0179231	12/2000	WO	N/A
0236688	12/2001	WO	N/A
0246238	12/2001	WO	N/A
02078683	12/2001	WO	N/A
2004050017	12/2003	WO	N/A
2004108078	12/2003	WO	N/A
2005060642	12/2004	WO	N/A
2007070175	12/2006	WO	N/A
2007115164	12/2006	WO	N/A
2007115175	12/2006	WO	N/A
2007115182	12/2006	WO	N/A
2008151257	12/2007	WO	N/A

2009006732	12/2008	WO	N/A
2009015011	12/2008	WO	N/A
2009036448	12/2008	WO	N/A
2009073527	12/2008	WO	N/A
2009142307	12/2008	WO	N/A
2009149278	12/2008	WO	N/A
2009149279	12/2008	WO	N/A
2010009319	12/2009	WO	N/A
2010065293	12/2009	WO	N/A
2010135541	12/2009	WO	N/A
2011069038	12/2010	WO	N/A
2012019237	12/2011	WO	N/A
2012088608	12/2011	WO	N/A
2012110842	12/2011	WO	N/A
2012118042	12/2011	WO	N/A
2013027680	12/2012	WO	N/A
2014115797	12/2013	WO	N/A
2015085055	12/2014	WO	N/A
2016033699	12/2015	WO	N/A
2016131943	12/2015	WO	N/A
2017082186	12/2016	WO	N/A
2017156310	12/2016	WO	N/A
2018034622	12/2017	WO	N/A
2020086406	12/2019	WO	N/A
2020236690	12/2019	WO	N/A
2020250159	12/2019	WO	N/A

OTHER PUBLICATIONS

Badri, et al., Optimization of radiation dosing schedules for proneural glioblastoma, J. Math. Biol., 2016, 1301-1336, 72. cited by applicant

Baylot, et al., Chapter 13: TCTP Has a Crucial Role in the Different Stages of Prostate Cancer Malignant Progression, TCTP/tpt1—Remodeling Signaling from Stem Cell to Disease, 2017, 255-261, 64. cited by applicant

Maslov, et al., The Role of Natriuretic Peptides and Erythropoietin in the Regulation of Cardiac Tolerance to the Impact of Ischemia and Reperfusion. Analysis of Experimental and Clinical Data, Russian Journal of Physiology, Feb. 1, 2019, 24-35, 105(1). cited by applicant

Muller, et al., Spliceosomal peptide P140 for immunotherapy of systemic lupus erythematosus: results of an early phase II clinical trial, Arthritis & Rheumatism, Dec. 2008, 3873-3883, 58(12). cited by applicant

Arora, P et al., Atrial natriuretic peptide is negatively regulated by microRNA-425, The Journal of Clinical Investigation, 2013, 3378-3382, 123(8). cited by applicant

Blech, M. et al., Structure of a Therapeutic Full-Length Anti-NPRA IgG4 Antibody: Dissecting Conformational Diversity, Biophysical Journal, May 1, 2019, 1637-1649, 11(9). cited by applicant

Cui, Y et al., Role of corin in trophoblast invasion and uterine spiral artery remodelling in pregnancy, Nature, 2012, 246-252, 484. cited by applicant

Dries, D. L. et al., Corin Gene Minor Allele Defined by 2 Missense Mutations Is Common in Blacks and Associated with High Blood Pressure and Hypertension, Circulation, 2005, 2403-2410, 112. cited by applicant

Ellmers L. J. et al., Npr1-regulated gene pathways contributing to cardiac hypertrophy and fibrosis, Journal of Molecular Endocrinology, Feb. 2007, 245-57, 38(1-2). cited by applicant

Hodgson-Zingman, D. M. et al., Atrial Natriuretic Peptide Frameshift Mutation in Familial Atrial Fibrillation, *The New England Journal of Medicine*, 2008, 158-165, 359(2). cited by applicant

Kitano K. et al., Production and characterization of monoclonal antibodies against human natriuretic peptide receptor-A or-B, *Immunol. Lett.*, Sep. 1995, 215-22, 47(3). cited by applicant

List, K. et al., Different mechanisms are involved in the antibody mediated inhibition of ligand binding to the urokinase receptor: a study based on biosensor technology, *Journal of Immunological Methods*, 1999, 125-133, 222. cited by applicant

Lowe D. G. et al., Human atrial natriuretic peptide receptor defines a new paradigm for second messenger signal transduction, *The EMBO Journal*, 1989, 1377-1384., 8(5). cited by applicant

Lowe D. G. et al., Human natriuretic peptide receptor-A guanylyl cyclase. Hormone cross-linking and antibody reactivity distinguish receptor glycoforms, *Journal Biol. Chem.*, Oct. 25, 1992, 21691-7., 267(30). cited by applicant

Lui, C. et al., Meta-analysis identifies common and rare variants influencing blood pressure and overlapping with metabolic trait loci, *Nature Genetics*, Oct. 2016, 1162-1170, 48(10). cited by applicant

Ma et al., Atrial natriuretic peptide/natriuretic peptide receptor A (ANP/NPRA) signaling pathway: A potential therapeutic target for allergic asthma, *Medical Hypotheses*, 2011, 832-33, 77. cited by applicant

Mezo, Adam R. et al., Atrial Natriuretic Peptide-Fc, ANP-Fc, Fusion Proteins: Semisynthesis, In Vitro Activity and Pharmacokinetics in Rats, *Bioconjugate Chemistry*, 2012, 518-526., 23(3). cited by applicant

Mumey, Brendan M. et al., A New Method for Mapping Discontinuous Antibody Epitopes to Reveal Structural Features of Proteins, *Journal of Computational Biology*, 2003, 555-567, 10(3-4). cited by applicant

Notification of Transmittal of The International Search Report and The Written Opinion of The International Searching Authority, or The Declaration, Mailed on Sep. 16, 2020, issued in International Patent Application No. PCT/IB2020/055468, filed Jun. 10, 2020. cited by applicant

Oliver P. M. et al., Hypertension, cardiac hypertrophy, and sudden death in mice lacking natriuretic peptide receptor A, *Proc. Natl. Acad. Sci. USA*, Dec. 23, 1997, 14730-5., 94(26). cited by applicant

Potter, L. et al., Natriuretic Peptides: Their Structures, Receptors, Physiologic Functions and Therapeutic Applications, *Handbook of Experimental Pharmacology*, 2009, 341-66., 191. cited by applicant

Rame, J. E. et al., Dysfunctional Corin I555(P568) Allele Is Associated With Impaired Brain Natriuretic Peptide Processing and Adverse Outcomes in Blacks With Systolic Heart Failure (Results From the Genetic Risk Assessment in Heart Failure Substudy), *Cir. Heart Fail.*, 2009, 541-548, 2. cited by applicant

Rondeau, Jean Jacques et al., Production of polyclonal antibody to the bovine adrenal atrial natriuretic factor-R1 receptor, *Journal of Receptor Research*, 1992, 485-505., 12(4). cited by applicant

Saito, Yoshihiko, Roles of atrial natriuretic peptide and its therapeutic use, *Journal of Cardiology*, 2010, 262-270, 56. cited by applicant

Vellaichamy, E. et al., Genetically Altered Mutant Mouse Models of Guanylyl Cyclase/Natriuretic Peptide Receptor-A Exhibit the Cardiac Expression of Proinflammatory Mediators in a Gene-Dose-Dependent Manner, *Endocrinology*, Mar. 2014, 1045-1056., 155(3). cited by applicant

Vellaichamy, et al., Reduced cGMP signaling activates NF- κ B in hypertrophied hearts of mice lacking natriuretic peptide receptor-A, *Biochemical and Biophysical Research Communications*, 327, 106-111, 2005. cited by applicant

U.S. Appl. No. 18/193,581, filed Mar. 30, 2023, Natriuretic Peptide Receptor 1 Antibodies and Methods of Use. cited by applicant

Background/Summary

RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/348,949, filed Jun. 16, 2021, which is a direct filing of U.S. patent application Ser. No. 16/897,935, filed Jun. 10, 2020, which claims the benefit of and priority to U.S. Provisional Application No. 62/860,508 filed Jun. 12, 2019, the entire contents of each of which are incorporated herein by reference in its entirety.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically herewith and is hereby incorporated by reference in its entirety. This sequence listing is named PAT058321-US-DIV02_SL.

BACKGROUND

(2) Heart failure is a major public health problem concerning more than 20 million patients around the world (Orso et al., 2014, Expert Opin Pharmacother. 15 (13): 1849-1861) and is associated with high morbidity (Ibebuogu et al., 2011, Circulation. Heart failure 4 (2): 114-120). Natriuretic Peptide Receptor 1 (NPR1; also known as NPRA) is a receptor guanylate cyclase, which is activated by Atrial Natriuretic Peptide (ANP) resulting in lowering of blood pressure and blood volume (Chen & Burnett, 2006, European Heart Journal Supplements 8 (Suppl E): E18-E25; Ibebuogu et al. 2011, supra; Mani et al. 2015, Bioscience Reports 35 (5): e00260). ANP binding induces dimerization and twisting of the receptor that induces activation of the guanylate cyclase domain and conversion of GTP into cGMP (Misono et al., 2011, The FEBS journal 278 (11): 1818-1829). ANP is cleared by NPR3, a natriuretic peptide receptor that lacks the guanylate cyclase domain, and degraded by Neutral Endopeptidase (NEP) (Chen & Burnett 2006, supra; Schmitt et al., 2003, Clin Sci (Lond). 105 (2): 141-160). Certain antibodies against NPR1 have been described, for example, in WO2010/065293 (including antibody 5591-IgG). However, these antibodies appeared to have no functional activity in the absence of ANP in vitro and no functional activity in vivo.

(3) It has been shown that an increase in ANP may be beneficial for patients with heart failure with reduced ejection fraction (outbound pumping of blood by heart). See McMurray et al., N. Engl. J. Med.; Vol. 371, No. 11, pp 993-1004 (2014); and Nougue et al., Eur J Heart Fail. 2019 May; 21 (5): 598-605. However, there is a need for further longer acting agents that have an alternative mode of action to supplement or replace existing therapies.

SUMMARY OF THE DISCLOSURE

(4) Herein is demonstrated that it is possible to activate NPR1 by the use of agonistic anti-NPR1 antibodies or antigen binding fragments thereof. Furthermore, the present disclosure demonstrates that there are two types of such antibodies. While one type binds to NPR1 and competes with ANP binding (yet still activates NPR1; hereinafter “ANP competitive” anti-NPR1 antibodies), the second type is able to bind and activate NPR1 while not competing with ANP (hereinafter “ANP non-competitive” anti-NPR1 antibodies). Such antibodies (e.g., ANP non-competitive anti-NPR1 antibodies) may be used to bolster the body's natural system and/or existing treatment rationales. Furthermore, certain NPR1 agonist antibodies that are able to activate NPR1 in the absence of ANP have been found to be functionally equivalent to ANP.

(5) The antibodies of the instant application show in vivo activity in both mouse and rat. Furthermore, the unique epitope binding of the antibodies described herein has been demonstrated using crystal structure data.

(6) Thus the disclosure provides anti-NPR1 antibodies (e.g., human monoclonal antibodies) or antigen-binding fragments thereof that (i) bind to natriuretic peptide receptor 1 (NPR1); and (ii) are capable of activating NPR1 in the absence of ANP. Such antibodies are agonistic anti-NPR1 antibodies. In some embodiments of the invention, the disclosure also provides anti-NPR1 antibodies or antigen binding fragments thereof that (i) bind to natriuretic peptide receptor 1 (NPR1); and (ii) activate NPR1 in the absence of ANP. In some embodiments of the invention, the disclosure also provides antibodies or antigen binding fragments thereof that (i) bind to natriuretic peptide receptor 1 (NPR1); and (ii) activate NPR1 in both the presence and absence of ANP. Also provided are nucleic acids encoding said antibodies, vectors comprising said nucleic acids, host cells comprising said nucleic acids and/or vectors, and methods of manufacture of said antibodies using said nucleic acids, vectors and/or host cells. Also provided are pharmaceutical compositions and combinations comprising said antibodies, nucleic acids, vectors or host cells, as well as methods of treatment using said antibodies, nucleic acids, vectors, host cells or pharmaceutical compositions. The use of said antibodies, nucleic acids, vectors, host cells or pharmaceutical compositions or combinations in treating disease is also disclosed herein.

(7) Thus, in one aspect of the invention, herein is provided an isolated antibody or antigen binding fragment that (i) binds to natriuretic peptide receptor 1 (NPR1); and (ii) is capable of activating NPR1 in the absence of atrial natriuretic peptide (ANP). In some embodiments of the invention, the isolated antibody or antigen binding fragment does not bind to and/or does not activate natriuretic peptide receptor 2 (NPR2) and/or natriuretic peptide receptor 3 (NPR3). In some embodiments of the invention, the isolated antibody or antigen binding fragment binds to (a) human NPR1; and (b) mouse NPR1 and/or rat NPR1.

(8) In some embodiments of the invention, the antibody or antigen binding fragment binds to (a) human NPR1; and (b) cyno NPR1. In some embodiments of the invention, the antibody or antigen binding fragment is ANP non-competitive. In some embodiments of the invention, the antibody or antigen binding fragment is ANP competitive. In some embodiments of the invention, the antibody or antigen binding fragment is capable of stabilizing the ANP-NPR1 complex.

(9) In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino acids 99-133 of SEQ ID NO: 1, e.g., within a region of human NPR1 encompassed by amino acids 99-133 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope comprising at least two amino acid residues within amino acids 99-133 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope comprising at least 3, 4, 5, 6, 7, or 8 amino acid residues within amino acids 99-133 of SEQ ID NO: 1.

(10) In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino acids 99-111 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino acids 99-103 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino acids 105-111 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope comprising at least 2, 3, or 4 amino acid residues within amino acids 105-111 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to a conformational epitope of human NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 99-103 of SEQ ID NO: 1, (ii) 105-111 of SEQ ID NO: 1, (iii) 131-134 of SEQ ID NO: 1, and additionally binds to amino acid 375 and/or 378 of SEQ ID NO: 1. In some embodiments of the invention, the epitope is a conformational epitope, and the conformational epitope additionally comprises at least one amino acid residue selected from the group consisting of amino acids 33, 34, 76, 82, and 104 of SEQ ID NO: 1. In some embodiments of the invention, the conformational epitope additionally comprises at least one amino acid residue selected from the group consisting of amino acids 33, 34, 76, 82, 104, 374, and 375 of SEQ ID NO: 1.

(11) In some embodiments of the invention, the antibody or antigen binding fragment binds to at least amino acids 82, 102, 103, 105, 106, 109, 132, and 375 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to at least amino acids 34, 82, 102, 103, 105, 106, 107, 109, 132, 133, 375, and 378 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to at least amino acids 79, 82, 99, 102, 103, 105, 106, 109, 131, 132, and 375 of SEQ ID NO: 1.

(12) In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino acids 188-219 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope comprising at least 2, 3, 4, 5, 6, or 7 amino acids within amino acids 188-219 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to a conformational epitope within NPR1, and the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) 201-208 of SEQ ID NO: 1, (iii) 215-238 of SEQ ID NO: 1, and (iv) 294-297 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to at least amino acids 188, 192, 194, 197, 201, 208, and 219 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to at least amino acids 188, 192, 194, 197, 201, 208, 219, and 295 of SEQ ID NO: 1.

(13) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in

X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in

Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y.sub.5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in

X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in

Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y 5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (III)

HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1SX.sub.2GX.sub.3Y (SEQ ID NO: 431), wherein X.sub.1 is S or E, X.sub.2 is D or K, or X.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1Y.sub.2Y.sub.3Y.sub.4PR (SEQ ID NO: 432); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is Y or W, Y.sub.3 is E, V, R, A, T, or M, and Y.sub.4 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in

IX.sub.1SX.sub.2GX.sub.3YX.sub.4 (SEQ ID NO: 433), wherein X.sub.1 is S or E, X.sub.2 is D or K, X.sub.3 is S or N, and X.sub.4 is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y.sub.5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1SX.sub.2GX.sub.3Y (SEQ ID NO: 431), wherein X.sub.1 is S or E, X.sub.2 is D or K, or X.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1WY.sub.2Y.sub.3PR (SEQ ID NO: 435); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in IX.sub.1SX.sub.2GX.sub.3YX.sub.4 (SEQ ID NO: 433), wherein X.sub.1 is S or E, X.sub.2 is D or K, X.sub.3 is S or N, and X.sub.4 is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 119; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists

of an amino acid sequence as set forth in SEQ ID NO: 119; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 120, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1WY.sub.2Y.sub.3PR (SEQ ID NO: 435); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 121, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (d) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THYIH (SEQ ID NO: 436), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5TY.sub.6YADSVKG (SEQ ID NO: 437), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, Y.sub.5 is S, N, or M, and Y.sub.6 is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5TY.sub.6YADSVKG (SEQ ID NO: 437), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, Y.sub.5 is S, N, or M, and Y.sub.6 is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1TH (SEQ ID NO: 438), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5 (SEQ ID NO: 439), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, and Y.sub.5 is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THY (SEQ ID NO: 440), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in IY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5T (SEQ ID NO: 441), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, and Y.sub.5 is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino

acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (e) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THYIH (SEQ ID NO: 436), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1SGY.sub.2Y.sub.3TYYADSVKG (SEQ ID NO: 442), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1SGY.sub.2Y.sub.3TYYADSVKG (SEQ ID NO: 442), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1TH (SEQ ID NO: 438), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SY.sub.1SGY.sub.2Y.sub.3 (SEQ ID NO: 443), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THY (SEQ ID NO: 440), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY.sub.1SGY.sub.2Y.sub.3T (SEQ ID NO: 444), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (f) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3X.sub.4X.sub.5 (SEQ ID NO: 445), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, X.sub.3 is W or Y, X.sub.4 is I or L, and X.sub.5 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2X.sub.3X.sub.4 (SEQ ID NO: 447), wherein X.sub.1 is S, K, or R, X.sub.2 is W or Y, X.sub.3 is I or L, and X.sub.4 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid

sequence as set forth in GFX.sub.1FSX.sub.2Y (SEQ ID NO: 448), wherein X.sub.1 is S or T, and X.sub.2 is S, K, or R, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4E (SEQ ID NO: 449), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, and Y.sub.4 is S, H, or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3 (SEQ ID NO: 450), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, and X.sub.3 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IY.sub.1QY.sub.2Y.sub.3Y.sub.4EY.sub.5 (SEQ ID NO: 451), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, Y.sub.4 is S, H, or L, and Y.sub.5 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (g) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3X.sub.4X.sub.5 (SEQ ID NO: 445), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, X.sub.3 is W or Y, X.sub.4 is I or L, and X.sub.5 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2X.sub.3X.sub.4 (SEQ ID NO: 447), wherein X.sub.1 is S, K, or R, X.sub.2 is W or Y, X.sub.3 is I or L, and X.sub.4 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2Y (SEQ ID NO: 448), wherein X.sub.1 is S or T, and X.sub.2 is S, K, or R, HCDR2 comprises or consists of an amino acid sequence as set forth in HQY.sub.1Y.sub.2Y.sub.3E (SEQ ID NO: 456), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, and Y.sub.3 is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3 (SEQ ID NO: 450), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, and X.sub.3 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IHQY.sub.1Y.sub.2Y.sub.3EY.sub.4 (SEQ ID NO: 458), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, and Y.sub.4 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (h) (I) HCDR1 comprises or consists of an amino acid

sequence as set forth in GFTFSX.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 452), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 454), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1Y (SEQ ID NO: 455), wherein X.sub.1 is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4E (SEQ ID NO: 449), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, and Y.sub.4 is S, H, or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2 (SEQ ID NO: 457), wherein X.sub.1 is S or R, and X.sub.2 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IY.sub.1QY.sub.2Y.sub.3Y.sub.4EY.sub.5 (SEQ ID NO: 451), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, Y.sub.4 is S, H, or L, and Y.sub.5 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; or (i) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 452), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 454), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3

comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1Y (SEQ ID NO: 455), wherein X.sub.1 is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in HQY.sub.1Y.sub.2Y.sub.3E (SEQ ID NO: 456), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, and Y.sub.3 is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2 (SEQ ID NO: 457), wherein X.sub.1 is S or R, and X.sub.2 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IHQY.sub.1Y.sub.2Y.sub.3EY.sub.4 (SEQ ID NO: 458), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, and Y.sub.4 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239.

(14) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in GX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 459), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in GNSNRPY.sub.1 (SEQ ID NO: 460), wherein Y.sub.1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7V (SEQ ID NO: 461), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 229, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in GX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 459), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in GNSNRPY.sub.1 (SEQ ID NO: 460), wherein Y.sub.1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7V (SEQ ID NO: 461), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313, HCDR3 comprises or consists of an amino acid sequence as set forth in GX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 459), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in YZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7 (SEQ ID NO: 462), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z, is P, S, or V; or (IV) HCDR1 comprises or

consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in ARGX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 463), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7V (SEQ ID NO: 461), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YAX.sub.3X.sub.4 (SEQ ID NO: 464), wherein X.sub.1 is S or G, X.sub.2 is S or T, X.sub.3 is I or M, and X.sub.4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1ISY.sub.2Y.sub.3GY.sub.4Y.sub.5Y.sub.6Y.sub.7YAY.sub.8SVKG (SEQ ID NO: 465), wherein Y.sub.1 is A or S, Y.sub.2 is A, S, or G, Y.sub.3 is S or H, Y.sub.4 is G or Y, Y.sub.5 is S or Y, Y.sub.6 is T or A, Y.sub.7 is Y, R, or N, and Y.sub.8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YAX.sub.2X.sub.3 (SEQ ID NO: 466), wherein X.sub.1 is S or T, X.sub.2 is I or M, and X.sub.3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1ISY.sub.2Y.sub.3GY.sub.4Y.sub.5Y.sub.6Y.sub.7YAY.sub.8SVKG (SEQ ID NO: 465), wherein Y.sub.1 is A or S, Y.sub.2 is A, S, or G, Y.sub.3 is S or H, Y.sub.4 is G or Y, Y.sub.5 is S or Y, Y.sub.6 is T or A, Y.sub.7 is Y, R, or N, and Y.sub.8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2Y (SEQ ID NO: 467), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SY.sub.1Y.sub.2GY.sub.3Y.sub.4 (SEQ ID NO: 468), wherein Y.sub.1 is A, S, or G, Y.sub.2 is S or H, Y.sub.3 is G or Y, and Y.sub.4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YA (SEQ ID NO: 469), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY.sub.1Y.sub.2GY.sub.3Y.sub.4T (SEQ ID NO: 470), wherein Y.sub.1 is S or G, Y.sub.2 is S or H, Y.sub.3 is G or Y, and Y.sub.4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; or (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YAX.sub.3X.sub.4 (SEQ ID NO: 464), wherein X.sub.1 is S or G, X.sub.2 is S or T, X.sub.3 is I or M, and X.sub.4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1Y.sub.2GYYY.sub.3Y.sub.4YAY 5SVKG (SEQ ID NO: 471), wherein Y.sub.1 is A or S, Y.sub.2 is S or H, Y.sub.3 is T or A, Y.sub.4 is R or N, and Y.sub.5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331,

LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YAX.sub.2X.sub.3 (SEQ ID NO: 466), wherein X.sub.1 is S or T, X.sub.2 is I or M, and X.sub.3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1Y.sub.2GYYY.sub.3Y.sub.4YAY.sub.5SVKG (SEQ ID NO: 471), wherein Y.sub.1 is A or S, Y.sub.2 is S or H, Y.sub.3 is T or A, Y.sub.4 is R or N, and Y.sub.5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2Y (SEQ ID NO: 467), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SY.sub.1Y.sub.2GYY (SEQ ID NO: 472), wherein Y.sub.1 is A or S, and Y.sub.2 is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YA (SEQ ID NO: 469), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY.sub.1Y.sub.2G (SEQ ID NO: 473), wherein Y.sub.1 is A, S, or G, and Y.sub.2 is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

(15) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 33, 120, and 191, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 46, 127, 135, 146, 173, 179, and 185; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34,

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acid sequence as set forth in any one of SEQ ID NOs: 231, 373, and 384, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239.

(16) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 321 and 354, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 229, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 321 and 354, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 325, 356, and 362; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 315 and 349, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; or (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 270 and 407, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 271, 389, and 408, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 273 and 409, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 271, 389, and 408, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 32 and 410, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 274, 390, and 411, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino

acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 275 and 412, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 276, 391, and 413, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

(17) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (a) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3); (b) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3); (c) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3); (d) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3); (e) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3); (f) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119

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(t) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3); (u) (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 9 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 11 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3); (v) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3); (w) (I) SEQ ID NO: 367 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 369 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 370 (HCDR1), SEQ ID NO: 371 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 372 (HCDR1), SEQ ID NO: 373 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (x) (I) SEQ ID NO: 378 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 380 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 381 (HCDR1), SEQ ID NO: 382 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 383 (HCDR1), SEQ ID NO: 384 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (y) (I) SEQ ID NO: 226 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 230 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 231 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (z) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 272

(HCDR3), SEQ ID NO: 284 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 285 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 277 (HCDR3), SEQ ID NO: 286 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 283 (LCDR3); or (aa) (I) SEQ ID NO: 291 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (II) SEQ ID NO: 294 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (III) SEQ ID NO: 295 (HCDR1), SEQ ID NO: 296 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 305 (LCDR3); or (IV) SEQ ID NO: 297 (HCDR1), SEQ ID NO: 298 (HCDR2), SEQ ID NO: 299 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 304 (LCDR3).

(18) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from:

(a) (I) SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (II) SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (III) SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3); or (IV) SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3); (b) (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3); (c) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 362 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 361 (LCDR3); (d) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 390 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 391 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); (e) (I) SEQ ID NO: 407 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 409 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 410 (HCDR1), SEQ ID NO: 411 (HCDR2), SEQ ID NO: 331

(HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 412 (HCDR1), SEQ ID NO: 413 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); (f) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 325 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 315 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 322 (LCDR3); (g) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); or (h) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 356 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 355 (LCDR3).

(19) In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 128; (d) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 128; (e) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 147; (f) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 147; (g) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174; (h) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174; (i) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 180; (j) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 180; (k) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region

region comprising an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 363; (d) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 392, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344; (e) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 414, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344; (f) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 316, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 327; (g) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 333, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344; or (h) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 357.

(21) In some embodiments of the invention, the antibody or antigen binding fragment comprises:

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an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (z) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (aa) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (bb) a heavy chain comprising an amino acid sequence of SEQ ID NO: 195, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (cc) a heavy chain comprising an amino acid sequence of SEQ ID NO: 195, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (dd) a heavy chain comprising an amino acid sequence of SEQ ID NO: 15, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (ee) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 50; (ff) a heavy chain comprising an amino acid sequence of SEQ ID NO: 376, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (gg) a heavy chain comprising an amino acid sequence of SEQ ID NO: 387, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (hh) a heavy chain comprising an amino acid sequence of SEQ ID NO: 235, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (ii) a heavy chain comprising an amino acid sequence of SEQ ID NO: 280, and a light chain comprising an amino acid sequence of SEQ ID NO: 289; or (jj) a heavy chain comprising an amino acid sequence of SEQ ID NO: 302, and a light chain comprising an amino acid sequence of SEQ ID NO: 308.

(22) In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 63, and a light chain comprising an amino acid sequence of SEQ ID NO: 74; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 87, and a light chain comprising an amino acid sequence of SEQ ID NO: 98; (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 365; (d) a heavy chain comprising an amino acid sequence of SEQ ID NO: 394, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; (e) a heavy chain comprising an amino acid sequence of SEQ ID NO: 416, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; (f) a heavy chain comprising an amino acid sequence of SEQ ID NO: 318, and a light chain comprising an amino acid sequence of SEQ ID NO: 329; (g) a heavy chain comprising an amino acid sequence of SEQ ID NO: 335, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; or (h) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 359.

(23) In some embodiments of the invention, the antibody or antigen binding fragment is an antigen binding fragment selected from the group consisting of a Fab, Fab', F(ab').sub.2, Fv, single domain antibody (dAb), and a single chain variable fragment (scFv). In some embodiments of the invention, the antibody or antigen binding fragment is an antigen binding fragment selected from the group consisting of a Fab, Fab', Fv, single domain antibody (dAb), and a single chain variable fragment (scFv).

(24) In some embodiments of the invention, the antibody or antigen binding fragment is monoclonal. In some embodiments of the invention, the antibody or antigen binding fragment is fully human. In some embodiments of the invention, the antibody or antigen binding fragment is an IgG antibody. In some embodiments of the invention, the antibody or antigen binding fragment is an IgG1 antibody. In some embodiments of the invention, the antibody or antigen binding fragment is an IgG1 antibody having a kappa light chain. In some embodiments of the invention, the antibody or antigen binding fragment is a fully human antibody of the IgG1 isotype and has a kappa light chain.

(25) In some embodiments of the invention, the antibody or antigen binding fragment additionally has mutations in the Fc region according to the EU index of Kabat, wherein the mutations comprise at least D265A and P329A.

(26) In some embodiments of the invention, the antibody or antigen binding fragment additionally has mutations in the Fc region according to the EU index of Kabat, wherein the mutations comprise at least L234A and L235A.

(27) In some embodiments of the invention, the antibody or antigen binding fragment is therapeutic.

(28) In some embodiments of the invention, the antibody or antigen binding fragment binds to the same epitope on human NPR1 as any of the antibodies or antigen binding fragments or groups defined herein (e.g., XX16). In some embodiments of the invention, the antibody or antigen binding fragment competes for binding to human NPR1 with any of the antibodies or antigen binding fragments or groups defined herein (e.g., XX16).

(29) In one aspect of the invention, provided herein is an isolated nucleic acid or nucleic acids encoding the amino acid sequence of any of the antibodies or antigen binding fragments or groups defined herein. In one aspect of the invention, provided herein is a vector comprising the isolated nucleic acid(s). In one aspect of the invention, provided herein is a host cell comprising the isolated nucleic acid(s) or the vector.

(30) In one aspect of the invention, provided herein is a method of producing any of the antibodies or antigen binding fragments described herein, comprising culturing the host cell described herein under conditions suitable to produce the antibody or antigen binding fragment. In some embodiments of the invention, the method additionally comprises purification of the antibody or antigen binding fragment.

(31) In one aspect of the invention, provided herein is pharmaceutical composition comprising a purified antibody or antigen binding fragment produced by the method described herein and a pharmaceutically acceptable carrier.

(32) In one aspect of the invention, provided herein is pharmaceutical composition comprising any of the antibodies or antigen binding fragments described herein and a pharmaceutically acceptable carrier.

(33) In one aspect of the invention, provided herein is pharmaceutical composition comprising: a) means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and b) a pharmaceutically acceptable excipient. In some embodiments of the invention, said means for binding and activating is ANP non-competitive. In some embodiments of the invention, said means for binding and activating is ANP competitive. In some embodiments of the invention, said means for binding and activating is additionally capable of stabilizing the ANP-NPR1 complex. In some embodiments of the invention, the composition further comprises an additional therapeutic agent.

(34) In some embodiments of the invention, the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is an angiotensin receptor-neprilysin inhibitor (ARNi).

(35) In some embodiments of the invention, the additional therapeutic agent is selected from a corticosteroid, a leukotriene modifier, a bronchodilator, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline and ipratropium, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from salmeterol, formoterol, albuterol, and levalbuterol, and combinations thereof.

(36) In some embodiments of the invention, the additional therapeutic agent is selected from a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil and ripasudil, and combinations thereof.

(37) In one aspect of the invention, provided herein is a method of treating a disorder or a disease associated with natriuretic peptide receptor activity in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein or a pharmaceutical composition or combination as described herein.

(38) In one aspect of the invention, provided herein is a method of treating a cardiovascular disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein.

(39) In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(40) In one aspect of the invention, provided herein is a method of treating heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein.

(41) In some embodiments of the invention, the subject has heart failure, wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

(42) In one aspect of the invention, provided herein is a method of treating a kidney disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein. In some

embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

(43) In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

(44) In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of a cardiovascular disorder in a subject in need of such treatment.

(45) In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(46) In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

(47) In some embodiments of the invention, the subject has heart failure, and the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

(48) In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of a kidney disorder in a subject in need of such treatment. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

(49) In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

(50) In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of a cardiovascular disorder in a subject in need of such treatment.

(51) In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(52) In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

(53) In some embodiments of the invention, the subject has heart failure, and the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

(54) In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of a kidney disorder in a subject in need of such treatment. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

(55) In one aspect of the invention, provided herein is a method of treating a disorder or a disease associated with natriuretic peptide receptor activity in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient.

(56) In one aspect of the invention, provided herein is a method of treating a cardiovascular disorder in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient. In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(57) In one aspect of the invention, provided herein is a method of treating heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or

cytokine release syndrome in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient. In some embodiments of the invention, the subject has heart failure, wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

(58) In one aspect of the invention, provided herein is a method of treating a kidney disorder in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD). In some embodiments of the invention, said means for binding and activating is ANP non-competitive. In some embodiments of the invention, said means for binding and activating is ANP competitive. In some embodiments of the invention, said means for binding and activating is additionally capable of stabilizing the ANP-NPR1 complex.

(59) In one aspect of the invention, herein is provided an isolated anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope of human NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 99-103 of SEQ ID NO: 1, (ii) 105-111 of SEQ ID NO: 1, (iii) 131-134 of SEQ ID NO: 1, and additionally binds to amino acid 375 and/or 378 of SEQ ID NO: 1.

(60) In one aspect of the invention, herein is provided an isolated anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope within NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) 201-208 of SEQ ID NO: 1, (iii) 215-238 of SEQ ID NO: 1, and (iv) 294-297 of SEQ ID NO: 1, optionally wherein the antibody or antigen binding fragment thereof binds to at least amino acids 188, 192, 194, 197, 201, 208, and 219 of SEQ ID NO: 1.

(61) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in

Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y 5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y.sub.5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1SX.sub.2GX.sub.3Y (SEQ ID NO: 431), wherein X.sub.1 is S or E, X.sub.2 is D or K, or X.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1Y.sub.2Y.sub.3Y.sub.4PR (SEQ ID NO: 432); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is Y or W, Y.sub.3 is E, V, R, A, T, or M, and Y.sub.4 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in IX.sub.1SX.sub.2GX.sub.3YX.sub.4 (SEQ ID NO: 433), wherein X.sub.1 is S or E, X.sub.2 is D or K, X.sub.3 is S or N, and X.sub.4 is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y.sub.5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YXX.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YXX.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1SX.sub.2GX.sub.3Y (SEQ ID NO: 431), wherein X.sub.1 is S or E, X.sub.2 is D or K, or

X.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1WY.sub.2Y.sub.3PR (SEQ ID NO: 435); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in IX.sub.1SX.sub.2GX.sub.3YX.sub.4 (SEQ ID NO: 433), wherein X.sub.1 is S or E, X.sub.2 is D or K, X.sub.3 is S or N, and X.sub.4 is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 119; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 119; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 120, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1WY.sub.2Y.sub.3PR (SEQ ID NO: 435); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 121, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (d) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THYIH (SEQ ID NO: 436), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5TY.sub.6YADSVKG (SEQ ID NO: 437), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, Y.sub.5 is S, N, or M, and Y.sub.6 is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II)

[illegible]

LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19.

(62) In some aspects and embodiments of the invention, herein is provided an isolated anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 33, 120, and 191, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 46, 127, 135, 146, 173, 179, and 185; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 35, 121, and 192, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184.

(63) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (a) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3); (b) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1),

[illegible]

[illegible]

33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3).

(64) In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3).

(65) In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; or (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 48.

(66) In some embodiments of the invention, the antibody or antigen binding fragment comprises: a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 85, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 96.

(67) In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; or (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 50.

(68) In some embodiments of the invention, the antibody or antigen binding fragment comprises: a heavy chain comprising an amino acid sequence of SEQ ID NO: 87, and a light chain comprising an amino acid sequence of SEQ ID NO: 98.

(69) In some embodiments of the invention, the antigen binding fragment is selected from the group consisting of a Fab, Fab', F(ab').sub.2, Fv, single domain antibody (dAb), and a single chain variable fragment (scFv). In some embodiments of the invention, the antibody or antigen binding fragment is therapeutic.

(70) In some aspects of the invention, provided herein is an isolated nucleic acid or nucleic acids encoding the amino acid sequence of any of the antibodies or antigen binding fragments described herein. In some aspects of the invention, provided herein is a vector comprising any of the isolated nucleic acid(s) described herein. In some aspects of the invention, provided herein is a host cell comprising any of the isolated nucleic acid(s) described herein. In some aspects of the invention, provided herein is a method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing any of the host cells described herein under conditions suitable for producing the antibody or antigen binding fragment.

(71) In some aspects of the invention, provided herein is a pharmaceutical composition comprising any of the antibodies or antigen binding fragments described herein and a pharmaceutically acceptable carrier. In some embodiments of the invention, the composition further comprises an additional therapeutic agent. In some embodiments of the invention, the additional therapeutic

agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), an angiotensin receptor-neprilysin inhibitor (ARNi), a corticosteroid, a leukotriene modifier, a bronchodilator, a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline, ipratropium, salmeterol, formoterol, albuterol, levalbuterol, timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil, ripasudil, and combinations thereof.

(72) In some aspects of the invention, provided herein is a method of treating a disorder or a disease associated with natriuretic peptide receptor activity in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein.

(73) In some aspects of the invention, provided herein is a method of treating a cardiovascular disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein. In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(74) In some aspects of the invention, provided herein is a method of treating heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein. In some embodiments of the invention, the subject has heart failure, and the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, and wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, and the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, and the hypertension is selected from resistant hypertension or hypertensive heart disease.

(75) In some aspects of the invention, provided herein is a method of treating a kidney disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically

effective amount of any of the antibodies or antigen binding fragments described herein. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a set of graphs displaying the results of antibody candidates WW01, WW02, WW03, WW04, and WW06 binding to the following antigens (ELISA analysis): human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target) both in the absence of and presence of a 250 fold molar excess of ANP.
- (2) FIG. 2 is a set of graphs displaying the results of flow cytometry analysis of antibody candidates WW01, WW02, WW03, WW04, and WW06 for binding to human NPR1 expressing CHO-K1 cells in the absence and presence of a saturating concentration of ANP and on parental CHO-K1 cells.
- (3) FIG. 3 is a graphical representation of a Fluorescence Resonance Energy Transfer (FRET)-based assay in which the NPR1-specific antibodies competed with ANP for binding to NPR1. In this FRET based assay, Eu-labeled Streptavidin (for measurement of IgGs) or Eu-labeled anti-hFc antibody (for measurement of FabCys) was used as an energy donor, while Cy5-labeled ANP was used as an acceptor.
- (4) FIG. 4 is a set of graphs displaying the results of ANP competition analyses of candidates WW01, WW02, WW03, WW04, and WW06 using the FRET-based assay depicted in FIG. 3.
- (5) FIG. 5 is a set of graphs depicting the results of functional activity analyses of candidates WW01, WW02, WW03, WW04, and WW06 in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.
- (6) FIG. 6 is a set of graphs depicting the results of functional activity analyses of candidate WW06 in IgG or FabCys format in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.
- (7) FIG. 7 demonstrates the results of SET screening (hNPR1 affinity) of 82 HCDR2 or LCDR3 unique improved HuCAL® derivatives based on the parental antibody.
- (8) FIG. 8 demonstrates the results of SET screening (hNPR1-ANP complex affinity) of 82 HCDR2 or LCDR3 unique improved HuCAL® derivatives based on the parental antibody.
- (9) FIG. 9 is a set of graphs displaying the results of antibody candidates XX01-XX08, XX10, and XX12 binding to the following antigens (ELISA analysis): human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target) in the absence of or in the presence of a 250 fold molar excess of ANP.
- (10) FIG. 10 is a set of graphs displaying the results of flow cytometry analysis of antibody candidates XX01-XX08, XX10, and XX12 for binding to human NPR1 expressing CHO-K1 cells in the absence or presence of a saturating concentration of ANP and on parental CHO-K1 cells.
- (11) FIG. 11 is a set of graphs displaying the results of ANP competition analyses of candidates XX01-XX07, XX10, and XX12 using the FRET-based assay depicted in FIG. 3.
- (12) FIG. 12 is a set of graphs depicting the results of functional activity analyses of candidates XX01-XX08, XX10, and XX12 in a cellular cGMP production assay using human NPR1

expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

(13) FIG. 13 is a set of graphs depicting the results of functional activity analyses of candidates XX06 and XX16 alongside natural ligand ANP in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

(14) FIG. 14 is a graph depicting the results of functional activity analyses of candidate XX16 alongside natural ligand ANP in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] as a function of either ANP or XX16.

(15) FIG. 15 is a graphical representation of the crystal structure of Fab06 (the WW06 antibody in Fab format) in complex with hNPR1 extracellular domain.

(16) FIG. 16 is a graphical representation of the conformation of the hNPR1 extracellular domain as it would be in complex with Fab06. The Fab06 were removed from this representation to more clearly reveal the conformation of hNPR1 induced by Fab binding. W74R is shown in space filling.

(17) FIG. 17 is a graphical representation of the crystal structure of hNPR1 extracellular domain in complex with ANP (left) and the crystal structure of Fab16 (the XX16 antibody in Fab format) in complex with hNPR1 extracellular domain (right) with W74 shown in space filling.

(18) FIG. 18 is a set of graphs depicting the results of functional activity analyses of candidate WW06 in a cellular cGMP production assay using when tested on CHO-K1 cells expressing hNPR1 W74R/C232T (constitutively active mutant; panel A) compared to WT hNPR1 (panel B).

(19) FIG. 19 is a set of graphs displaying the results of antibody candidates YY01-YY07 binding to the following antigens (ELISA analysis): human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target) in the absence of or presence of a 250 fold molar excess of ANP.

(20) FIG. 20 is a set of graphs displaying the results of flow cytometry analysis of antibody candidates YY01-YY07 for binding to human NPR1 expressing CHO-K1 cells in the absence or presence of a saturating concentration of ANP and on parental CHO K1 cells.

(21) FIG. 21 is a set of graphs displaying the results of ANP competition analyses of candidates YY01-YY07 using the FRET-based assay depicted in FIG. 3.

(22) FIG. 22 is a set of graphs depicting the results of functional activity analyses of candidates YY01-YY07 in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

(23) FIG. 23 is a set of graphs depicting the results of functional activity analyses of candidates YY05 and YY07 in IgG or FabCys format in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

(24) FIG. 24 demonstrates the results of SET screening (hNPR1 affinity) of 112 HCDR1/2 or LCDR3 unique improved Ylanthia® derivatives based on the parental antibody.

(25) FIG. 25 demonstrates the results of SET screening (hNPR1-ANP complex affinity) of 112 HCDR1/2 or LCDR3 unique improved Ylanthia® derivatives based on the parental antibody.

(26) FIG. 26 is a set of graphs representing the plasma cGMP (nM) concentration over time in hNPR1 Tg mice which were intravenously administered 1 mg/kg, 3 mg/kg, or 10 mg/kg of the WW06 Fab.

(27) FIG. 27 is a graph representing the concentration of antibody over time in hNPR1 Tg mice which were intravenously administered 1 mg/kg, 3 mg/kg, or 10 mg/kg of the WW06 Fab.

(28) FIG. 28 is a set of graphs demonstrating the results of subcutaneous administration of vehicle, 0.3 mg/kg XX16, or 3 mg/kg XX16 once every two weeks in wild type or ANP knockout (ANP KO) mice on heart weight/body weight ratio (left) and plasma NT-proBNP levels (pg/mL; right).

Measurements were taken two weeks after the second administration.

(29) FIG. 29 is a set of graphs demonstrating the results of a single intravenous administration of vehicle, 0.3 mg/kg XX16, or 1 mg/kg XX16 on blood pressure (mean arterial pressure; left) and urinary flow rate (right) in hypertensive rats (spontaneous hypertensive rat stroke prone, SHRsp). Measurements were taken three hours after the intravenous administration.

(30) FIG. 30 is a set of graphs demonstrating the results of a single subcutaneous administration of 0.1, 0.3, 1, 3, 10, or 30 mg/kg XX16 in telemetry implanted normal rats on mean arterial pressure (MAP; top) and plasma cGMP (bottom) over time.

(31) FIG. 31 is a set of graphs depicting the results of functional activity analyses of candidate WW03 and a comparator antibody (antibody 5591-IgG from PCT Application No. WO2010/065293A1) in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells or rat NPR1 expressing rat cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP (human or rat).

GENERAL DEFINITIONS

(32) In order that the present disclosure may be more readily understood, certain terms are first defined. Additional definitions are set forth throughout the detailed description as required.

(33) Throughout the description and claims of this specification, the words “comprise” and “contain” and variations of the words, for example “comprising” and “comprises”, mean “including but not limited to”, and do not exclude other components, integers, or steps. Moreover the singular encompasses the plural unless the context otherwise requires: in particular, where the indefinite article is used, the specification is to be understood as contemplating plurality as well as singularity, unless the context requires otherwise.

(34) As used herein, “NPR1” and “NPR1 protein” refers to Natriuretic Peptide Receptor 1. This protein is also known as Atrial natriuretic peptide receptor type A (ANP-A, ANPR-A or NPR-A) and Guanylate cyclase A (GC-A). In some embodiments, the NPR1 referred to is human NPR1. In some embodiments the human NPR1 has UniProt accession number P16066 or GenBank Accession number EAW53284.1 (SEQ ID NO: 1). In some embodiments, the NPR1 referred to is mouse (*Mus musculus*) NPR1. In some embodiments the mouse NPR1 has NCBI Reference Sequence number NP_032753.5 (SEQ ID NO: 2). In some embodiments, the NPR1 referred to is rat (*Rattus norvegicus*) NPR1. In some embodiments the rat NPR1 has NCBI Reference Sequence number NP_036745.1 (SEQ ID NO: 3). Exemplary NPR1 proteins are shown in Table 1. Where a constitutively active or W74R mutant is discussed herein, this mutant refers to Trp at amino acid 74 of the mature human NPR1 protein, which may also be referred to as the Trp at amino acid 106 of the hNPR1 protein shown in SEQ ID NO: 1.

(35) TABLE-US-00001 TABLE 1 NPR1 Protein Sequences SEQ ID NO: Animal Sequence (amino acid) 1 Human

MPGPRRPAGSRLRLLLLLLLPLLLLLRGSHAGNLTAVVVLPLANTSYPWS
WARVGPVELALAQVKARPDLLPGWTVRTVLGSSSENALGVCSDTAAPLA
AVDLKWEHNPAVFLGPGCVYAAAPVGRFTAHWRVPLLTAGAPALGFGV
KDEYALTTRAGPSYAKLGDFVAALHRRLGWERQALMLYAYRPGDEEHC
FFLVEGLFMRVRDRLNITVDHLEFAEDDLSHYTRLLRTMPRKGRVIYICSS
PDAFRTLMLLALEAGLCGEDYVFFHLDIFGQSLQGGQGPAPRRPW ERGDG
QDVSARQAFQAAKIITYKDPDNPEYLEFLKQLKHLAYEQFNFTMEDGLVN
TIPASFHDGLLLYIQAVTETLAHGGTVDGENITQRMWNR SFQGV TGYLK
IDSSGDRETDFSLWDMDPENGA FRVVLNYNGTSQELVAVSGRKLNWPLG
YPPPDIPKCGFDNEDPACNQDHLSTLEVLALVGSLSLLGILIVSFFIYRKM
QLEKELASELWRVRWEDVEPSSLERHLRSAGSRLTL SGRGSNYGSLLTTEG
QFQVFAKTAYYKGNLVAVKRVNRKRIELTRKVL FELKHMRDVQNEHLTR
FVGACTDPPNICILTEYCPRGSLQDILENESITLDWMFRYSLTNDIVKGM
LF LHNGAICSHGNLKSSNCVVDGRFVLKITDYGLE SFRDLDP EQGHTVYAKK

LWTAPELLRMASPPARGSQAGDVYSFGIILQEIALRSGVFYVEGLDLSPKEI
IERVTRGEQPPFRPSLALQSHLEELGLLMQRCWAEDPQERPPFQQIRLTLR
KFNRENSNILDNLLSRMEQYANNLEELVEERTQAYLEEKRAEALLYQI
LPHSVAEQLKRGETVQAEAFDSVTIYFSDIVGFTALSAESTPMQVVTLLND
LYTCFDAVIDNFDVYKVETIGDAYMVVSGLPVRNGRLHACEVARMALAL
LDAVRSFRIRHRPQEQLRLRIGIHTGPVCAGVVGLKMPRYCLFGDTVNTA
SRMESNGEALKIHLSSETKAVLEEFGGFELELRGDVEMKGKGKVRTYWL LGERGSSTRG

2 Mouse MPGSRRVRPRLRALLLLPLLLLRSGHASDLTVAVVLPLTNTSYPWSWAR
VGPAVELALGRVKARPDLLPGWTVRMVLGSSENAAGVCSDTAAPLA
DLKWEHSPAVFLGPGCVYSAAPVGRFTHWVRVPLLTAGAPALGIGVKDE
YALTTRTGPSHVKLGDFTALHRRLGWEHQALVLYADRLGDDRPCFFIV
EGLYMRVRERLNTVNHQEFVEGDPDHYTKLLRTVQRKGRVIYICSSPDA
FRNLMLLALDAGLTGEDYVFFHLDVFGQSLQGAQGPVPRKPWERDDGQ
DRRARQAFQAAKIITYKEPDNPEYLEFLKQLKLLADKKFNFTMEDGLKNII
PASFHDGLLLYVQAVTETLAQGGTVTDGENITQRMWNRSFQGVGTGYLKI
DRNGDRDITDFSLWDMDPETGAFRVVLNFNNGTSQELMAVSEHRLYWPLG
YPPPDIPKCGFDNEDPACNQDHFSTLEVLALVGSLSLVSLIVSFFIYRKMQ
LEKELVSELWRVRWEDLQPSSLERHLRSAGSRLTLSGRGSNYGSLLTTEG
QFQVFAKTAYYKGNLVAVKRVNRKRIELTRKVL FELKHMRDVQNEHLTR
FVGACTDPPNICILTEYCPRGSLQDILENESITLDWMFRYSLTNDIVKGMLF
LHNGAIGSHGNLKSSNCVVDGRFVLKITDYGLESFRDPEPEQGHTLFAKK
LWTAPELLRMASPPARGSQAGDVYSFGIILQEIALRSGVFYVEGLDLSPKEI
IERVTRGEQPPFRPSMDLQSHLEELGQLMQRCWAEDPQERPPFQQIRLALR
KFNKENSSNILDNLLSRMEQYANNLEELVEERTQAYLEEKRAEALLYQI
LPHSVAEQLKRGETVQAEAFDSVTIYFSDIVGFTALSAESTPMQVVTLLND
LYTCFDAVIDNFDVYKVETIGDAYMVVSGLPVRNGQLHAREVARMALAL
LDAVRSFRIRHRPQEQLRLRIGIHTGPVCAGVVGLKMPRYCLFGDTVNTA
SRMESNGEALRIHLSSETKAVLEEFDGFELELRGDVEMKGKGKVRTYWL LGERGCSTRG

3 Rat MPGSRRVRPRLRALLLLPLLLLRRGGHASDLTVAVVLPLTNTSYPWSWAR
VGPAVELALARVKARPDLLPGWTVRMVLGSSENAAGVCSDTAAPLA
DLKWEHSPAVFLGPGCVYSAAPVGRFTHWVRVPLLTAGAPALGIGVKDE
YALTTRTGPSHVKLGDFTALHRRLGWEHQALVLYADRLGDDRPCFFIV
EGLYMRVRERLNTVNHQEFVEGDPDHYPKLLRAVRRKGRVIYICSSPDA
FRNLMLLALNAGLTGEDYVFFHLDVFGQSLKSAQGLVPQKPWERGDGQD
RSARQAFQAAKIITYKEPDNPEYLEFLKQLKLLADKKFNFTVEDGLKNIIP
ASFHDGLLLYVQAVTETLAQGGTVTDGENITQRMWNRSFQGVGTGYLKID
RNGDRDITDFSLWDMDPETGAFRVVLNNGTSQELMAVSEHKLYWPLGY
PPPDVPKCGFDNEDPACNQDHFSTLEVLALVGSLSLISFLIVSFFIYRKMQ
EKELVSELWRVRWEDLQPSSLERHLRSAGSRLTLSGRGSNYGSLLTTEGQ
FQVFAKTAYYKGNLVAVKRVNRKRIELTRKVL FELKHMRDVQNEHLTRF
VGACTDPPNICILTEYCPRGSLQDILENESITLDWMFRYSLTNDIVKGMLFL
HNGAICSHGNLKSSNCVVDGRFVLKITDYGLESFRDPEPEQGHTLFAKKL
WTAPELLRMASPPARGSQAGDVYSFGIILQEIALRSGVFYVEGLDLSPKEI
ERVTRGEQPPFRPSMDLQSHLEELGQLMQRCWAEDPQERPPFQQIRLALR
KFNKENSSNILDNLLSRMEQYANNLEELVEERTQAYLEEKRAEALLYQI
LPHSVAEQLKRGETVQAEAFDSVTIYFSDIVGFTALSAESTPMQVVTLLND
LYTCFDAVIDNFDVYKVETIGDAYMVVSGLPVRNGQLHAREVARMALAL
LDAVRSFRIRHRPQEQLRLRIGIHTGPVCAGVVGLKMPRYCLFGDTVNTA
SRMESNGEALKIHLSSETKAVLEEFDGFELELRGDVEMKGKGKVRTYWL LGERGCSTRG

(36) In various embodiments, the anti-NPR1 antibodies and antigen binding fragments disclosed

herein are capable of binding to NPR1 and activating NPR1 in the absence of ANP. By virtue of this activity, the disclosed anti-NPR1 antibodies and antigen binding fragments may be useful in treating undesirable conditions, diseases and disorders including cardiovascular disorders (e.g., hypertension, peripheral vascular disease, heart failure (including but not limited to heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure), coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy (e.g., ventricular hypertrophy), diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, or myocardial infarction (MI)), hypertension (e.g., resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, or pulmonary arterial hypertension), preeclampsia, asthma, glaucoma, cytokine release syndrome, and/or a kidney disorder (e.g., diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD)).

(37) The term “antibody” as used herein refers to a whole antibody or antigen binding fragment thereof. A whole antibody is a glycoprotein comprising at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds. Each heavy chain is comprised of a heavy chain variable region (abbreviated herein as VH) and a heavy chain constant region. The heavy chain constant region is comprised of three domains, CH1, CH2 and CH3. Each light chain is comprised of a light chain variable region (abbreviated herein as VL) and a light chain constant region. The light chain constant region is comprised of one domain, CL. The VH and VL regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDR), interspersed with regions that are more conserved, termed framework regions (FR). Each VH and VL is composed of three CDRs and four FRs arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The variable regions of the heavy and light chains contain a binding domain that interacts with an antigen. The constant regions of the antibodies may mediate the binding of the immunoglobulin to host tissues or factors, including various cells of the immune system (e.g., effector cells) and the first component (C1q) of the classical complement system. The term “antibody” includes, but is not limited to, monoclonal antibodies, human antibodies, humanized antibodies, camelised antibodies, and chimeric antibodies. The antibodies can be of any isotype/class (e.g., IgG, IgE, IgM, IgD, IgA and IgY) or subclass (e.g., IgG.sub.1, IgG.sub.2, IgG.sub.3, IgG.sub.4, IgA.sub.1, and IgA.sub.2).

(38) The term “antigen binding fragment” refers to a fragment of an intact antibody that retains the ability to specifically bind to a given antigen (e.g., NPR1) and/or provide a function of the intact antibody. Such fragments include Fab fragments, Fab' fragments, monovalent fragments consisting of the VL, VH, CL and CHI domains; F (ab')₂ fragments, bivalent fragments comprising two Fab fragments linked by a disulfide bridge at the hinge region; an Fd fragment consisting of the VH and CHI domains; an Fv fragment consisting of the VL and VH domains, a single chain Fv fragment (scFv) consisting of the VL and VH domains connected by a linker sequence; and a single domain antibody (dAb) fragment (Ward et al., 1989 Nature 341:544-546), which consists of a VH domain or a VL domain.

(39) The term “single chain antibody”, “single chain Fv” or “scFv” is refers to a molecule comprising an antibody heavy chain variable domain (or region; VH) and an antibody light chain variable domain (or region; VL) connected by a linker. Such scFv molecules can have the general

structures: NH2-VL-linker-VH-COOH or NH2-VH-linker-VL-COOH. Any suitable linker may be used. A non-limiting set of linkers that can be used in such single chain antibodies are described by Holliger et al. (1993), Proc. Natl. Acad. Sci. USA 90:6444-6448, Alfthan et al. (1995), Protein Eng. 8:725-731, Choi et al. (2001), Eur. J. Immunol. 31:94-106, Hu et al. (1996), Cancer Res. 56:3055-3061, Kipriyanov et al. (1999), J. Mol. Biol. 293:41-56 and Roovers et al. (2001), Cancer Immunol; the contents of each of which are herein incorporated by reference for this purpose. Such single chain antibodies are also intended to be encompassed within the term “antigen binding fragment” of an antibody. These antibody fragments are obtained using techniques known to those of skill in the art, and the fragments are screened for utility in the same manner as are intact antibodies. Without limitation, an antigen binding fragment can be produced by any suitable method known in the art. For instance, the various antigen binding fragments described herein can be produced by enzymatic or chemical modification of intact antibodies, synthesized de novo using recombinant DNA methodologies (e.g., single chain Fv), or identified using phage display libraries (see, e.g., Pini and Bracci, Curr Protein Pept Sci 2000; 1(2):155-69, the contents of which are herein incorporated by reference for this purpose). Antigen binding fragments are screened for utility (e.g., binding affinity, activity) in the same manner as are intact antibodies.

(40) Antigen binding fragments can also be incorporated into single domain antibodies, maxibodies, minibodies, intrabodies, diabodies, triabodies, tetrabodies, v-NAR and bis-scFv (see, e.g., Hollinger and Hudson, 2005, Nature Biotechnology, 23, 9, 1126-1136, the contents of which are herein incorporated by reference for this purpose). Antigen binding portions of antibodies can be grafted into scaffolds based on polypeptides such as Fibronectin type III (Fn3) (see e.g., U.S. Pat. No. 6,703,199, which describes fibronectin polypeptide monobodies, the contents of which are herein incorporated by reference for this purpose).

(41) Antigen binding fragments can be incorporated into single chain molecules comprising a pair of tandem Fv segments (VH-CH1-VH-CH1) which, together with complementary light chain polypeptides, form a pair of antigen binding regions (see Zapata et al., 1995 Protein Eng. 8 (10): 1057-1062; and U.S. Pat. No. 5,641,870; the contents of each of which are herein incorporated by reference for this purpose).

(42) The term “isolated” means throughout this specification, that the immunoglobulin, antibody or polynucleotide, as the case may be, exists in a physical milieu distinct from that in which it may occur in nature. For example, a naturally-occurring polynucleotide or polypeptide present in a living organism is not isolated, but the same polynucleotide or polypeptide separated from some or all of the coexisting materials in the living organism, is isolated.

(43) The term “isolated antibody,” as used herein, refers to an antibody that has been identified and separated from one or more (e.g., the majority) of the components (by weight) of its source environment, e.g., from the components of a hybridoma cell culture or a different cell culture that was used for its production (e.g., producer cells including but not limited to the exemplary host cells described herein that recombinantly express the antibody). The separation is performed such that it sufficiently removes components that may otherwise interfere with the suitability of the antibody for the desired applications (e.g., for therapeutic use of an anti-NPR1 antibody). Methods for preparing isolated antibodies are known in the art and include Protein A chromatography, anion exchange chromatography, cation exchange chromatography, virus retentive filtration, and ultrafiltration.

(44) Throughout this specification, complementarity determining regions (“CDR”) are defined according to the Kabat definition unless specified that the CDR are defined according to another definition. The precise amino acid sequence boundaries of a given CDR can be determined using any of a number of well-known schemes, including those described by Kabat et al. (1991), “Sequences of Proteins of Immunological Interest,” 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD (“Kabat” numbering scheme), Al-Lazikani et al., (1997) JMB 273, 927-948 (“Chothia” numbering scheme) and ImMunoGenTics (IMGT) numbering (Lefranc,

M.-P., *The Immunologist*, 7, 132-136 (1999); Lefranc, M.-P. et al., *Dev. Comp. Immunol.*, 27, 55-77 (2003) (“IMGT” numbering scheme); the contents of each of which are herein incorporated by reference for this purpose. For example, for classic formats, under Kabat, the CDR amino acid residues in the heavy chain variable domain (VH) are numbered 31-35 (HCDR1), 50-65 (HCDR2), and 95-102 (HCDR3); and the CDR amino acid residues in the light chain variable domain (VL) are numbered 24-34 (LCDR1), 50-56 (LCDR2), and 89-97 (LCDR3). Under Chothia the CDR amino acids in the VH are numbered 26-32 (HCDR1), 52-56 (HCDR2), and 95-102 (HCDR3); and the amino acid residues in VL are numbered 26-32 (LCDR1), 50-52 (LCDR2), and 91-96 (LCDR3). By combining the CDR definitions of both Kabat and Chothia, the CDRs consist of amino acid residues 26-35 (HCDR1), 50-65 (HCDR2), and 95-102 (HCDR3) in human VH and amino acid residues 24-34 (LCDR1), 50-56 (LCDR2), and 89-97 (LCDR3) in human VL. Under IMGT the CDR amino acid residues in the VH are numbered approximately 26-35 (CDR1), 51-57 (CDR2) and 93-102 (CDR3), and the CDR amino acid residues in the VL are numbered approximately 27-32 (CDR1), 50-52 (CDR2), and 89-97 (CDR3) (numbering according to “Kabat”). Under IMGT, the CDR regions of an antibody can be determined using the program IMGT/DomainGap Align.

(45) By convention, the CDR regions in the heavy chain are typically referred to as HCDR1, HCDR2 and HCDR3 and in the light chain as LCDR1, LCDR2 and LCDR3. They are numbered sequentially in the direction from the amino terminus to the carboxy terminus.

(46) The term “antibody framework” as used herein refers to the part of the variable domain, either VL or VH, which serves as a scaffold for the antigen binding loops (CDRs) of this variable domain. In essence, it is the variable domain without the CDRs.

(47) The terms “constant region” or “constant domain” refer to a carboxy terminal portion of the light and heavy chain which is not directly involved in binding of the antibody to antigen but exhibits various effector functions, such as interaction with the Fc receptor. The terms refer to the portion of an immunoglobulin molecule having a more conserved amino acid sequence relative to the other portion of the immunoglobulin, the variable domain, which contains the antigen binding site. The constant domain contains the CH1, CH2 and CH3 domains of the heavy chain and the CHL domain of the light chain.

(48) The term “epitope” or “antigenic determinant” refers to a site on an antigen to which an immunoglobulin or antibody specifically binds (e.g., a specific site on the target molecule). An epitope typically includes at least 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 consecutive or non-consecutive amino acids in a unique spatial conformation. See, e.g., *Epitope Mapping Protocols in Methods in Molecular Biology*, Vol. 66, G. E. Morris, Ed. (1996), the contents of which are herein incorporated by reference for this purpose. In addition, as used herein, an epitope can comprise one or more monosaccharide units of a polysaccharide to which an antibody specifically binds. In specific aspects, an epitope can be a conformational epitope. See, e.g., Thompson et al., 2009, *J. of Biol. Chem.* 51:35621-35631, the contents of which are herein incorporated by reference for this purpose.

(49) The terms “monoclonal antibody” or “monoclonal antibody composition” as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies produced by a particular cell or cell line, wherein the individual antibodies comprising the population are essentially identical in sequence except for possible naturally-occurring mutations that may be present in minor amounts. A monoclonal antibody preparation displays a single binding specificity and affinity for a particular epitope. In contrast, conventional (polyclonal) antibody preparations typically include a multitude of antibodies directed against or specific for different epitopes. The modifier “monoclonal” indicates the character of the antibody as being obtained from a substantially homogeneous population of antibodies and is not to be construed as requiring production of the antibody by any particular method. Monoclonal antibodies (mAbs) can be produced by a variety of techniques, including conventional methodology, e.g., the standard

somatic cell hybridization technique of Kohler and Milstein (Nature 1975; 256(5517):495-7), the contents of which are herein incorporated by reference for this purpose. A monoclonal antibody may also be obtained from other suitable methods, including phage display techniques such as those described in Clackson et al. (Nature 1991; 352(6336):624-8) or Marks et al. (J Mol Biol 1991; 222(3):581-97), the contents of each of which are herein incorporated by reference for this purpose. The term “monoclonal antibody” is also not limited to antibody sequences from particular species of origin or from one single species of origin. Thus, the meaning of the term “monoclonal antibody” encompasses chimeric monoclonal antibodies such as humanized monoclonal antibodies.

(50) The term “chimeric antibody,” as used herein, refers to antibodies in which (a) the constant region is altered, replaced, or exchanged such that the antigen binding site (variable region) is linked to a constant region of a different or altered class, effector function, and/or species; or (b) the variable region, or a portion thereof, is altered, replaced, or exchanged with a variable region, or a portion thereof, having a different or altered antigen specificity. To create a chimeric antibody, the variable region sequences from a non-human donor antibody (e.g., a mouse, rabbit, or rat donor antibody) can be linked to human constant regions using methods known in the art (see, e.g., U.S. Pat. No. 4,816,567 (Cabilly et al.), the contents of which are herein incorporated by reference for this purpose). For instance, a mouse anti-NPR1 antibody can be modified by replacing its constant region with the constant region from a human immunoglobulin. Due to the replacement with a human constant region, the chimeric antibody can retain its specificity in recognizing human NPR1 while having reduced immunogenicity in human as compared to the original mouse antibody.

(51) As used herein, the term “humanized antibody” refers to forms of antibodies that contain at least some human sequence and at least some non-human sequence. Typically, the antibody contains human sequences and a minor portion of non-human sequences which confer binding specificity to the target antigen. Such antibodies are chimeric antibodies which contain minimal sequence derived from a non-human immunoglobulin and retain the reactivity of a non-human antibody while being less immunogenic in humans. Typically, humanized antibodies are generated by replacing hypervariable region sequences from a human acceptor antibody with hypervariable region sequences from a non-human donor antibody (e.g., a mouse, rabbit, or rat donor antibody) that binds to an antigen of interest (e.g., NPR1). In some cases, framework region sequences of the acceptor antibody may also be replaced with the corresponding sequences of the donor antibody (e.g., via affinity maturation). In addition to the sequences derived from the donor and acceptor antibodies, the humanized antibody can be further modified by the substitution of residues, either in the framework region and/or within the replaced non-human residues to refine and optimize antibody specificity, affinity, and/or activity, as discussed herein. Methods to generate humanized antibodies are known in the art. See, e.g., Riechmann et al. (Nature 1988; 332(6162):323-7); Jones et al. (Nature 1986; 321(6069):522-5); U.S. Pat. No. 5,225,539 (Winter); and U.S. Pat. Nos. 5,530,101; 5,585,089; 5,693,762, and 6,180,370 (Queen et al.), the contents of each of which are herein incorporated by reference for this purpose.

(52) The term “human antibody”, as used herein, is intended to include antibodies having variable regions in which both the framework and CDR regions are derived from sequences of human origin. Furthermore, if the antibody contains a constant region, the constant region also is derived from such human sequences, e.g., human germline sequences, or mutated versions of human germline sequences or antibody containing consensus framework sequences derived from human framework sequences analysis, for example, as described in Knappik, et al., (2000) J Mol Biol; 296:57-86, the contents of which are herein incorporated by reference for this purpose). Human antibodies may include amino acid residues not encoded by human sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo).

(53) The antibodies or antigen binding fragments of the disclosure may include amino acid residues not encoded by human sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo). However, the term “human antibody”, as

used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

(54) The terms “peptide,” “polypeptide,” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms encompass amino acid polymers in which one or more amino acid residues is an artificial chemical mimetic of a corresponding naturally-occurring amino acid, as well as naturally-occurring amino acid polymers and non-naturally-occurring amino acid polymers. Unless otherwise indicated, a particular polypeptide sequence also implicitly encompasses conservatively modified variants thereof.

(55) The term “conservatively modified variant” applies to both amino acid and nucleic acid sequences. For nucleic acid sequences, conservatively modified variants refer to those nucleic acids which encode identical or essentially identical amino acid sequences, or where the nucleic acid does not encode an amino acid sequence, to essentially identical sequences. Because of the degeneracy of the genetic code, a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG, and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations,” which are one species of conservatively modified variations. Every nucleic acid sequence herein which encodes a polypeptide also describes every possible silent variation of the nucleic acid. One of skill will recognize that each codon in a nucleic acid (except AUG, which is ordinarily the only codon for methionine, and TGG, which is ordinarily the only codon for tryptophan) can be modified to yield a functionally identical molecule.

Accordingly, each silent variation of a nucleic acid that encodes a polypeptide is implicit in each described sequence. For polypeptide sequences, conservatively modified variants include individual substitutions, deletions, or additions to a polypeptide sequence which result in the substitution of an amino acid with a chemically similar amino acid. Conservative substitution tables providing functionally similar amino acids are well known in the art. The following eight groups contain amino acids that are conservative substitutions for one another: 1) Alanine (A), Glycine (G); 2) Aspartic acid (D), Glutamic acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V); 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W); 7) Serine(S), Threonine (T); and 8) Cysteine (C), Methionine (M).

(56) The term “identity” or “homology” refers to a relationship between the sequences of two or more polypeptides, as determined by comparing the sequences. “Identity” also means the degree of sequence relatedness between polypeptides, as determined by the number of matches between strings of two or more amino acid residues. The percent “identity” between the two sequences is a function of the number of identical positions shared by the sequences (i.e., percent identity equals number of identical positions/total number of positions $\times$ 100), taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences. The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. For sequence comparison, typically one sequence acts as a reference sequence, to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. Default program parameters can be used, or alternative parameters can be designated. The sequence comparison algorithm then calculates the percent sequence identities for the test sequences relative to the reference sequence, based on the program parameters.

Additionally, or alternatively, the protein sequences of the present disclosure can further be used as a “query sequence” to perform a search against public databases to, for example, identify related sequences. For example, such searches can be performed using the BLAST program of Altschul et

al. (J Mol Biol 1990; 215(3):403-10), the contents of which are herein incorporated by reference for this purpose.

(57) Two sequences are “substantially identical” if two sequences have a specified percentage of amino acid residues or nucleotides that are the same (e.g., 60% identity, 65% identity, 70% identity, 75% identity, 80% identity, 85% identity, 90% identity, 95% identity, or 99% identity over a specified region, or, when not specified, over the entire sequence), when compared and aligned for maximum correspondence over a comparison window, or designated region as measured using one of the following sequence comparison algorithms or by manual alignment and visual inspection. Optionally, the identity exists over a region that is at least about 50 nucleotides (or 10 amino acids) in length, or exists over a region that is 100 to 500 or 1000 or more nucleotides (or 20, 50, 200 or more amino acids) in length.

(58) Binding “affinity” refers to the strength of interaction between antibody and antigen at single antigenic sites. Within each antigenic site, the variable region of the antibody “arm” interacts through weak non-covalent forces with antigen at numerous sites. In general, the more interactions, the stronger the affinity. Generally, such determinations can be made using a cell-based assay.

(59) The term “Kassoc” or “K_a”, as used herein, is intended to refer to the association rate of a particular binding molecule-antigen interaction, whereas the term “Kdis” or “K_d,” as used herein, is intended to refer to the dissociation rate of a particular binding molecule-antigen interaction. The term “K_D”, as used herein, is intended to refer to the equilibrium dissociation constant, which is obtained from the ratio of K_d to K_a (i.e., K_d/K_a) and is expressed as a molar concentration (M). K_{sub.D} values for antibodies can be determined using methods well established in the art. A method for determining the K_{sub.D} of an antibody is by using surface plasmon resonance, such as a Biacore® system, or solution equilibrium titration (SET) (see Friguet et al., (1985) J. Immunol. Methods, 77 (2): 305-319, and Hanel et al., (2005) Anal. Biochem., 339 (1): 182-184), the contents of each of which are herein incorporated by reference for this purpose.

(60) As used herein, the term “specific,” “specifically binds,” and “binds specifically” refers to a binding reaction between an antibody or antigen binding fragment (e.g., an anti-NPR1 antibody) and a target antigen (e.g., NPR1) in a heterogeneous population of proteins and other biologics. Antibodies can be tested for specificity of binding by comparing binding to an appropriate antigen to binding to an irrelevant antigen or antigen mixture under a given set of conditions. If the antibody binds to the appropriate antigen with at least 2, 5, 7, and preferably 10 or more times more affinity than to the irrelevant antigen or antigen mixture, then it is considered to be specific. A “specific antibody” or a “target-specific antibody” is one that only binds the target antigen (e.g., NPR1), but does not bind (or exhibits minimal binding) to other antigens. In certain embodiments, an antibody or antigen binding fragment that specifically binds the target antigen (e.g., NPR1) has a K_{sub.D} of less than 1×10^{sup.-6} M, less than 1×10^{sup.-7} M, less than 1×10^{sup.-8} M, less than 1×10^{sup.-9} M, less than 1×10^{sup.-10} M, less than 1×10^{sup.-11} M, less than 1×10^{sup.-12} M, or less than 1×10^{sup.-13} M. In certain embodiments, the K_{sub.D} is about 1 pM to about 600 pM. In certain embodiments, the K_{sub.D} is between 600 pM to 1 μM, 1 μM to 100 nM, or 100 mM to 10 nM (inclusive).

(61) In some embodiments, the antibodies or antigen binding fragments thereof act as non-competitive agonists. A “non-competitive agonist” refers to a molecule which binds to an enzyme or receptor at a site distant from the binding sites of its natural ligands. The non-competitive or allosteric agonism is generally independent of the association or concentration of the natural ligands for the enzyme or receptor. Such non-competitive agonists can, for example, provide for a level of activation that can be substantially independent of natural ligands. In a specific embodiment, the anti-NPR1 antibodies or antigen binding fragments described herein are ANP non-competitive, meaning that the antibody or antigen binding fragment acts as an agonist which binds at site away from ANP binding site of NPR1 and effect agonistic activity regardless of whether or not NPR1 is bound to ANP.

(62) In some embodiments, the antibodies or antigen binding fragments thereof act as competitive agonists. A “competitive agonist” refers to an agonist which interferes or competes with a natural ligand for its binding site on an enzyme or receptor. In a specific embodiment, the anti-NPR1 antibodies or antigen binding fragments described herein are ANP competitive, meaning that the antibody or antigen binding fragment acts as an agonist which competes with ANP at the ANP binding site of NPR1.

(63) In some embodiments, the activation of NPR1 by an antibody or antigen binding fragment may be determined by any suitable assay. An exemplary assay for determination of NPR1 activation is the production of cGMP by mammalian cells (e.g., CHO cells or a human cell line) expressing hNPR1.

(64) In some embodiments, the stabilization of the ANP-NPR1 complex may be determined by any suitable assay. An exemplary assay for determination of the stability of the ANP-NPR1 complex is the FRET assay described herein (see, e.g., FIG. 3).

(65) The term “about” in relation to a numerical value x means, for example, $x \pm 10\%$.

Antibodies of the Disclosure

(66) Below are disclosed certain specific anti-NPR1 antibody sequences of the disclosure. As used herein, the term “anti-NPR1 antibody” or “antibody that binds to NPR1” refers to any form of an antibody or antigen binding fragment that specifically binds to NPR1, e.g., those binding with a $K_{sub.D}$ of less than 1×10^{-8} M, as determined by, e.g., surface plasmon resonance (SPR) spectroscopy (using Biacore™) or solution equilibrium titration (SET). The term encompasses monoclonal antibodies (including intact monoclonal antibodies), polyclonal antibodies, and biologically functional antigen binding fragments so long as they specifically bind to NPR1.

(67) Amino acid and nucleic acid sequences of exemplary anti-NPR1 antibodies of the present disclosure are set forth in Table 2. In some embodiments, the antibody has the heavy and light chain CDRs, VH and VL sequence, and/or the heavy and light chain sequence of any of the antibodies described in Table 2. In some embodiments, the anti-NPR1 antibody is a four-chain antibody (also referred to as an intact antibody), comprising two heavy chains and two light chains. In some embodiments, the anti-NPR1 antibody is an antigen binding fragment of an intact antibody, e.g., a functional fragment of an intact antibody selected from any of those set forth in Table 2 that retains the ability to bind NPR1 and/or provide a function of the intact antibody (e.g., activating NPR1 in the absence of ANP). In some embodiments, the anti-NPR1 antibody is an antibody having the CDRs of any heavy chain variable region and light chain variable region pair shown in Table 2. In some embodiments, the anti-NPR1 antibody is an antibody having the CDRs of any heavy and light chain pair shown in Table 2.

(68) TABLE-US-00002 TABLE 2 Exemplary anti-NPR1 antibody sequences

WW01_	LALA SEQ ID	HCDR1	GFTFNTHYIH NO:	4 (Combined)	SEQ ID	HCDR2
SISGSGSNTYYADSVKG	NO:	5 (Combined)	SEQ ID	HCDR3	ERGYVYYHMFDP	NO:
NO:	6 (Combined)	SEQ ID	HCDR1	THYIH NO:	7 (Kabat)	SEQ ID
SISGSGSNTYYADSVKG	NO:	5 (Kabat)	SEQ ID	HCDR3	ERGYVYYHMFDP	NO:
(Kabat)	SEQ ID	HCDR1	GFTFNTH NO:	8 (Chothia)	SEQ ID	HCDR2
SGSGSN	NO:	9 (Chothia)	SEQ ID	HCDR3	ERGYVYYHMFDP	NO:
6 (Chothia)	SEQ ID	HCDR1	GFTFNTHY NO:	10 (IMGT)	SEQ ID	HCDR2
ISGSGSNT	NO:	11 (IMGT)	SEQ ID	HCDR3	ARERGYVYYHMFDP	NO:
12 (IMGT)	SEQ ID	VH	QVQLLES	GGGLVQPGGSLRLS	CAASGFTFNTHYIHWVRQAPGKGLEWVSSI	NO:
13	SGSGSNTYYADSVKGRFTISRDN	SKNTLYLQMNSLRAEDTAVYYCARERG	YVYYHMFDPWGQGT	LVTVSS	SEQ ID	DNA
VH	caggtgcaattgctggaaagcggcggtggcctggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg	NO:	14	gattcacctttaacactcattacatccattgggtgcgccaggccccggcaaaggtctcagtggttctctatctctggttctggttctaacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacaccctgta	tctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgaacgtggttacgtttactaccatat	

gttcgatccgtggggccaaggcacccctgggtgactgttagctca SEQ ID Heavy
QVQLLESGLVQPGGSLRLSCAASGFTFNTYIHVWRQAPGKGLEWVSSI NO: 15
Chain SGSGSNTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARERG
YVYYHMFDPWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYI
CNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT
LMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
GSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHYTQKSLSLSPGK SEQ ID
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NO: 16 Heavy

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SEQ ID LCDR2 GASSRAT NO: 18 (Combined) SEQ ID LCDR3 QQHSMYPRT
NO: 19 (Combined) SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Kabat) SEQ ID
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QSITRNY NO: 23 (IMGT) SEQ ID LCDR2 GAS NO: 21 (IMGT) SEQ ID
LCDR3 QQHSMYPRT NO: 19 (IMGT) SEQ ID VL
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ID DNA VL

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SEQ ID DNA

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(Combined) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Combined) SEQ
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SYWMN NO: 31 (Kabat) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Kabat)
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
MQSYEKPRT NO: 43 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat)
SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 MQSYEKPRT
NO: 43 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2
TAS NO: 45 (Chothia) SEQ ID LCDR3 SYEKPR NO: 46 (Chothia) SEQ ID
LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID
LCDR3 MQSYEKPRT NO: 43 (IMGT) SEQ ID VL
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(Chothia) SEQ ID HCDR2 YPDVSY NO: 57 (Chothia) SEQ ID HCDR3
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SEQ ID LCDR2 GDNDRPS NO: 66 (Combined) SEQ ID LCDR3
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(Kabat) SEQ ID LCDR2 GDNDRPS NO: 66 (Kabat) SEQ ID LCDR3
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NO: 69 (IMGT) SEQ ID LCDR3 GTYTLLFTSKV NO: 67 (IMGT) SEQ ID VL
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(Combined) SEQ ID LCDR3 QSYDLQKSSRV NO: 91 (Combined) SEQ ID
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NO: 19 (Combined) SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Kabat) SEQ ID
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LCDR3 QQHSMYPRT NO: 19 (IMGT) SEQ ID VL
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SEQ ID LCDR2 GASSRAT NO: 18 (Combined) SEQ ID LCDR3 QQHSMYPRT
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NO: 19 (Combined) SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Kabat) SEQ ID
LCDR2 GASSRAT NO: 18 (Kabat) SEQ ID LCDR3 QQHSMYPRT NO: 19 (Kabat)
SEQ ID LCDR1 QSITRNY NO: 20 (Chothia) SEQ ID LCDR2 GAS NO: 21
(Chothia) SEQ ID LCDR3 HSMYPR NO: 22 (Chothia) SEQ ID LCDR1
QSITRNY NO: 23 (IMGT) SEQ ID LCDR2 GAS NO: 21 (IMGT) SEQ ID LCDR3
QQHSMYPRT NO: 19 (IMGT) SEQ ID VL
DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA NO: 24
SSRATGIPARFSGSGSGTDFTLTISSELPEDFAVYYCQQHSMYPRTFGQGTK VEIK SEQ
ID DNA VL
gacatcgtgctgactcagtcctcgcactctgagcctgtcaccggggagaacggggccaccctctcttgcgcgcctcc NO: 110
caatccattactcggaactacctggcctggatcagcagaagccaggacaggcccctaggcttctgatctacggggcc
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SEQ ID Light
DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA NO: 26 Chain
SSRATGIPARFSGSGSGTDFTLTISSELPEDFAVYYCQQHSMYPRTFGQGTK
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SEQ ID DNA
gacatcgtgctgactcagtcctcgcactctgagcctgtcaccggggagaacggggccaccctctcttgcgcgcctcc NO: 111
Light caatccattactcggaactacctggcctggatcagcagaagccaggacaggcccctaggcttctgatctacggggcc Chain
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(Combined) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Combined) SEQ
ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
SYWMN NO: 31 (Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat)
SEQ ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1
GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ
ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1
GFTFSSYW NO: 34 (IMGT) SEQ ID HCDR2 IESKGNYY NO: 121 (IMGT) SEQ
ID HCDR3 ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
EVQLLES GGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 122

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EVQLLES GGGGLVQP GGSRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 124
Chain VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
MQSYEKPRT NO: 43 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat)
SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 MQSYEKPRT
NO: 43 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2
TAS NO: 45 (Chothia) SEQ ID LCDR3 SYEKPR NO: 46 (Chothia) SEQ ID
LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID
LCDR3 MQSYEKPRT NO: 43 (IMGT) SEQ ID VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSYLA WYQQKPGKAPKLLIYTAS NO: 48
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ID DNA VL
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ID Light DIQMTQSPSSLSASVGDRVTITCRASQGISSYLA WYQQKPGKAPKLLIYTAS
NO: 50 Chain TLQSGVPSRFS GSGSGTDFTLTIS SLQPEDFATYYCMQSYEKPRTFGQG TKV

EIKRTVAAPFPPPSDEQLKSGTASVVCLLNNFYPPREAKVQWVKVDNALQ
SGNSQESVTEQDSKDYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID DNA

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(Combined) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Combined) SEQ
ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
SYWMN NO: 31 (Kabat) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Kabat)
SEQ ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1
GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 SSDGSY NO: 33 (Chothia) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSYW
NO: 34 (IMGT) SEQ ID HCDR2 ISSDGSYT NO: 35 (IMGT) SEQ ID HCDR3
ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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NO: 40 Heavy

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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
QQEWVKPRT NO: 126 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat)
SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 QQEWVKPRT
NO: 126 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2
TAS NO: 45 (Chothia) SEQ ID LCDR3 EWVKPR NO: 127 (Chothia) SEQ ID
LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID
LCDR3 QQEWVKPRT NO: 126 (IMGT) SEQ ID VL
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ID Light DIQMTQSPSSLSASVGDRTITCRASQGISSYLA WYQQKPGKAPKLLIYTAS
NO: 130 Chain
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SEQ ID DNA
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Light gccaggggtatttcttcttacctggcttggtaccagcagaaaccggggcaaagcgccgaaactattaatctacactgcttcta Chain
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(Combined) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Combined) SEQ
ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
SYWMN NO: 31 (Kabat) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Kabat)
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GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 SSDGSY NO: 33 (Chothia) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSY
NO: 34 (IMGT) SEQ ID HCDR2 ISSDGSYT NO: 35 (IMGT) SEQ ID HCDR3
ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
QQTWRKPRT NO: 134 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41
(Kabat) SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3
QQTWRKPRT NO: 134 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ
ID LCDR2 TAS NO: 45 (Chothia) SEQ ID LCDR3 TWRKPR NO: 135 (Chothia)
SEQ ID LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT)
SEQ ID LCDR3 QQTWRKPRT NO: 134 (IMGT) SEQ ID VL
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TAS NO: 45 (Chothia) SEQ ID LCDR3 TWRKPRT NO: 135 (Chothia) SEQ ID
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LCDR3 QQTWRKPRT NO: 134 (IMGT) SEQ ID VL
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145 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2 TAS
NO: 45 (Chothia) SEQ ID LCDR3 IWTVPRT NO: 146 (Chothia) SEQ ID LCDR1
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ID HCDR3 ERGYVYYHMFDP NO: 6 (Combined) SEQ ID HCDR1 THYIH NO: 7
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(Chothia) SEQ ID HCDR2 GGQGGM NO: 152 (Chothia) SEQ ID HCDR3
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NO: 160 Heavy

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SEQ ID LCDR2 GASSRAT NO: 18 (Combined) SEQ ID LCDR3 QQHSMYPRT
NO: 19 (Combined) SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Kabat) SEQ ID
LCDR2 GASSRAT NO: 18 (Kabat) SEQ ID LCDR3 QQHSMYPRT NO: 19 (Kabat)
SEQ ID LCDR1 QSITRNY NO: 20 (Chothia) SEQ ID LCDR2 GAS NO: 21
(Chothia) SEQ ID LCDR3 HSMYPR NO: 22 (Chothia) SEQ ID LCDR1
QSITRNY NO: 23 (IMGT) SEQ ID LCDR2 GAS NO: 21 (IMGT) SEQ ID
LCDR3 QQHSMYPRT NO: 19 (IMGT) SEQ ID VL

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ID DNA VL

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SEQ ID Light

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SEQ ID DNA

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SEQ ID HCDR3 ERGYVYYHMFDP NO: 6 (Combined) SEQ ID HCDR1 THYIH
NO: 7 (Kabat) SEQ ID HCDR2 SIGGQGGMTLYADSVKG NO: 151 (Kabat) SEQ

ID CDR3 ERGYVYYHMFDP NO: 6 (Kabat) SEQ ID HCDR1 GFTFSTH NO: 113 (Chothia) SEQ ID HCDR2 GGQGGM NO: 152 (Chothia) SEQ ID HCDR3 ERGYVYYHMFDP NO: 6 (Chothia) SEQ ID HCDR1 GFTFSTHY NO: 114 (IMGT) SEQ ID HCDR2 IGGQGGMT NO: 153 (IMGT) SEQ ID HCDR3 ARERGYVYYHMFDP NO: 12 (IMGT) SEQ ID VH EVQLLES GGGLVQPGGSLRLSCAASGFTFSTHYIHWVRQAPGKGLEWVSSI NO: 161 GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYYHMFDPWGQGTLVTVSS SEQ ID DNA VH gaagtcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgcatgtgccgcgagcg NO: 162 ggtttactttctccaccactacatccactgggtccgccaagctcccggaaggactcgaatgggtgtcctccattggt ggacagggcgcatgaccctttacgcgtagcgtgaaggggaggttcacatctcccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccaggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacctgaggacaggaacgctggtcaccgtgtcctcc SEQ ID Heavy EVQLLES GGGLVQPGGSLRLSCAASGFTFSTHYIHWVRQAPGKGLEWVSSI NO: 163 Chain GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYYHMFDPWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTY ICNVNHKPSNTK VDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDT LMISRTPEVTCV VAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHYTQKSLSLSPGK SEQ ID DNA gaagtcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgcatgtgccgcgagcg NO: 164 Heavy ggtttactttctccaccactacatccactgggtccgccaagctcccggaaggactcgaatgggtgtcctccattggt Chain ggacagggcgcatgaccctttacgcgtagcgtgaaggggaggttcacatctcccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccaggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacctgaggacaggaacgctggtcaccgtgtcctccgctccactaagggcccgtcagtggt cccgctggctccatcgtcgaagtcacctccggaggaaccgcagcactcggttgctggtcaaggactacttcctga gccagtgaccgtgtcgtggaacagcggagccctgacttcggcgtgcacactttcccgcggtgctgcagtcctccggt ctgtactcccttcgtccgtggtcaccgtgccgtcgttagcctgggcacccagacctacatctgcaacgtgaaccacaa gccgtccaacaccaaagtggataagcgggtggagccgaagtcctgcgataagacacacacgtgcccgccatgtcca gcgcctgaattgcttggcggaccttcgtgttctgttcccgccctaagcccaaggacacctgatgattagccggactcc cgaagtcacctgtgtggtggcagtgccccagaggaccccgagggtcaagtttaattgttacgtggacggcgctcga agtgcacaacgccaaactaagccccgggaggaacagtacaacagcacctaccgggtcgtgtccgtgctgaccgtg ctgcaccaggactggctgaatgggaaagagtacaagtgc aaagtgtccaacaaggccttgccgctcctatcgaaaaa actatcagcaaggctaagggacagccgagggaaacccaagtctacacctgcccccttcacgcgaagagatgacca agaatacgaatgtcgtgacctgcctcgtcaagggttctacccctccgacattgcgggtggagtgaggagtgcaacggcc agcccgagaacaactacaagactactccgcccgtgctggactccgacggcagcttcttctgtattccaagctgaccgt ggacaagtccccggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagccctgcacaaccactatacgca gaagtcctgtccttgagccccggggaaa SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Combined) SEQ ID LCDR2 GASSRAT NO: 18 (Combined) SEQ ID LCDR3 QQHSMYPRT NO: 19 (Combined) SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Kabat) SEQ ID LCDR2 GASSRAT NO: 18 (Kabat) SEQ ID LCDR3 QQHSMYPRT NO: 19 (Kabat) SEQ ID LCDR1 QSITRNY NO: 20 (Chothia) SEQ ID LCDR2 GAS NO: 21 (Chothia) SEQ ID LCDR3 HSMYPR NO: 22 (Chothia) SEQ ID LCDR1 QSITRNY NO: 23 (IMGT) SEQ ID LCDR2 GAS NO: 21 (IMGT) SEQ ID LCDR3 QQHSMYPRT NO: 19 (IMGT) SEQ ID VL DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA NO: 24 SSRATGIPARFSGSGSGTDFTLTISSELPEDFAVYYCQQHSMYPRTFGQGTK VEIK SEQ

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SEQ ID Light

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HCDR1 THYIH NO: 7 (Kabat) SEQ ID HCDR2 SIGGQGGMTLYADSVKG NO: 151
(Kabat) SEQ ID HCDR3 ERGYVYYHMFDP NO: 6 (Kabat) SEQ ID HCDR1
GFTFQTH NO: 166 (Chothia) SEQ ID HCDR2 GGQGGM NO: 152 (Chothia)
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GFTFQTHY NO: 167 (IMGT) SEQ ID HCDR2 IGGQGGMT NO: 153 (IMGT) SEQ
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SEQ ID LCDR2 GASSRAT NO: 18 (Combined) SEQ ID LCDR3 QQHSMYPRT
NO: 19 (Combined) SEQ ID LCDR1 RASQSITRNYLA NO: 17 (Kabat) SEQ ID
LCDR2 GASSRAT NO: 18 (Kabat) SEQ ID LCDR3 QQHSMYPRT NO: 19 (Kabat)
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(Chothia) SEQ ID LCDR3 HSMYPR NO: 22 (Chothia) SEQ ID LCDR1
QSITRNY NO: 23 (IMGT) SEQ ID LCDR2 GAS NO: 21 (IMGT) SEQ ID
LCDR3 QQHSMYPRT NO: 19 (IMGT) SEQ ID VL
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(Combined) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Combined) SEQ
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SYWMN NO: 31 (Kabat) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Kabat)
SEQ ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1
GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 SSDGSY NO: 33 (Chothia) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSYW
NO: 34 (IMGT) SEQ ID HCDR2 ISSDGSYT NO: 35 (IMGT) SEQ ID HCDR3

ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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A I S S D G S Y T Y Y A D S V K G R F T I S R D N S K N T L Y L Q M N S L R A E D T A V Y Y C A R D
R Y S M I Y S Y G A G A F D Y W G Q G T L V T V S S SEQ ID DNA VH
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QVQLLES G G G L V Q P G G S L R L S C A A S G F T F S S Y W M N W V R Q A P G K G L E W V S NO: 39 Chain
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G C L V K D Y F P E P V T V S W N S G A L T S G V H T F P A V L Q S S G L Y S L S S V V T V P S S S L
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P K P K D T L M I S R T P E V T C V V V D V S H E D P E V K F N W Y V D G V E V H N A K T K P R E E
Q Y N S T Y R V V S V L T V L H Q D W L N G K E Y K C K V S N K A L P A P I E K T I S K A K G Q P R
E P Q V Y T L P P S R E E M T K N Q V S L T C L V K G F Y P S D I A V E W E S N G Q P E N N Y K T T P
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
QQEWAKPRT NO: 172 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat)
SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 QQEWAKPRT
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TAS NO: 45 (Chothia) SEQ ID LCDR3 EWAKPR NO: 173 (Chothia) SEQ ID
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ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
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TAS NO: 45 (Chothia) SEQ ID LCDR3 SWTRPR NO: 179 (Chothia) SEQ ID
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SYWMN NO: 31 (Kabat) SEQ ID HCDR2 AISSDGSYTYADSVKG NO: 29 (Kabat)
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GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 SSDGSY NO: 33 (Chothia) SEQ ID
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
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TAS NO: 45 (Chothia) SEQ ID LCDR3 IW MAPR NO: 185 (Chothia) SEQ ID
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DNA caagttcagctccttgagctctggggggggcctggtgcaacctgggggctctctgcggctttcatgtgcggcctcagggt
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SEQ ID LCDR1 RASQGISSYLA NO: 41 (Combined) SEQ ID LCDR2 TASTLQS
NO: 42 (Combined) SEQ ID LCDR3 QQTWRKPRT NO: 134 (Combined) SEQ ID
LCDR1 RASQGISSYLA NO: 41 (Kabat) SEQ ID LCDR2 TASTLQS NO: 42
(Kabat) SEQ ID LCDR3 QQTWRKPRT NO: 134 (Kabat) SEQ ID LCDR1
SQGISSY NO: 44 (Chothia) SEQ ID LCDR2 TAS NO: 45 (Chothia) SEQ ID
LCDR3 TWRKPRT NO: 135 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT)
SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQTWRKPRT NO: 134
(IMGT) SEQ ID VL
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ID DNA VL
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138 Chain TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV
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AISSKGSYTYADSVKG NO: 190 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31
(Kabat) SEQ ID HCDR2 AISSKGSYTYADSVKG NO: 190 (Kabat) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1 GFTFSSY NO:
32 (Chothia) SEQ ID HCDR2 SSKGSY NO: 191 (Chothia) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSY NO: 34
(IMGT) SEQ ID HCDR2 ISSKGSY NO: 192 (IMGT) SEQ ID HCDR3
ARDYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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QVQLLES G GGLVQP G GSLRLS CAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 195
Chain AISSKGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD

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DNA caagttcagctccttgagctctggggggggcctggtgcaacctgggggctctctgcggcttcatgtgcggcctcagggt NO:
196 Heavy tcactttcagctcatactggatgaattgggtacgccaagctccaggcaaaggactcgaatgggtaagcgctatatccagc
Chain aaaggagctatactattacgcggattccgtaagggcaggttcactatatccccgcgacaactccaaaaatactttgtat
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NO: 42 (Combined) SEQ ID LCDR3 QQEWAKPRT NO: 172 (Combined) SEQ ID
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SQGISSY NO: 44 (Chothia) SEQ ID LCDR2 TAS NO: 45 (Chothia) SEQ ID
LCDR3 EWAKPR NO: 173 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT)
SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQEWAKPRT NO: 172
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SEQ ID HCDR1 GFTFSSYWMN NO: 28 (Combined) SEQ ID HCDR2
VIESKGNIFYADSVKG NO: 119 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31
(Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1 GFTFSSY NO:
32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSYW NO: 34
(IMGT) SEQ ID HCDR2 IESKGNYY NO: 121 (IMGT) SEQ ID HCDR3
ARDYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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ID LCDR1 RASQGISSYLA NO: 41 (Combined) SEQ ID LCDR2 TASTLQS NO:
42 (Combined) SEQ ID LCDR3 QQEWVKPRT NO: 126 (Combined) SEQ ID
LCDR1 RASQGISSYLA NO: 41 (Kabat) SEQ ID LCDR2 TASTLQS NO: 42
(Kabat) SEQ ID LCDR3 QQEWVKPRT NO: 126 (Kabat) SEQ ID LCDR1
SQGISSY NO: 44 (Chothia) SEQ ID LCDR2 TAS NO: 45 (Chothia) SEQ ID

LCDR3 EWWVKPRT NO: 127 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT)
SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQEWVKPRT NO: 126
(IMGT) SEQ ID VL
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NO: 130 Chain
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DAPA SEQ ID HCDR1 GFTFSSYWMN NO: 28 (Combined) SEQ ID HCDR2
VIESKGNIFYADSVKG NO: 119 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31
(Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1 GFTFSSY NO:
32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSY NO: 34
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ARDYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 122
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42 (Combined) SEQ ID LCDR3 QQTWRKPRT NO: 134 (Combined) SEQ ID
LCDR1 RASQGISSYLA NO: 41 (Kabat) SEQ ID LCDR2 TASTLQS NO: 42
(Kabat) SEQ ID LCDR3 QQTWRKPRT NO: 134 (Kabat) SEQ ID LCDR1
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
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ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
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ID LCDR1 RASQGISSYLA NO: 41 (Combined) SEQ ID LCDR2 TASTLQS NO:
42 (Combined) SEQ ID LCDR3 QQIWTVPRT NO: 145 (Combined) SEQ ID
LCDR1 RASQGISSYLA NO: 41 (Kabat) SEQ ID LCDR2 TASTLQS NO: 42
(Kabat) SEQ ID LCDR3 QQIWTVPRT NO: 145 (Kabat) SEQ ID LCDR1
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LCDR3 IWTVPR NO: 146 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT)
SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQIWTVPRT NO: 145
(IMGT) SEQ ID VL
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VIESKGNIFYADSVKG NO: 119 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31
(Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1 GFTFSSY NO:
32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSY NO: 34
(IMGT) SEQ ID HCDR2 IESKGNYY NO: 121 (IMGT) SEQ ID HCDR3
ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH

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NO: 209 Heavy

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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
QQIWTVPRT NO: 145 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat)
SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 QQIWTVPRT NO:
145 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2 TAS
NO: 45 (Chothia) SEQ ID LCDR3 IWTVPR NO: 146 (Chothia) SEQ ID LCDR1
QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3
QQIWTVPRT NO: 145 (IMGT) SEQ ID VL
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(Combined) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Combined) SEQ
ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
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ID HCDR3 ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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SEQ ID HCDR1 GFTFSSYWMN NO: 28 (Combined) SEQ ID HCDR2
VIESKGNIFYADSVKG NO: 119 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31
(Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat) SEQ ID
HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1 GFTFSSY NO:
32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSY NO: 34
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EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 122
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EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 208
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
QQEWAKPRT NO: 172 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat)
SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 QQEWAKPRT

NO: 172 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2
TAS NO: 45 (Chothia) SEQ ID LCDR3 EWAKPR NO: 173 (Chothia) SEQ ID
LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID
LCDR3 QQEWAKPRT NO: 172 (IMGT) SEQ ID VL
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(Combined) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Combined) SEQ
ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
SYWMN NO: 31 (Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat)
SEQ ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1
GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ
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GFTFSSYW NO: 34 (IMGT) SEQ ID HCDR2 IESKGNFY NO: 121 (IMGT) SEQ
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42 (Combined) SEQ ID LCDR3 QQSWTRPRT NO: 178 (Combined) SEQ ID
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LCDR3 SWTRPR NO: 179 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT)
SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQSWTRPRT NO: 178
(IMGT) SEQ ID VL

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ID DNA VL

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DAPA SEQ ID HCDR1 GFTFSSYWMN NO: 28 (Combined) SEQ ID HCDR2
VIESKGNFYADSVKG NO: 119 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31

(Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKGN NO: 119 (Kabat) SEQ ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Kabat) SEQ ID HCDR1 GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Chothia) SEQ ID HCDR1 GFTFSSYW NO: 34 (IMGT) SEQ ID HCDR2 IESKGNYY NO: 121 (IMGT) SEQ ID HCDR3 ARDRYSMIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH EVQLES GGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 122 VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGT LTVVSS SEQ ID DNA VH gaagtgcagctgctggaatccggcgaggctgtgtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg NO: 207 attcaccttttcgtcgtactggatgaactgggtcagacaggctcctggaaagggcctggaatgggtgtctgtgattgaatc caaggggaactacatcttctacgcggacagcgtgaagggccggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcgatcgtactcgatgatctac agctatgggtgccggagcgttcgattactggggacaggggaaccctcgtgaccgtcagctcc SEQ ID Heavy EVQLES GGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS NO: 208 Chain VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGT LTVVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTPETCVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSDGSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHYTQKSLSLSPGK SEQ ID DNA gaagtgcagctgctggaatccggcgaggctgtgtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg NO: 209 Heavy attcaccttttcgtcgtactggatgaactgggtcagacaggctcctggaaagggcctggaatgggtgtctgtgattgaatc Chain caaggggaactacatcttctacgcggacagcgtgaagggccggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcgatcgtactcgatgatctac agctatgggtgccggagcgttcgattactggggacaggggaaccctcgtgaccgtcagctccgcctcaaccaagggccc gtcagtgttcccgtgggtccatcgtcgaagtccacctccggaggaaccgcagcactcgggtgcctgggtcaaggactac ttccctgagccagtgaccgtgtcgtggaacagcggagccctgacttccggcgtgcacacttttccgcgggtgtcaggt cctccgggtgtactcccttcgtccgtgggtcaccgtgccgtcgtctagcctgggcacccagacctacatctgcaacgtg aaccacaagccgtccaacaccaaagtggataagcgggtggagccgaagtcctgcgataagacacacacgtgcccgc catgtccagcgctgaattgcttgccggaccttccgtgttctgttccgcctaagcccaaggacaccttgatgattagcc ggactcccgaagtcacctgtgtgggtgggtgaggtgtccacagaggaccccgaggtcaagttaattggtacgtggacg gcgtcgaagtgcacaacgccaagactaagccccgggaggaacagtacaacagcacctaccgggtcgtgtccgtgtc gaccgtgctgcaccaggactgggtgaatgggaaagagtacaagtgtcaaaagtgtccaacaaggccttgccgctcctat cgaaaaaactatcagcaaggctaagggacagccgaggggaaccccaagtctacacctgcccccttcacgcgaagag atgaccaagaatcaagtgtcgtgacctgcctcgtcaagggtattaccctccgacattgcgggtggagtgggagtgcca acggccagcccgagaacaactacaagactactccgcccgtgtggtgactccgacggcagcttcttctgtattccaagct gaccgtggacaagtccccgtggcagcaaggaaacgtgttctcctgctcgggtcatgcacgaagccctgcacaaccacta tacgcagaagtcctgtccttgagcccggggaaa SEQ ID LCDR1 RASQGISSYLA NO: 41 (Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3 QQSWTRPRT NO: 178 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41 (Kabat) SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3 QQSWTRPRT NO: 178 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia) SEQ ID LCDR2 TAS NO: 45 (Chothia) SEQ ID LCDR3 SWTRPR NO: 179 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQSWTRPRT NO: 178 (IMGT) SEQ ID VL DIQMTQSPSSLSASVGDRVTITCRASQGISSYLA WYQQKPGKAPKLLIYTAS NO: 180

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ID HCDR3 DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1
SYWMN NO: 31 (Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat)
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GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ
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42 (Combined) SEQ ID LCDR3 QQIWMAPRT NO: 184 (Combined) SEQ ID
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(Kabat) SEQ ID LCDR3 QQIWMAPRT NO: 184 (Kabat) SEQ ID LCDR1
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LCDR3 IWMAPR NO: 185 (Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT)
SEQ ID LCDR2 TAS NO: 45 (IMGT) SEQ ID LCDR3 QQIWMAPRT NO: 184
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VIESKGNIFYADSVKG NO: 119 (Combined) SEQ ID HCDR3
DRYSMIYSYGAGAFDY NO: 30 (Combined) SEQ ID HCDR1 SYWMN NO: 31
(Kabat) SEQ ID HCDR2 VIESKGNIFYADSVKG NO: 119 (Kabat) SEQ ID
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32 (Chothia) SEQ ID HCDR2 ESKGNY NO: 120 (Chothia) SEQ ID HCDR3
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(IMGT) SEQ ID HCDR2 IESKGNYY NO: 121 (IMGT) SEQ ID HCDR3

ARDRYMSIYSYGAGAFDY NO: 36 (IMGT) SEQ ID VH
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(Combined) SEQ ID LCDR2 TASTLQS NO: 42 (Combined) SEQ ID LCDR3
QQIWMAPRT NO: 184 (Combined) SEQ ID LCDR1 RASQGISSYLA NO: 41
(Kabat) SEQ ID LCDR2 TASTLQS NO: 42 (Kabat) SEQ ID LCDR3
QQIWMAPRT NO: 184 (Kabat) SEQ ID LCDR1 SQGISSY NO: 44 (Chothia)
SEQ ID LCDR2 TAS NO: 45 (Chothia) SEQ ID LCDR3 IWMAPR NO: 185
(Chothia) SEQ ID LCDR1 QGISSY NO: 47 (IMGT) SEQ ID LCDR2 TAS NO:
45 (IMGT) SEQ ID LCDR3 QQIWMAPRT NO: 184 (IMGT) SEQ ID VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSYLA WYQQKPGKAPKLLIYTAS NO: 186
TLQSGVPSRFS GSGSGTDFTLTIS SLQPEDFATYYCQQIWMAPRTFGQGTKV EIK SEQ
ID DNA VL
gacatccagatgacctcgtctcctcctcctgtccgcatcagtgggggacagagtgacctacactgtcgggcctccc NO: 224
aaggcatctcgtcatactggcctgggtatcagcagaaacccggaaaggctccaaagctgctcatctacaccgctcga
ctctgcaatccggagtgccttcccgtcttccggatccgggttcgggaaccgacttcacctcaccattagcagccttcag

ccggaagatttcgcgacactactgcagcaaatctggatggcccccaggacatttggccaggggacaggtcgag attaaag SEQ ID Light DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWEYQQKPGKAPKLLIYTAS NO: 188 Chain

TLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQIWMAPRTFGQGTKV
EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ
SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID DNA

gacatccagatgacccagtcctccgtcctcctgtccgcatcagtgccgggacagagtgaccatcactgtcgggcctccc NO: 225
Light aaggcatctcgtcatacctggcctggtatcagcagaaacccggaaaggctccaaagctgctcatctacaccgcctcga Chain
ctctgcaatccggagtgcttcccgtcttcggatccgggtcgggaaccgacttcaccctcaccattagcagccttcag
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agagcttcaacaggggagcagtg YY01_LALA SEQ ID HCDR1 GFTFSSYWIS NO: 226
(Combined) SEQ ID HCDR2 NIKQSGSETYYVESVKG NO: 227 (Combined) SEQ ID
HCDR3 SLRRRSTEHAGFDV NO: 228 (Combined) SEQ ID HCDR1 SYWIS NO:
229 (Kabat) SEQ ID HCDR2 NIKQSGSETYYVESVKG NO: 227 (Kabat) SEQ ID
HCDR3 SLRRRSTEHAGFDV NO: 228 (Kabat) SEQ ID HCDR1 GFTFSSY NO: 32
(Chothia) SEQ ID HCDR2 KQSGSE NO: 230 (Chothia) SEQ ID HCDR3
SLRRRSTEHAGFDV NO: 228 (Chothia) SEQ ID HCDR1 GFTFSSYW NO: 34
(IMGT) SEQ ID HCDR2 IKQSGSET NO: 231 (IMGT) SEQ ID HCDR3
ARSLRRRSTEHAGFDV NO: 232 (IMGT) SEQ ID VH
EVQLVESGGGLVQPGLSLRLSCAASGFTFSSYWISWVRQAPGKGLEWVAN NO: 233
IKQSGSETYYVESVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARSLR
RRSTEHAGFDVWGQGLTVTVSS SEQ ID DNA VH

gaagtgcagctggtggaaagcggcggtggcctggtgcagccaggtggtagcctgcgcctgagctgcgccgccagc NO: 234
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gcactgagcacgcaggattcgacgtttggggccaggggcaccctggttactgtctcgagc SEQ ID Heavy
EVQLVESGGGLVQPGLSLRLSCAASGFTFSSYWISWVRQAPGKGLEWVAN NO: 235
Chain IKQSGSETYYVESVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARSLR
RRSTEHAGFDVWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV
KDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTY
ICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT
LMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPDSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
GSFFLYSLKLTVDKSRWQQGNVVFSCSVMHEALHNHYTQKSLSLSPGK SEQ ID
DNA gaagtgcagctggtggaaagcggcggtggcctggtgcagccaggtggtagcctgcgcctgagctgcgccgccagc
NO: 236 Heavy

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ggccctgcacaaccactacaccagaagtcctgagcctgagccccggcaag SEQ ID LCDR1 RASQGISNYLA
NO: 237 (Combined) SEQ ID LCDR2 AASTLQS NO: 238 (Combined) SEQ ID
LCDR3 QQADKFPYT NO: 239 (Combined) SEQ ID LCDR1 RASQGISNYLA NO:
237 (Kabat) SEQ ID LCDR2 AASTLQS NO: 238 (Kabat) SEQ ID LCDR3
QQADKFPYT NO: 239 (Kabat) SEQ ID LCDR1 SQGISNY NO: 240 (Chothia) SEQ
ID LCDR2 AAS NO: 241 (Chothia) SEQ ID LCDR3 ADKFPY NO: 242 (Chothia)
SEQ ID LCDR1 QGISNY NO: 243 (IMGT) SEQ ID LCDR2 AAS NO: 241
(IMGT) SEQ ID LCDR3 QQADKFPYT NO: 239 (IMGT) SEQ ID VL
DIQMTQSPSSLSASVGDRVITTCRASQGISNYLAWYQQKPGKVPKLLIYAAS NO: 244
TLQSGVPSRFSGSGSGTDFTLTISLQPEDVATYYCQQADKFPYTFGQGTKV EIK SEQ
ID DNA VL
gatattcagatgacccagagcccagcagcctgagcgcgaagcgtgggcgatcgctgaccattacctgccgcgcca NO: 245
gccagggcattagcaactatctggcctggtatcagcagaaaccggggcaaagtgccgaaactgctgatctatgccgcca
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SEQ ID Light
DIQMTQSPSSLSASVGDRVITTCRASQGISNYLAWYQQKPGKVPKLLIYAAS NO: 246
Chain TLQSGVPSRFSGSGSGTDFTLTISLQPEDVATYYCQQADKFPYTFGQGTKV
EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ
SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID DNA
gatattcagatgacccagagcccagcagcctgagcgcgaagcgtgggcgatcgctgaccattacctgccgcgcca NO: 247
Light gccagggcattagcaactatctggcctggtatcagcagaaaccggggcaaagtgccgaaactgctgatctatgccgcca Chain
gcaccctgcaaagcggcgtgccaagtcgcttagcggcagcggtagcggcaccgattcacctgaccattagcagc
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(Combined) SEQ ID HCDR2 SSSSSSYIYYADSVKG NO: 249 (Combined) SEQ ID
HCDR3 SGYRGVYGFYD NO: 250 (Combined) SEQ ID HCDR1 SYSMN NO: 251
(Kabat) SEQ ID HCDR2 SSSSSSYIYYADSVKG NO: 249 (Kabat) SEQ ID
HCDR3 SGYRGVYGFYD NO: 250 (Kabat) SEQ ID HCDR1 GFTFSSY NO: 32
(Chothia) SEQ ID HCDR2 SSSSY NO: 252 (Chothia) SEQ ID HCDR3
SGYRGVYGFYD NO: 250 (Chothia) SEQ ID HCDR1 GFTFSSYS NO: 253 (IMGT)
SEQ ID HCDR2 ISSSSSYI NO: 254 (IMGT) SEQ ID HCDR3
ARSGYRGVYGFYD NO: 255 (IMGT) SEQ ID VH
EVQLVESGGGLVKPGGSLRLSCAASGFTFSSYSMNWVRQAPGKGLEWVSS NO: 256
ISSSSSYIYYADSVKGRFTISRDNALNSLYLQMNSLRAEDTAVYYCARSGY
RGVYGFYDYGWGQGLTVTVSS SEQ ID DNA VH

gaagtgcagctggtggaaagcggcggtggcctggtgaaaccaggcggtagcctgcgcctgagctgcgccgccagc NO: 257
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tacggatttgattattggggccaggggcacctgggttactgtctcgagc SEQ ID Heavy
EVQLVESGGGLVKPGLSLRSCAASGFTFSSYSMNWVRQAPGKGLEWVSS NO: 258
Chain ISSSSSYIYYADSVKGRFTISRDNALNSLYLQMNSLRADTAVYYCARSGY
RGVYGFQDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKD
YFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYIC
NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTL
MISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR
VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLR
PSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGS
FFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQKSLSLSPGK SEQ ID DNA
gaagtgcagctggtggaaagcggcggtggcctggtgaaaccaggcggtagcctgcgcctgagctgcgccgccagc NO: 259
Heavy ggctttaccttttagcagctatagcatgaactgggttcgccaggccccaggcaaaaggcctggaatgggtagcagcatca
Chain gcagcagtagcagctatatctattacgccgatagcgtgaaaggccgctttaccattagccgcgataacgccccaaacag
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(Combined) SEQ ID LCDR2 AASSLQS NO: 261 (Combined) SEQ ID LCDR3
QQYYHSPLT NO: 262 (Combined) SEQ ID LCDR1 RASQGISSWLA NO: 260
(Kabat) SEQ ID LCDR2 AASSLQS NO: 261 (Kabat) SEQ ID LCDR3
QQYYHSPLT NO: 262 (Kabat) SEQ ID LCDR1 SQGISSW NO: 263 (Chothia) SEQ
ID LCDR2 AAS NO: 241 (Chothia) SEQ ID LCDR3 YYHSPL NO: 264 (Chothia)
SEQ ID LCDR1 QGISSW NO: 265 (IMGT) SEQ ID LCDR2 AAS NO: 241
(IMGT) SEQ ID LCDR3 QQYYHSPLT NO: 262 (IMGT) SEQ ID VL
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ID DNA VL
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SEQ ID Light
DIQMTQSPSSVSASVGDRVTITCRASQGISSWLAWYQQKPGKAPKLLIYAA NO: 268
Chain SSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYYHSPLTFGQGGTK
VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL

QSGNSQESVTEQDSKDYSLSTLTLSKADYEEKHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID DNA
gatattcagatgacccagagccccgagcagcgtagcgccagcggtggcgatcgcgtagaccattacctgccgcgccag NO: 269
Light tcagggcattagcagctggctggcctggatcagcagaaaccgggcaaagccccgaaactgctgatctatgccgcca Chain
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caagagcttcaaccggggcgagtgt YY03_LALA SEQ ID HCDR1 GFTFSSY AIS NO: 270
(Combined) SEQ ID HCDR2 AISGSGGSTYYAESVKG NO: 271 (Combined) SEQ
ID HCDR3 ESGYVYYLKFDY NO: 272 (Combined) SEQ ID HCDR1 SYAIS
NO: 273 (Kabat) SEQ ID HCDR2 AISGSGGSTYYAESVKG NO: 271 (Kabat) SEQ
ID HCDR3 ESGYVYYLKFDY NO: 272 (Kabat) SEQ ID HCDR1 GFTFSSY NO:
32 (Chothia) SEQ ID HCDR2 SGSGGS NO: 274 (Chothia) SEQ ID HCDR3
ESGYVYYLKFDY NO: 272 (Chothia) SEQ ID HCDR1 GFTFSSY NO: 275 (IMGT)
SEQ ID HCDR2 ISGSGGST NO: 276 (IMGT) SEQ ID HCDR3
ARESGYVYYLKFDY NO: 277 (IMGT) SEQ ID VH
EVQLLES GGGGLVQPGGSLRLSCAASGFTFSSY AISWVRQAPGKGLEWVSAI NO: 278
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YVYYLKFDYWGQGT LTVSS SEQ ID DNA VH
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EVQLLES GGGGLVQPGGSLRLSCAASGFTFSSY AISWVRQAPGKGLEWVSAI NO: 280
Chain SGSGGSTYYAESVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARESG
YVYYLKFDYWGQGT LTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYI
CNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT
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ccactacaccagaagtccctgagcctgagccccggcaag SEQ ID LCDR1 RASQSISSYLN NO: 282
(Combined) SEQ ID LCDR2 AASSLQS NO: 261 (Combined) SEQ ID LCDR3
QQHVRVPIT NO: 283 (Combined) SEQ ID LCDR1 RASQSISSYLN NO: 282
(Kabat) SEQ ID LCDR2 AASSLQS NO: 261 (Kabat) SEQ ID LCDR3
QQHVRVPIT NO: 283 (Kabat) SEQ ID LCDR1 SQSISSY NO: 284 (Chothia) SEQ
ID LCDR2 AAS NO: 241 (Chothia) SEQ ID LCDR3 HVRVPI NO: 285 (Chothia)
SEQ ID LCDR1 QSISSY NO: 286 (IMGT) SEQ ID LCDR2 AAS NO: 241
(IMGT) SEQ ID LCDR3 QQHVRVPIT NO: 283 (IMGT) SEQ ID VL
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAAS NO: 287
SLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQHVRVPITFGQGGTKVE IK SEQ ID
DNA VL gatattcagatgacccagagccccgagcagcctgagcgccagcgtgggtgatcgctgaccattacctgtcgcgcaag
NO: 288 ccagagcattagcagctatctgaactggtatcagcagaaaccaggcaaagccccaaaactgctgatttatgccgcaag
cagcctgcaaagcgggtgtgccgagccgcttagcggcagcggtagcggcaccgattttaccctgaccattagtagcct
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SEQ ID Light
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAAS NO: 289
Chain SLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQHVRVPITFGQGGTKVE
IKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQS
GNSQESVTEQDSKSTYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTK SFNRGEC
SEQ ID DNA
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Light ccagagcattagcagctatctgaactggtatcagcagaaaccaggcaaagccccaaaactgctgatttatgccgcaag Chain
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caagagcttcaaccggggcgagtgt YY04_LALA SEQ ID HCDR1 GFTFSNYWIS NO: 291
(Combined) SEQ ID HCDR2 RIKSKTYGGTTDYAEPVKG NO: 292 (Combined) SEQ
ID HCDR3 EKYSIRARGHGDYGFVDV NO: 293 (Combined) SEQ ID HCDR1
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(Kabat) SEQ ID HCDR3 EKYSIRARGHGDYGFVDV NO: 293 (Kabat) SEQ ID
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(Chothia) SEQ ID HCDR3 EKYSIRARGHGDYGFVDV NO: 293 (Chothia) SEQ ID
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NO: 302 Chain

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(Combined) SEQ ID LCDR3 QQGYHAPFT NO: 304 (Combined) SEQ ID LCDR1
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SEQ ID LCDR3 QQGYHAPFT NO: 304 (Kabat) SEQ ID LCDR1 SQGISNY
NO: 240 (Chothia) SEQ ID LCDR2 AAS NO: 241 (Chothia) SEQ ID LCDR3
GYHAPF NO: 305 (Chothia) SEQ ID LCDR1 QGISNY NO: 243 (IMGT) SEQ ID
LCDR2 AAS NO: 241 (IMGT) SEQ ID LCDR3 QQGYHAPFT NO: 304 (IMGT)
SEQ ID VL
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(Chothia) SEQ ID HCDR2 YPGTSY NO: 313 (Chothia) SEQ ID HCDR3
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320 (Combined) SEQ ID LCDR2 GNSNRPS NO: 321 (Combined) SEQ ID

LCDR3 QSYYTSSHGPV NO: 322 (Combined) SEQ ID LCDR1
TGSSSNIGAGYDVH NO: 320 (Kabat) SEQ ID LCDR2 GNSNRPS NO: 321
(Kabat) SEQ ID LCDR3 QSYYTSSHGPV NO: 322 (Kabat) SEQ ID LCDR1
SSSNIGAGYD NO: 323 (Chothia) SEQ ID LCDR2 GNS NO: 324 (Chothia) SEQ
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NO: 270 (Combined) SEQ ID HCDR2 AISGSGGSTYYAESVKG NO: 271 (Combined)
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GFTFSSY NO: 32 (Chothia) SEQ ID HCDR2 SGSGGS NO: 274 (Chothia) SEQ ID
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NO: 337 (Combined) SEQ ID LCDR2 EGSKRPS NO: 338 (Combined) SEQ ID
LCDR3 SSYGFHIVVVV NO: 339 (Combined) SEQ ID LCDR1 TGTSSDVGSYNLVS
NO: 337 (Kabat) SEQ ID LCDR2 EGSKRPS NO: 338 (Kabat) SEQ ID LCDR3
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NO: 341 (IMGT) SEQ ID LCDR3 SSYGFHIVVVV NO: 339 (IMGT) SEQ ID VL
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NO: 310 (Combined) SEQ ID HCDR2 IYPGTSYTRYSPSFQG NO: 311 (Combined)
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80 (Chothia) SEQ ID HCDR2 YPGTSY NO: 313 (Chothia) SEQ ID HCDR3
GSLPGLLGFDH NO: 348 (Chothia) SEQ ID HCDR1 GYSFTSYW NO: 82 (IMGT)
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320 (Combined) SEQ ID LCDR2 GNSNRPN NO: 354 (Combined) SEQ ID
LCDR3 QSYDSPTSSSV NO: 355 (Combined) SEQ ID LCDR1 TGSSSNIGAGYDVH
NO: 320 (Kabat) SEQ ID LCDR2 GNSNRPN NO: 354 (Kabat) SEQ ID LCDR3
QSYDSPTSSSV NO: 355 (Kabat) SEQ ID LCDR1 SSSNIGAGYD NO: 323 (Chothia)
SEQ ID LCDR2 GNS NO: 324 (Chothia) SEQ ID LCDR3 YDSPTSSS NO: 356
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80 (Chothia) SEQ ID HCDR2 YPGTSY NO: 313 (Chothia) SEQ ID HCDR3
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320 (Combined) SEQ ID LCDR2 GNSNRPN NO: 354 (Combined) SEQ ID
LCDR3 QSYGAFPRFVV NO: 361 (Combined) SEQ ID LCDR1
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NO: 237 (Combined) SEQ ID LCDR2 AASTLQS NO: 238 (Combined) SEQ ID
LCDR3 QQADKFPYT NO: 239 (Combined) SEQ ID LCDR1 RASQGISNYLA
NO: 237 (Kabat) SEQ ID LCDR2 AASTLQS NO: 238 (Kabat) SEQ ID
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(Chothia) SEQ ID LCDR2 AAS NO: 241 (Chothia) SEQ ID LCDR3 ADKFPY
NO: 242 (Chothia) SEQ ID LCDR1 QGISNY NO: 243 (IMGT) SEQ ID LCDR2
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NO: 383 (IMGT) SEQ ID HCDR2 IHQHGLE NO: 384 (IMGT) SEQ ID
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NO: 237 (Combined) SEQ ID LCDR2 AASTLQS NO: 238 (Combined) SEQ ID
LCDR3 QQADKFPYT NO: 239 (Combined) SEQ ID LCDR1 RASQGISNYLA NO:
237 (Kabat) SEQ ID LCDR2 AASTLQS NO: 238 (Kabat) SEQ ID LCDR3
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NO: 241 (IMGT) SEQ ID LCDR3 QQADKFPYT NO: 239 (IMGT) SEQ ID VL
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ID HCDR3 PYLGDRRSYGFDH NO: 331 (Combined) SEQ ID HCDR1 SYAIS
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NO: 32 (Chothia) SEQ ID HCDR2 SSHGYT NO: 390 (Chothia) SEQ ID
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Chain SHGYTRYAESVKGRFTISRDNSTNTLYLQMNSLRAEDTAVYYCARPYLG
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NO: 337 (Combined) SEQ ID LCDR2 EGSKRPS NO: 338 (Combined) SEQ ID
LCDR3 SSYGFHIVVVV NO: 339 (Combined) SEQ ID LCDR1 TGTSSDVGSYNLVS
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SEQ ID LCDR2 EGS NO: 341 (Chothia) SEQ ID LCDR3 YGFHIVVV NO: 342
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Chain GSKRPSGVSNRSTFSLTASLTISGLQAEDHYKSSYGFHIVVVFGG
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Chain SGSGVYTYAESVKG RFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL
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NO: 337 (Combined) SEQ ID LCDR2 EGSKRPS NO: 338 (Combined) SEQ ID
LCDR3 SSYGFHIVVVV NO: 339 (Combined) SEQ ID LCDR1 TGTSSDVGSYNLVS
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340 (Chothia) SEQ ID LCDR2 EGS NO: 341 (Chothia) SEQ ID LCDR3
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Heavy gggttacattcggcacctatgcaatgacttgggtgcgccaagcaccaggcaaaggcctggaatgggtgagtagcattag
Chain cgcacccgatattacgctaactacgcaggcagcgtcaaaggccgcttaccattagtcgcgataacagcaaaaacacc
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cctctggccccagcagcaagagcacctctggcggaacagccgcctgggctgcctgggtcaaggactacttccccga
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cacaagcccagcaacacaaaggtggacaagcgggtggaaccaagagctgcgacaagaccacacactgtcccccc
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cggacccccgaagtgcctgctgggtggacgtgtccacagaggacctgaagtgaagttaattggtacgtggac
ggcgtggaagtgcacaacccaagaccaagcccagagaggaacagtacaacagcacctaccgggtggtgtccgtg
ctgaccgtgctgcaccaggactggctgaacggcaaagagtacaagtgaaggtgtccaacaaggccctgcctgcccc
catcgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccggga
agagatgaccaagaaccaggtgtccctgacctgcctcgtgaagggttctaccccagcgacattgccgtggaatggga
gagcaacggccagcccgagaacaactacaagaccacccccctgtgctggacagcgacggctcattcttctgtaca
gcaagctgaccgtggacaagagccggtggcagcagggcaacgtgttcagctgctccgtgatgcacgaggccctgca
caaccactacaccagaagtcctgagcctgagccccggcaag SEQ ID LCDR1 TGTSSDVGSYNLVS
NO: 337 (Combined) SEQ ID LCDR2 EGSKRPS NO: 338 (Combined) SEQ ID
LCDR3 SSYGFHIVVVV NO: 339 (Combined) SEQ ID LCDR1 TGTSSDVGSYNLVS
NO: 337 (Kabat) SEQ ID LCDR2 EGSKRPS NO: 338 (Kabat) SEQ ID LCDR3
SSYGFHIVVVV NO: 339 (Kabat) SEQ ID LCDR1 TSSDVGSYNL NO: 340 (Chothia)
SEQ ID LCDR2 EGS NO: 341 (Chothia) SEQ ID LCDR3 YGFHIVVV NO: 342
(Chothia) SEQ ID LCDR1 SSDVGSYNL NO: 343 (IMGT) SEQ ID LCDR2 EGS
NO: 341 (IMGT) SEQ ID LCDR3 SSYGFHIVVVV NO: 339 (IMGT) SEQ ID VL
QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSQYQHPGKAPKLMIE NO: 344
GSKRPSGVSNRFSGSKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFGG GTKLTVL
SEQ ID DNA VL
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gcagcgacgtgggcagctataacctggttagctggtatcagcagcatccgggcaaagccccgaaactgatgatctatg
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tagcggcctgcaagccgaagacgaagccgattattactgtcctcttacgggttccatcgtgtgtgtgtgtgtggcggc
ggtaccaagctgaccgtgctg SEQ ID Light
QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSQYQHPGKAPKLMIE NO: 346
Chain GSKRPSGVSNRFSGSKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFGG
GTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA
DSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGST
VEKTVAPTECS SEQ ID DNA
cagagcgccctgaccagccagccagcgttagcggtagcccaggccagagcattaccattagctgcaccggcacca NO: 347

Light gcagcgacgtgggcagctataacctgggttagctgggtatcagcagcatccgggcaagccccgaactgatgatg Chain
aaggcagcaaagcggcgagcggttagcaaccgcttagtgaggcagcaaaagcggcaacaccgcccagcctgacat
tagcggcctgcaagccgaagacgaagccgattactgtcctcttacgggttccatcatcggtgtgtgtgtgtgtggcggc
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agctacctgagcctgacccccgagcagtggaagtcacacagatcctacagctgccaggtcacacacgagggcagca
ccgtggaaaagaccgtggccccaccgagtgagc ZZ17_ LALA SEQ ID HCDR1 GFTFSDY AIS
NO: 418 (Combined) SEQ ID HCDR2 SISGGGYHTQYAGSVKG NO: 419
(Combined) SEQ ID HCDR3 PYLGDRRSYGFDH NO: 331 (Combined) SEQ ID
HCDR1 DY AIS NO: 420 (Kabat) SEQ ID HCDR2 SISGGGYHTQYAGSVKG NO:
419 (Kabat) SEQ ID HCDR3 PYLGDRRSYGFDH NO: 331 (Kabat) SEQ ID
HCDR1 GFTFSDY NO: 421 (Chothia) SEQ ID HCDR2 SGGGYH NO: 422
(Chothia) SEQ ID HCDR3 PYLGDRRSYGFDH NO: 331 (Chothia) SEQ ID
HCDR1 GFTFSDY NO: 423 (IMGT) SEQ ID HCDR2 ISGGGYHT NO: 424
(IMGT) SEQ ID HCDR3 ARPYLGDRRSYGFDH NO: 332 (IMGT) SEQ ID VH
EVQLLES GGGGLVQPGGSLRLSCAASGFTFSDY AISWVRQAPGKGLEWVSSI NO: 425
SGGGYHTQYAGSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL
GDRRSYGFDHWGQGT LVT VSS SEQ ID DNA VH
gaagtgcagctgctggaaagcgggtggcgggtcgtggtgcagccaggtggttagcctgcgcctgagctgtgccgcaagcg NO: 426
gctttacctttccgactatgcaatcagctgggtgcgccaagcaccaggcaaaaggcctggaatgggtgagcagcatttc
cgggggggggtatcatacacaatatgcaggatccgtgaaaggccgctttaccattagtcgcgataacagcaaaaacac
cctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttattattgcgcgcgaccttatctgggtgaccgtc
gtagctatggtttcaccactggggccagggcaccctggttactgtctcgagc SEQ ID Heavy
EVQLLES GGGGLVQPGGSLRLSCAASGFTFSDY AISWVRQAPGKGLEWVSSI NO: 427
Chain SGGGYHTQYAGSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL
GDRRSYGFDHWGQGT LVT VSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYI
CNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT
LMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY
RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSD
GSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK SEQ ID DNA
gaagtgcagctgctggaaagcgggtggcgggtcgtggtgcagccaggtggttagcctgcgcctgagctgtgccgcaagcg NO: 428
Heavy gctttacctttccgactatgcaatcagctgggtgcgccaagcaccaggcaaaaggcctggaatgggtgagcagcatttc Chain
cgggggggggtatcatacacaatatgcaggatccgtgaaaggccgctttaccattagtcgcgataacagcaaaaacac
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agagcaacggccagcccagagaacaactacaagaccacccccctgtgctggacagcgacggctcatttctctgtac
agcaagctgaccgtggacaagagccggtggcagcagggcaacgtgttcagctgctccgtgatgcagaggccctgc

acaaccactacaccagaagtcctgagcctgagccccggcaag SEQ ID LCDR1 TGTSSDVGSYNLVS
NO: 337 (Combined) SEQ ID LCDR2 EGSKRPS NO: 338 (Combined) SEQ ID
LCDR3 SSYGFHIVVVV NO: 339 (Combined) SEQ ID LCDR1 TGTSSDVGSYNLVS
NO: 337 (Kabat) SEQ ID LCDR2 EGSKRPS NO: 338 (Kabat) SEQ ID LCDR3
SSYGFHIVVVV NO: 339 (Kabat) SEQ ID LCDR1 TSSDVGSYNL NO: 340 (Chothia)
SEQ ID LCDR2 EGS NO: 341 (Chothia) SEQ ID LCDR3 YGFHIVVV NO: 342
(Chothia) SEQ ID LCDR1 SSDVGSYNL NO: 343 (IMGT) SEQ ID LCDR2 EGS
NO: 341 (IMGT) SEQ ID LCDR3 SSYGFHIVVVV NO: 339 (IMGT) SEQ ID VL
QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSQYQHPGKAPKLMIYE NO: 344
GSKRPSGVSNRFSGSKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFVG GTKLTVL
SEQ ID DNA VL

SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, WW01_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 13, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, WW01_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 15, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(70) For example, WW03_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3).

In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 48. In some embodiments, WW03_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 50.

(71) For example, WW05_LALA may be defined as having three heavy chain complementarity regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (II) SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (III) SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3); or (IV) SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57

(HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 61, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 72. In some embodiments, WW05_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 63, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 74.

(72) For example, WW06_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3). In some embodiments, WW06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3). In some embodiments, WW06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3). In some embodiments, WW06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3). In some embodiments, WW06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3). In some embodiments, WW06_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 85, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 96. In some embodiments, WW06_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 87, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 98.

(73) For example, XX01_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some

embodiments, XX01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 103, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX01_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 105, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(74) For example, XX01_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 103, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX01_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 108, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(75) For example, XX01_N30S_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 112 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 113 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 114 (HCDR1), SEQ ID NO: 102

(HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 112 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 113 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX01_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 114 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_N30S_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 115, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX01_N30S_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 117, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(76) For example, XX03_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, XX03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, XX03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, XX03_LALA may be defined as ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3). In some embodiments, XX03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, XX03_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 48. In some embodiments, XX03_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 124, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 50.

(77) For example, XX04_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29

(HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3). In some embodiments, XX04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX04_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 128. In some embodiments, XX04_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 130.

(78) For example, XX06_LALA may be defined as having three heavy chain complementarity regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3). In some embodiments, XX06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136. In some embodiments, XX06_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 138.

(79) For example, XX06_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity

determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3). In some embodiments, XX06_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX06_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136. In some embodiments, XX06_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 141, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 138.

(80) For example, XX07_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3). In some embodiments, XX07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX07_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 147. In some embodiments, XX07_LALA may be defined as comprising or having

a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 149.

(81) For example, XX08_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 4 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 8 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX08_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 10 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 154, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX08_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 156, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(82) For example, XX08_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 4 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 8 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX08_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 10 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2),

and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 154, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX08_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 159, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(83) For example, XX08_N30S_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 112 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 113 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 114 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 112 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 113 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 114 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 161, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX08_N30S_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 163, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(84) For example, XX08_N30Q_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 165 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 166 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 167 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30Q_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 165 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30Q_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and

SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30Q_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 166 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX08_N30Q_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 167 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30Q_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 168, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, XX08_N30Q_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 170, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

(85) For example, XX09_LALA may be defined as having three heavy chain complementarity regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX09_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX09_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX09_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3). In some embodiments, XX09_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX09_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 174. In some embodiments, XX09_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 176.

(86) For example, XX11_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3),

SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 180. In some embodiments, XX11_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 182.

(87) For example, XX12_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3). In some embodiments, XX12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX12_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 186. In some embodiments, XX12_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 188.

(88) For example, XX13_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135

(LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3). In some embodiments, XX13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX13_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 193, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136. In some embodiments, XX13_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 195, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 138.

(89) For example, XX14_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 193, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 174. In some embodiments, XX14_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 195, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 176.

(90) For example, XX15_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO:

42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_LALA may be defined as ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3). In some embodiments, XX15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 128. In some embodiments, XX15_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 130.

(91) For example, XX15_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3). In some embodiments, XX15_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3). In some embodiments, XX15_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 128. In some embodiments, XX15_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 208, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 130.

(92) For example, XX16_LALA may be defined as having three heavy chain complementarity

regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136. In some embodiments, XX16_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 138.

(93) For example, XX16_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3). In some embodiments, XX16_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136. In some embodiments, XX16_DAPA may be defined as comprising

or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 208, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 138.

(94) For example, XX17_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3). In some embodiments, XX17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 147. In some embodiments, XX17_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 149.

(95) For example, XX17_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3). In some embodiments, XX17_DAPA may be defined as comprising or having amino acid sequences of (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO:

45 (LCDR2), and SEQ ID NO: 145 (LCDR3). In some embodiments, XX17_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 147. In some embodiments, XX17_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 208, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 149.

(96) For example, XX18_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3). In some embodiments, XX18_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 174. In some embodiments, XX18_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 176.

(97) For example, XX18_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_DAPA may be defined as ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45

(LCDR2), and SEQ ID NO: 173 (LCDR3). In some embodiments, XX18_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX18_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 174. In some embodiments, XX18_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 208, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 176.

(98) For example, XX19_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3). In some embodiments, XX19_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 180. In some embodiments, XX19_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 182.

(99) For example, XX19_DAPA may be defined as having three heavy chain complementarity regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42

(LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3). In some embodiments, XX19_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX19_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 180. In some embodiments, XX19_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 208, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 182.

(100) For example, XX20_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3). In some embodiments, XX20_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 186. In some embodiments, XX20_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 188.

(101) For example, XX20_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_DAPA may be defined as comprising or having amino acid

sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3). In some embodiments, XX20_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3). In some embodiments, XX20_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 186. In some embodiments, XX20_DAPA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 208, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 188.

(102) For example, YY01_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 226 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 230 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 231 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, YY01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 226 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, YY01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 229 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, YY01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 230 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3). In some embodiments, YY01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 231 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, YY01_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 233, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 244. In some embodiments, YY01_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 235, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 246.

(103) For example, YY02_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 248 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3); (II) SEQ ID NO: 251 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261

(LCDR2), and SEQ ID NO: 262 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 252 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 263 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 264 (LCDR3); or (IV) SEQ ID NO: 253 (HCDR1), SEQ ID NO: 254 (HCDR2), SEQ ID NO: 255 (HCDR3), SEQ ID NO: 265 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 248 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 251 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 252 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 263 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 264 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 253 (HCDR1), SEQ ID NO: 254 (HCDR2), SEQ ID NO: 255 (HCDR3), SEQ ID NO: 265 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 256, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 266. In some embodiments, YY02_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 258, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 268.

(104) For example, YY03_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 284 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 285 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 277 (HCDR3), SEQ ID NO: 286 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 283 (LCDR3). In some embodiments, YY03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3). In some embodiments, YY03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3). In some embodiments, YY03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 284 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 285 (LCDR3). In some embodiments, YY03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 277 (HCDR3), SEQ ID NO: 286 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 283 (LCDR3). In some embodiments, YY03_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 278, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 287. In some embodiments, YY03_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 280, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 289.

(105) For example, YY04_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 291 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (II) SEQ ID NO: 294 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (III) SEQ ID NO: 295 (HCDR1), SEQ ID NO: 296 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 305 (LCDR3); or (IV) SEQ ID NO: 297 (HCDR1), SEQ ID NO: 298 (HCDR2), SEQ ID NO: 299 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 304 (LCDR3). In some embodiments, YY04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 291 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3). In some embodiments, YY04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 294 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3). In some embodiments, YY04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 295 (HCDR1), SEQ ID NO: 296 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 305 (LCDR3). In some embodiments, YY04_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 297 (HCDR1), SEQ ID NO: 298 (HCDR2), SEQ ID NO: 299 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 304 (LCDR3). In some embodiments, YY04_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 300, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 306. In some embodiments, YY04_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 302, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 308.

(106) For example, YY05_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 325 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 315 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 322 (LCDR3). In some embodiments, YY05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3). In some embodiments, YY05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3). In some embodiments, YY05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 325 (LCDR3). In some embodiments, YY05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 315 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 322 (LCDR3).

(LCDR3). In some embodiments, YY05_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 316, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 327. In some embodiments, YY05_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 318, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 329.

(107) For example, YY06_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, YY06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, YY06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, YY06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3). In some embodiments, YY06_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, YY06_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 333, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344. In some embodiments, YY06_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 335, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346.

(108) For example, YY07_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 356 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 355 (LCDR3). In some embodiments, YY07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3). In some embodiments, YY07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3).

(LCDR3). In some embodiments, YY07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 356 (LCDR3). In some embodiments, YY07_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 355 (LCDR3). In some embodiments, YY07_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 357. In some embodiments, YY07_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 352, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 359.

(109) For example, ZZ05_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 362 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 361 (LCDR3). In some embodiments, ZZ05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3). In some embodiments, ZZ05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3). In some embodiments, ZZ05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 362 (LCDR3). In some embodiments, ZZ05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 361 (LCDR3). In some embodiments, ZZ05_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 363. In some embodiments, ZZ05_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 352, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 365.

(110) For example, ZZ12_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 367 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 369 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 370 (HCDR1), SEQ ID NO: 371 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 372 (HCDR1), SEQ ID NO: 373 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID

NO: 239 (LCDR3). In some embodiments, ZZ12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 367 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 369 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 370 (HCDR1), SEQ ID NO: 371 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3). In some embodiments, ZZ12_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 372 (HCDR1), SEQ ID NO: 373 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ12_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 374, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 244. In some embodiments, ZZ12_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 376, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 246.

(111) For example, ZZ13_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 378 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 380 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 381 (HCDR1), SEQ ID NO: 382 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 383 (HCDR1), SEQ ID NO: 384 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 378 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 380 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ13_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 381 (HCDR1), SEQ ID NO: 382 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3). In some embodiments, ZZ13_LALA may be defined as comprising or having amino acid sequences of (IV) SEQ ID NO: 383 (HCDR1), SEQ ID NO: 384 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3). In some embodiments, ZZ13_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 385, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 244. In some embodiments, ZZ13_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 387, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 246.

(112) For example, ZZ14_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1),

SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 390 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 391 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 270 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 273 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 390 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3). In some embodiments, ZZ14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 275 (HCDR1), SEQ ID NO: 391 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ14_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 392, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344. In some embodiments, ZZ14_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 394, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346.

(113) For example, ZZ15_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 396 (HCDR1), SEQ ID NO: 397 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 398 (HCDR1), SEQ ID NO: 397 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 399 (HCDR1), SEQ ID NO: 400 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 401 (HCDR1), SEQ ID NO: 402 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 396 (HCDR1), SEQ ID NO: 397 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 398 (HCDR1), SEQ ID NO: 397 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 399 (HCDR1), SEQ ID NO: 400 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3). In some embodiments, ZZ15_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 401 (HCDR1), SEQ ID NO: 402 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ15_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 403, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344. In some embodiments, ZZ15_LALA may be defined as comprising or having a heavy chain

comprising or having an amino acid sequence of SEQ ID NO: 405, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346.

(114) For example, ZZ16_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 407 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 409 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 410 (HCDR1), SEQ ID NO: 411 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 412 (HCDR1), SEQ ID NO: 413 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 407 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 409 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 410 (HCDR1), SEQ ID NO: 411 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3). In some embodiments, ZZ16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 412 (HCDR1), SEQ ID NO: 413 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ16_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 414, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344. In some embodiments, ZZ16_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 416, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346.

(115) For example, ZZ17_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 418 (HCDR1), SEQ ID NO: 419 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 420 (HCDR1), SEQ ID NO: 419 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 421 (HCDR1), SEQ ID NO: 422 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 423 (HCDR1), SEQ ID NO: 424 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 418 (HCDR1), SEQ ID NO: 419 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 420 (HCDR1), SEQ ID NO: 419 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ17_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 421 (HCDR1), SEQ ID NO: 422 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3). In some embodiments, ZZ17_LALA may be defined as comprising or having amino

acid sequences of SEQ ID NO: 423 (HCDR1), SEQ ID NO: 424 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3). In some embodiments, ZZ17_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 425, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344. In some embodiments, ZZ17_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 427, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346.

(116) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 13 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 13, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 24. In some other embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 15 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 15, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 26.

(117) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 48 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 48. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 50 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 50.

(118) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 61 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 61, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 72 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 72. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 63 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 63, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 74 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 74.

(119) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 85 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 85, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 96 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 96. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 87 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 87, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 98 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 98.

(120) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 103 or a sequence

having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 403, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 344. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 405 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 405, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 346.

(161) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 414 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 414, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 344. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 416 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 416, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 346.

(162) In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 425 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 425, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 344. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 427 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 427, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 346.

(163) Groups of exemplary anti-NPR1 antibodies of the present disclosure are set forth in Table 3 or Table 4 by CDR (i.e., numerical values in these tables represent sequence identifiers such that “28” represents “SEQ ID NO: 28”) or amino acid consensus sequences. When multiple numerical values are presented in Table 3, they may be used in the alternative for that CDR (i.e., when SEQ ID Nos: 29, 119, and 190 are listed for HCDR2, they may be used in the alternative for that CDR). In Table 4, those amino acids presented in parentheses and separated by a slash represent alternative amino acids in that position (e.g., “(A/V)” represents a position at which the amino acid may be alanine or valine). In some embodiments, the antibody has the heavy and light chain CDRs of any of the antibodies described in Table 3 or Table 4. In some embodiments, the anti-NPR1 antibody is a four-chain antibody (also referred to as an intact antibody), comprising two heavy chains and two light chains. In some embodiments, the anti-NPR1 antibody is an antigen binding fragment of an intact antibody, e.g., a functional fragment of an intact antibody selected from any of those set forth in Table 3 or Table 4 that retains the ability to bind NPR1 and/or provide a function of the intact antibody (e.g., activating NPR1 in the absence of ANP).

(164) TABLE-US-00003 TABLE 3 Exemplary anti-NPR1 antibody groups by CDR HCDR1 HCDR2 HCDR3 LCDR1 LCDR2 LCDR3 ANP non- competitive group 1 Combined 28 29 30 41 42 43 119 126 190 134 145 172 178 184 Kabat 31 29 30 41 42 43 119 126 190 134 145 172 178 184 Chothia 32 33 30 44 45 46 120 127 191 135 146 173 179 185 IMGT 34 35 36 47 45 43 121 126 192 134 145 172 178 184 ANP non- competitive group 2 Combined 28 119 30 41 42 126 134 145 172 178 184 Kabat 31 119 30 41 42 126 134 145 172 178 184 Chothia 32 120 30 44 45 127 135 146 173 179 185 IMGT 34 121 36 47 45 126 134 145 172 178 184 ANP non- competitive group 3 Combined 4 5 6 17 18 19 112 100 165 151 Kabat 7 5 6 17 18 19 100 151 Chothia 8 9 6 20

21 22 101 113 166 152 IMGT 10 11 12 23 21 19 114 102 167 153 ANP non- competitive group 4
Combined 4 5 6 17 18 19 112 100 165 Kabat 7 5 6 17 18 19 100 Chothia 8 9 6 20 21 22 113 101
166 IMGT 10 11 12 23 21 19 114 102 167 ANP non- competitive group 5 Combined 226 227 228
237 238 239 367 368 378 379 Kabat 229 227 228 237 238 239 369 368 380 379 Chothia 32 230
228 240 241 242 370 371 381 382 IMGT 34 231 232 243 241 239 372 373 383 384 ANP non-
competitive group 6 Combined 226 227 228 237 238 239 378 368 379 Kabat 229 227 228 237 238
239 380 368 379 Chothia 32 230 228 240 241 242 381 371 382 IMGT 34 231 232 243 241 239
383 373 384 ANP non- competitive group 7 Combined 226 368 228 237 238 239 367 379 378
Kabat 229 368 228 237 238 239 369 379 380 Chothia 32 371 228 240 241 242 370 382 381 IMGT
34 373 232 243 241 239 372 384 383 ANP non- competitive group 8 Combined 226 368 228 237
238 239 378 379 Kabat 229 368 228 237 238 239 380 379 Chothia 32 371 228 240 241 242 381
382 IMGT 34 373 232 243 241 239 383 384 ANP competitive group 1 Combined 310 311 312 320
321 322 348 354 355 361 Kabat 229 311 312 320 321 322 348 354 355 361 Chothia 80 313 312
323 324 325 348 356 362 IMGT 82 314 315 326 324 322 349 355 361 ANP competitive group 2
Combined 270 271 331 337 338 339 407 389 408 Kabat 273 271 331 337 338 339 409 389 408
Chothia 32 274 331 340 341 342 410 390 411 IMGT 275 276 332 343 341 339 412 391 413 ANP
competitive group 3 Combined 270 389 331 337 338 339 407 408 Kabat 273 389 331 337 338 339
409 408 Chothia 32 390 331 340 341 342 410 411 IMGT 275 276 332 343 341 339 412 391 413
(165) TABLE-US-00004 TABLE 4 Exemplary anti-NPR1 antibody groups by consensus sequence
HCDR1 HCDR2 HCDR3 LCDR1 LCDR2 LCDR3 ANP non- competitive group A Combined 28
(A/V)I(S/E)S(D/K) 30 41 42 (M/Q)Q(S/E/T/I) G(S/N)Y(I/T)(Y/F) (Y/W)(E/V/R/A/T/M)
YADSVKG (K/V/R/A)PRT (SEQ ID NO: 429) (SEQ ID NO: 430) Kabat 31 (A/V)I(S/E)S(D/K)
30 41 42 (M/Q)Q(S/E/T/I) G(S/N)Y(I/T)(Y/F) (Y/W)(E/V/R/A/T/M) YADSVKG (K/V/R/A)PRT
(SEQ ID NO: 429) (SEQ ID NO: 430) Chothia 32 (E/S)S(D/K) 30 44 45 (S/E/T/I)(Y/W) G(S/N)Y
(E/V/R/A/T/M) (SEQ ID NO: 431) (K/V/R/A)PR (SEQ ID NO: 432) IMGT 34 I(S/E)S(D/K)G 36
47 45 (M/Q)Q(S/E/T/I) (S/N)Y(I/T) (Y/W)(E/V/R/A/T/M) (SEQ ID NO: 433) (K/V/R/A)PRT
(SEQ ID NO: 430) ANP non- competitive group B Combined 28 (A/V)I(S/E)S(D/K) 30 41 42
QQ(S/E/T/I)W G(S/N)Y(I/T)(Y/F) (V/R/A/T/M) YADSVKG (K/V/R/A)PRT (SEQ ID NO: 429)
(SEQ ID NO: 434) Kabat 31 (A/V)I(S/E)S(D/K) 30 41 42 QQ(S/E/T/I)W G(S/N)Y(I/T)(Y/F)
(V/R/A/T/M) YADSVKG (K/V/R/A)PRT (SEQ ID NO: 429) (SEQ ID NO: 434) Chothia 32
(E/S)S(D/K) 30 44 45 (S/E/T/I)W G(S/N)Y (V/R/A/T/M) (SEQ ID NO: 431) (K/V/R/A)PR (SEQ
ID NO: 435) IMGT 34 I(S/E)S(D/K) 36 47 45 QQ(S/E/T/I)W G(S/N)Y(I/T) (V/R/A/T/M) (SEQ
ID NO: 433) (K/V/R/A)PRT (SEQ ID NO: 434) ANP non- competitive group C Combined 28 119
30 41 42 QQ(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434) Kabat 31 119 30 41 42
QQ(S/E/T/I) W(V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434) Chothia 32 120 30 44 45
(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PR (SEQ ID NO: 435) IMGT 34 121 36 47 45 QQ(S/E/T/I)
W(V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434) ANP non- competitive group D Combined
GFTF(N/S/Q) SI(S/G)(S/G) 6 17 18 19 THYIH (S/Q)G(S/Q/G) (SEQ ID NO: 436) (S/N/M)T(Y/L)
YADSVKG (SEQ ID NO: 437) Kabat 7 SI(S/G)(S/G)(S/Q) 6 17 18 19 G(S/Q/G)(S/N/M)T
(Y/L)YADSVKG (SEQ ID NO: 437) Chothia GFTF (S/G)(S/G)(S/Q) 6 20 21 22 (N/S/Q)TH
G(S/Q/G)(S/N/M) (SEQ ID NO: 438) (SEQ ID NO: 439) IMGT GFTF I(S/G)(S/G) 12 23 21 19
(N/S/Q)THY (S/Q)G(S/Q/G) (SEQ ID NO: 440) (S/N/M)T (SEQ ID NO: 441) ANP non-
competitive group E Combined GFTF(N/S/Q) SIS(S/G)SG 6 17 18 19 THYIH (S/Q)(S/N) (SEQ
ID NO: 436) TYYADSVKG (SEQ ID NO: 442) Kabat 7 SIS(S/G)SG 6 17 18 19 (S/Q)(S/N)
TYYADSVKG (SEQ ID NO: 442) Chothia GFTF S(S/G)SG 6 20 21 22 (N/S/Q)TH (S/Q)(S/N)
(SEQ ID NO: 438) (SEQ ID NO: 443) IMGT GFTF IS(S/G)SG 12 23 21 19 (N/S/Q)THY (S/Q)
(S/N)T (SEQ ID NO: 440) (SEQ ID NO: 444) ANP non- competitive group F Combined
GF(S/T)FS (N/S)I(K/H)Q 228 237 238 239 (S/K/R)Y(W/Y) (S/Q/H)(G/A) (I/L)(S/N) (S/H/L)E
(SEQ ID NO: 445) (T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446) Kabat (S/K/R)Y(W/Y)
(N/S)I(K/H)Q(S/Q/H) 228 237 238 239 (I/L)(S/N) (G/A)(S/H/L)E (SEQ ID NO: 447) (T/K)

(Y/K/R) YVESVKG (SEQ ID NO: 446) Chothia GF(S/T)FS (K/H)Q(S/Q/H) 228 240 241 242 (S/K/R)Y (G/A)(S/H/L)E (SEQ ID NO: 448) (SEQ ID NO: 449) IMGT GF(S/T)FS I(K/H)Q(S/Q/H) 232 243 241 239 (S/K/R)Y(W/Y) (G/A)(S/H/L)E(T/K) (SEQ ID NO: 450) (SEQ ID NO: 451) ANP non- competitive group G Combined GFTFS(S/R)Y (N/S)I(K/H)Q 228 237 238 239 (W/Y)I(S/N) (S/Q/H)(G/A)(S/H/L) (SEQ ID NO: 452) E(T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446) Kabat (S/R)Y (N/S)I(K/H)Q 228 237 238 239 (W/Y)I(S/N) (S/Q/H)(G/A)(S/H/L) (SEQ ID NO: 454) E(T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446) Chothia GFTFS(S/R)Y (K/H)Q(S/Q/H) 228 240 241 242 (SEQ ID NO: 455) (G/A)(S/H/L)E (SEQ ID NO: 449) IMGT GFTFS(S/R) I(K/H)Q(S/Q/H) 232 243 241 239 Y(W/Y) (G/A)(S/H/L)E(T/K) (SEQ ID NO: 457) (SEQ ID NO: 451) ANP non- competitive group H Combined GF(S/T)FS SIHQ(Q/H)(G/A) 228 237 238 239 (S/K/R)Y(W/Y) (H/L)E(T/K)(K/R) (I/L)(S/N) YVESVKG (SEQ ID NO: 445) (SEQ ID NO: 453) Kabat (S/K/R)Y SIHQ(Q/H)(G/A) 228 237 238 239 (W/Y)(I/L)(S/N) (H/L)E(T/K) (K/R) (SEQ ID NO: 447) YVESVKG (SEQ ID NO: 453) Chothia GF(S/T)FS HQ(Q/H) 228 240 241 242 (S/K/R)Y (G/A)(H/L)E (SEQ ID NO: 448) (SEQ ID NO: 456) IMGT GF(S/T)FS IHQ(Q/H)(G/A) 232 243 241 239 (S/K/R)Y(W/Y) (H/L)E(T/K) (SEQ ID NO: 450) (SEQ ID NO: 458) ANP non- competitive group I Combined GFTFS(S/R) SIHQ(Q/H)(G/A) 228 237 238 239 Y(W/Y)I(S/N) (H/L)E(T/K)(K/R) (SEQ ID NO: 452) YVESVKG (SEQ ID NO: 453) Kabat (S/R)Y(W/Y) SIHQ(Q/H)(G/A) 228 237 238 239 I(S/N) (H/L)E(T/K)(K/R) (SEQ ID NO: 454) YVESVKG (SEQ ID NO: 453) Chothia GFTFS(S/R)Y HQ(Q/H) 228 240 241 242 (SEQ ID NO: 455) (G/A)(H/L)E (SEQ ID NO: 456) IMGT GFTFS(S/R) IHQ(Q/H)(G/A) 232 243 241 239 Y(W/Y) (H/L)E(T/K) (SEQ ID NO: 457) (SEQ ID NO: 458) ANP competitive group A Combined 310 311 G(A/S)(V/L) 320 GNSNRP QSY(Y/D/G) (A/P)G(Q/L)L (S/N) (T/S/A)(S/P/F) GFDH (SEQ ID NO: 460) (S/T/P)(H/S/R) (SEQ ID NO: 459) (G/S/F)(P/S/V)V (SEQ ID NO: 461) Kabat 229 311 G(A/S)(V/L) 320 GNSNRP QSY(Y/D/G)(T/S/A) (A/P)G(Q/L) (S/N) (S/P/F)(S/T/P) (H/S/R) LGFDH (SEQ ID NO: 460) (G/S/F)(P/S/V)V (SEQ ID NO: 459) (SEQ ID NO: 461) Chothia 80 313 G(A/S)(V/L) 323 324 Y(Y/D/G)(T/S/A) (A/P)G(Q/L) (S/P/F)(S/T/P)(H/S/R) LGFDH (G/S/F)(P/S/V) (SEQ ID NO: 459) (SEQ ID NO: 462) IMGT 82 314 ARG(A/S)(V/L) 326 324 QSY(Y/D/G)(T/S/A) (A/P)G(Q/L) (S/P/F)(S/T/P)(H/S/R) LGFDH (G/S/F)(P/S/V)V (SEQ ID NO: 463) (SEQ ID NO: 461) ANP competitive group B Combined GFTF(S/G) (A/S)IS(A/S/G) 331 337 338 339 (S/T)YA(I/M)(S/T) (S/H)G(G/Y)(S/Y) (SEQ ID NO: 464) (T/A)(Y/R/N) YA(E/G)SVKG (SEQ ID NO: 465) Kabat (S/T)YA (A/S)IS(A/S/G) 331 337 338 339 (I/M)(S/T) (S/H)G(G/Y)(S/Y) (SEQ ID NO: 466) (T/A)(Y/R/N) YA(E/G)SVKG (SEQ ID NO: 465) Chothia GFTF(S/G) S(A/S/G)(S/H) 331 340 341 342 (S/T)Y G(G/Y)(S/Y) (SEQ ID NO: 467) (SEQ ID NO: 468) IMGT GFTF(S/G) IS(S/G)(S/H) 332 343 341 339 (S/T)YA G(G/Y)(S/Y)T (SEQ ID NO: 469) (SEQ ID NO: 470) ANP competitive group C Combined GFTF(S/G)(S/T) SIS(A/S) (S/H)GY 331 337 338 339 YA(I/M)(S/T) (T/A)(R/N)YA (SEQ ID NO: 464) (E/G)SVKG (SEQ ID NO: 471) Kabat (S/T)YA(I/M) SIS(A/S)(S/H)GY 331 337 338 339 (S/T) (T/A)(R/N) (SEQ ID NO: 466) YA(E/G)SVKG (SEQ ID NO: 471) Chothia GFTF(S/G)(S/T)Y S(A/S)(S/H)GY 331 340 341 342 (SEQ ID NO: 467) (SEQ ID NO: 472) IMGT GFTF(S/G) IS(A/S/G)(S/H)G 332 343 341 339 (S/T)YA (SEQ ID NO: 473) (SEQ ID NO: 469)

(166) In some embodiments, an antibody or antigen-binding fragment thereof as provided herein binds to (a) human NPR1; and (b) mouse NPR1 and/or rat NPR1.

(167) In some embodiments, an antibody or antigen-binding fragment thereof as provided herein binds to (a) human NPR1; and (b) cyno NPR1. In some embodiments, the antibody or antigen binding fragment thereof is therapeutic. A therapeutic antibody, as defined herein, is an antibody that is both efficacious and stable.

(168) Antibodies that Bind to the Same Epitope as Anti-NPR1 Antibodies of the Disclosure

(169) In another embodiment, the disclosure provides antibodies or antigen-binding fragments thereof that bind to the same epitope as one or more of the anti-NPR1 antibodies described herein (e.g., WW06). Such antibodies: (i) bind NPR1; (ii) are agonists of NPR1; (iii) are ANP

competitive; and bind the same epitope in NPR1 as antibody WW06.

(170) In another embodiment, the disclosure provides antibodies or antigen-binding fragments thereof that bind to the same epitope as one or more of the anti-NPR1 antibodies described herein (e.g., XX16). Such antibodies: (i) bind NPR1; (ii) are agonists of NPR1; (iii) are ANP non-competitive; and bind the same epitope in NPR1 as XX16.

(171) In another embodiment, the disclosure provides antibodies or antigen-binding fragments thereof that bind to the same epitope as one or more of the anti-NPR1 antibodies described herein (e.g., WW03). Such antibodies: (i) bind NPR1; (ii) are agonists of NPR1; (iii) are ANP non-competitive; and bind the same epitope in NPR1 as WW03.

(172) Following the crystallisation and structure determination, the binding regions of the preferred antibodies of the disclosure have been more clearly defined. Such binding is defined herein as being inclusive of both covalent and non-covalent bonds.

(173) Thus, the disclosure provides an ANP competitive antibody that binds the same epitope as WW06. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 188, 192, 194, 197, 201, 208, and 219. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 188, 192, 194, 197, 201, 208, 219, and 295. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 188-198 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, or 4 amino acid residues within amino acid numbers 188-198 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 201-208 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2 amino acids within amino acid numbers 201-208 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, or 4 amino acid residues within amino acid numbers 188-198 of SEQ ID NO: 1, and binds to at least 2 amino acids within amino acid numbers 201-208 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope comprising at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) amino acids 201-208 of SEQ ID NO: 1, (iii) amino acids 215-238 of SEQ ID NO: 1, and (iv) amino acids 294-297 of SEQ ID NO: 1.

(174) The disclosure provides an ANP non-competitive antibody that binds the same epitope as WW03. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 82, 102, 103, 105, 106, 109, 132, and 375. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 79, 82, 99, 102, 103, 105, 106, 109, 131, 132, and 375. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 99-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 99-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, 6, 7, or 8 amino acid residues within amino acid numbers 99-133 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 99-111 of SEQ ID NO: 1, and binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope comprising at least one amino acid residue within each of (i) amino acids 99-111 of SEQ ID NO: 1, (ii) amino acids 131-134 of SEQ ID NO: 1, and (iii) amino acids 374-375 of SEQ ID NO: 1. Optionally, the antibody may additionally bind to amino acids 79 and/or 82 of SEQ ID NO:

1.

(175) The disclosure additionally provides an ANP non-competitive antibody that binds the same epitope as XX16. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 82, 102, 103, 105, 106, 109, 132, and 375. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 34, 82, 102, 103, 105, 106, 107, 109, 132, 133, 375, and 378. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 102-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 102-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 102-111 of SEQ ID NO: 1, and binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope comprising at least one amino acid residue within each of (i) amino acids 102-111 of SEQ ID NO: 1, (ii) amino acids 131-134 of SEQ ID NO: 1, and (iii) amino acids 374-378 of SEQ ID NO: 1. Optionally, the antibody may additionally bind to amino acids 34, 76, and/or 82 of SEQ ID NO: 1.

(176) Additional antibodies can be identified based on their ability to cross-compete (e.g., to competitively inhibit the binding of, in a statistically significant manner) with other antibodies of the disclosure in standard NPR1 binding assays (e.g., XX16, WW06, or WW03). The ability of a test antibody to inhibit the binding of antibodies of the present disclosure to human NPR1 demonstrates that the test antibody can compete with that antibody for binding to human NPR1; such an antibody may, according to non-limiting theory, bind to the same or a related (e.g., a structurally similar or spatially proximal) epitope on human NPR1 as the antibody with which it competes. In a certain embodiment, the antibody that binds to the same epitope on human NPR1 as the antibodies of the present disclosure is a human antibody (e.g., a human monoclonal antibody or antigen binding fragment thereof). Such antibodies can be prepared and isolated as described herein.

(177) Engineered and Modified Antibodies

(178) An antibody of the disclosure can be prepared using an antibody having one or more of the VH and/or VL sequences shown herein as starting material to engineer a modified antibody, which modified antibody may have altered properties from the starting antibody. An antibody can be engineered by modifying one or more residues within one or both variable regions (i.e., VH and/or VL), for example within one or more CDR regions and/or within one or more framework regions. Additionally or alternatively, an antibody can be engineered by modifying residues within the constant region(s), for example to alter the effector function(s) of the antibody.

(179) One type of variable region engineering that can be performed is antibody binding region/paratope or CDR grafting. Because paratope sequences are responsible for most antibody-antigen interactions, it is possible to express recombinant antibodies that mimic the properties of specific naturally occurring antibodies by constructing expression vectors that include CDR/paratope sequences from the specific naturally occurring antibody grafted onto framework sequences from a different antibody with different properties (see, e.g., Riechmann, L. et al., 1998 Nature 332:323-327; Jones, P. et al., 1986 Nature 321:522-525; Queen, C. et al., 1989 Proc. Natl. Acad. Sci. U.S.A. 86:10029-10033; U.S. Pat. Nos. 5,225,539, and 5,530,101; 5,585,089; 5,693,762 and 6,180,370; the contents of each of which are herein incorporated by reference for this purpose).

(180) Accordingly, another embodiment of the disclosure pertains to an isolated anti-NPR1 antibody, or a antigen-binding fragment thereof, comprising an antigen binding portion thereof,

comprising a heavy chain variable region comprising the CDR sequences of an antibody or group of antibodies shown in Table 2, Table 3, or Table 4. Thus, such antibodies contain the VH and VL CDR sequences of monoclonal antibodies, yet may contain different framework sequences from these antibodies.

(181) Such framework sequences can be obtained from public DNA databases or published references that include germline antibody gene sequences. For example, germline DNA sequences for human heavy and light chain variable region genes can be found in the “VBase” human germline sequence database (available on the Internet at [www \[dot\] mrc-cpe \[dot\] cam \[dot\] ac \[dot\] uk/vbase](http://www.mrc-cpe.cam.ac.uk/vbase)), as well as in Kabat, E. A., et al., 1991 Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242; Tomlinson, I. M., et al., 1992 J. Mol. Biol. 227:776-798; and Cox, J. P. L. et al., 1994 Eur. J. Immunol. 24:827-836; the contents of each of which are herein incorporated by reference for this purpose.

(182) An example of framework sequences for use in the antibodies or antigen binding fragments of the disclosure are those that are structurally similar to the framework sequences used by selected antibodies of the disclosure, e.g., consensus sequences and/or framework sequences used by the antibodies or antigen binding fragments of the disclosure. The VH CDR1, 2 and 3 sequences, and the VL CDR1, 2 and 3 sequences, can be grafted onto framework regions that have an identical sequence to that found in the germline immunoglobulin gene from which the framework sequence derive, or the CDR sequences can be grafted onto framework regions that contain one or more mutations as compared to the germline sequences. For example, it has been found that in certain instances it is beneficial to mutate residues within the framework regions to maintain or enhance the antigen binding ability of the antibody (see e.g., U.S. Pat. Nos. 5,530,101; 5,585,089; 5,693,762 and 6,180,370; the contents of each of which are herein incorporated by reference for this purpose).

(183) Another type of variable region modification is to mutate amino acid residues within the VH and/or VL CDR1, CDR2 and/or CDR3 regions to thereby improve one or more binding properties (e.g., affinity) of the antibody of interest, known as “affinity maturation.” Site-directed mutagenesis or PCR-mediated mutagenesis can be performed to introduce the mutation(s) and the effect on antibody binding, or other functional property of interest, can be evaluated in in vitro or in vivo assays as described herein. Conservative modifications (as discussed above) can also be introduced. The mutations may be amino acid substitutions, additions or deletions. Moreover, typically no more than one, two, three, four or five residues within a CDR region are altered.

(184) Grafting Antigen Binding Domains into Alternative Frameworks or Scaffolds

(185) A wide variety of antibody/immunoglobulin frameworks or scaffolds can be employed so long as the resulting polypeptide includes at least one binding region which specifically binds to NPR1. Such frameworks or scaffolds include the 5 main idiotypes of human immunoglobulins, or fragments thereof (such as those disclosed elsewhere herein), and include immunoglobulins of other animal species, preferably having humanized aspects. Single heavy-chain antibodies such as those identified in camelids are of particular interest in this regard. Novel frameworks, scaffolds and fragments continue to be discovered and developed by those skilled in the art.

(186) In one aspect, the disclosure pertains to generating non-immunoglobulin based antibodies using non-immunoglobulin scaffolds onto which CDRs of the disclosure can be grafted. Known or future non-immunoglobulin frameworks and scaffolds may be employed, as long as they comprise a binding region specific for NPR1, e.g., such as those disclosed for an antibody described herein including, but not limited to, XX16, WW03, or WW06. Such compounds are known herein as “polypeptides comprising a target-specific binding region”. Examples of non-immunoglobulin framework are further described in the sections below (camelid antibodies and non-antibody scaffold).

(187) Camelid Antibodies

(188) Antibody proteins obtained from members of the camel and dromedary (*Camelus bactrianus*

and *Camelus dromaderius*) family including new world members such as llama species (*Lama paccos*, *Lama glama*, and *Lama vicugna*) have been characterized with respect to size, structural complexity and antigenicity for human subjects. Certain IgG antibodies from this family of mammals as found in nature lack light chains, and are thus structurally distinct from the typical four chain quaternary structure having two heavy and two light chains, for antibodies from other animals, see WO94/04678, the contents of which are herein incorporated by reference for this purpose.

(189) A region of the camelid antibody which is the small single variable domain identified as VHH can be obtained by genetic engineering to yield a small protein having high affinity for a target, resulting in a low molecular weight antibody-derived protein known as a “camelid nanobody”. See U.S. Pat. No. 5,759,808; see also Stijlemans, B. et al., 2004 J Biol Chem 279:1256-1261; Dumoulin, M. et al., 2003 Nature 424:783-788; Pleschberger, M. et al. 2003 Bioconjugate Chem 14:440-448; Cortez-Retamozo, V. et al. 2002 Int J Cancer 89:456-62; and Lauwereys, M. et al. 1998 EMBO J 17:3512-3520; the contents of each of which are herein incorporated by reference for this purpose. Engineered libraries of camelid antibodies and antibody fragments are commercially available, for example, from Ablynx, Ghent, Belgium. As with other antibodies of non-human origin, an amino acid sequence of a camelid antibody can be altered recombinantly to obtain a sequence that more closely resembles a human sequence, i.e., the nanobody can be “humanized”. Thus the natural low antigenicity of camelid antibodies to humans can be further reduced.

(190) The camelid nanobody has a molecular weight approximately one-tenth that of a human IgG molecule, and the protein has a physical diameter of only a few nanometers. One consequence of the small size is the ability of camelid nanobodies to bind to antigenic sites that are functionally invisible to larger antibody proteins, i.e., camelid nanobodies are useful as reagents detect antigens that are otherwise cryptic using classical immunological techniques, and as possible therapeutic agents. Thus yet another consequence of small size is that a camelid nanobody can inhibit as a result of binding to a specific site in a groove or narrow cleft of a target protein, and hence can serve in a capacity that more closely resembles the function of a classical low molecular weight drug than that of a classical antibody.

(191) The low molecular weight and compact size further result in camelid nanobodies being extremely thermostable, stable to extreme pH and to proteolytic digestion, and poorly antigenic. Another consequence is that camelid nanobodies readily move from the circulatory system into tissues, and even cross the blood-brain barrier and can treat disorders that affect nervous tissue. Nanobodies can further facilitated drug transport across the blood brain barrier, see US2004/0161738, the contents of which are herein incorporated by reference for this purpose. These features combined with the low antigenicity to humans indicate great therapeutic potential. Further, these molecules can be fully expressed in prokaryotic cells such as *E. coli* and may be expressed as functional fusion proteins with bacteriophage.

(192) Accordingly, a feature of the present disclosure is a camelid antibody or nanobody having high affinity for NPR1. In one embodiment, the camelid antibody or nanobody is obtained by grafting the CDRs sequences of the heavy or light chain of the human antibodies of the disclosure into nanobody or single domain antibody framework sequences as described, for example, in WO94/04678 (the contents of which are herein incorporated by reference for this purpose).

(193) Framework or Fc Engineering

(194) Engineered antibodies of the disclosure include those in which modifications have been made to framework residues within VH and/or VL, e.g., to improve one or more properties of the antibody. Typically such framework modifications are made to decrease the immunogenicity of the antibody. Antibodies of the disclosure may be modified in one or more ways, including each of the ways described herein.

(195) For example, one approach is to “backmutate” one or more framework residues to the

corresponding germline sequence. More specifically, an antibody that has undergone somatic mutation may contain framework residues that differ from the germline sequence from which the antibody is derived. Such residues can be identified by comparing the antibody framework sequences to the germline sequences from which the antibody is derived. To return the framework region sequences to their germline configuration, the somatic mutations can be “backmutated” to the germline sequence by, for example, site-directed mutagenesis or PCR-mediated mutagenesis. Such “backmutated” antibodies and additional modifications described herein are also intended to be encompassed by the disclosure.

(196) Another type of framework modification involves mutating one or more residues within the framework region, or even within one or more CDR regions, to remove T-cell epitopes to thereby reduce the potential immunogenicity of the antibody. This approach is also referred to as “deimmunization” and is described in further detail in US2003/0153043, the contents of which are herein incorporated by reference for this purpose.

(197) In addition or alternative to modifications made within the framework or CDR regions, antibodies of the disclosure may be engineered to include modifications within the Fc region, typically to alter one or more functional properties of the antibody, such as serum half-life, complement fixation, Fc receptor binding, and/or antigen-dependent cellular cytotoxicity. Furthermore, an antibody of the disclosure may be chemically modified (e.g., one or more chemical moieties can be attached to the antibody) or be modified to alter its glycosylation, again to alter one or more functional properties of the antibody. Each of these embodiments is described in further detail below. The numbering of residues in the Fc region is that of the EU index of Kabat.

(198) In one embodiment, the hinge region of CHI is modified such that the number of cysteine residues in the hinge region is altered, e.g., increased or decreased. This approach is described further in U.S. Pat. No. 5,677,425, the contents of which are herein incorporated by reference for this purpose. The number of cysteine residues in the hinge region of CHI is altered to, for example, facilitate assembly of the light and heavy chains or to increase or decrease the stability of the antibody.

(199) In another embodiment, the Fc hinge region of an antibody is mutated to decrease the biological half-life of the antibody. More specifically, one or more amino acid mutations are introduced into the CH2-CH3 domain interface region of the Fc-hinge fragment such that the antibody has impaired Staphylococcal protein A (SpA) binding relative to native Fc-hinge domain SpA binding. This approach is described in further detail in U.S. Pat. No. 6,165,745, the contents of which are herein incorporated by reference for this purpose.

(200) In another embodiment, the antibody is modified to increase its biological half-life. Various approaches are possible. For example, one or more of the following mutations can be introduced: T252L, T254S, T256F, as described in U.S. Pat. No. 6,277,375, the contents of which are herein incorporated by reference for this purpose. Alternatively, to increase the biological half life, the antibody can be altered within the CHI or CL region to contain a salvage receptor binding epitope taken from two loops of a CH2 domain of an Fc region of an IgG, as described in U.S. Pat. Nos. 5,869,046 and 6,121,022, the contents of each of which are herein incorporated by reference for this purpose.

(201) In yet other embodiments, the Fc region is altered by replacing at least one amino acid residue with a different amino acid residue to alter the effector functions of the antibody. For example, one or more amino acids can be replaced with a different amino acid residue such that the antibody has an altered affinity for an effector ligand but retains the antigen-binding ability of the parent antibody. The effector ligand to which affinity is altered can be, for example, an Fc receptor or the C1 component of complement. This approach is described in further detail in U.S. Pat. Nos. 5,624,821 and 5,648,260, the contents of each of which are herein incorporated by reference for this purpose.

(202) In order to minimize the ADCC activity of an antibody, specific mutations in the Fc region

result in “Fc silent” antibodies that have minimal interaction with effector cells. In general, the “IgG Fc region” is used to define the C-terminal region of an immunoglobulin heavy chain, including native sequence Fc region and variant Fc regions. The human IgG heavy chain Fc region is generally defined as comprising the amino acid residue from position C226 or from P230 to the carboxyl-terminus of the IgG antibody. The numbering of residues in the Fc region is that of the EU index of Kabat. The C-terminal lysine (residue K447) of the Fc region may be removed, for example, during production or purification of the antibody.

(203) Silenced effector functions can be obtained by mutation in the Fc region of the antibodies. See, for example, LALA and N297A (Strohl, W., 2009, Curr. Opin. Biotechnol. vol. 20(6):685-691); and D265A (Baudino et al., 2008, J. Immunol. 181:6664-69) see also Heusser et al., WO2012065950; the contents of each of which are herein incorporated by reference for this purpose. In particular, residues 234 and/or 235 may be mutated, optionally to alanine. Thus, in one embodiment, an antibody according to the disclosure has a mutation in the Fc region at one or both of amino acids 234 and 235. Such substitution of both amino acids 234 and 235 results in reduced ADCC activity. One example of such a mutation is the LALA mutant comprising L234A and L235A mutation in the IgG1 Fc amino acid sequence. Another example of a silent IgG1 antibody is the DAPA (D265A, P329A) mutation (U.S. Pat. No. 6,737,056, the contents of which are herein incorporated by reference for this purpose). Another silent IgG1 antibody comprises the N297A mutation, which results in aglycosylated/non-glycosylated antibodies. Fc silent antibodies result in no or low ADCC activity, meaning that an Fc silent antibody exhibits an ADCC activity that is below 50% specific cell lysis. No ADCC activity means that the Fc silent antibody exhibits an ADCC activity (specific cell lysis) that is below 1%.

(204) In another embodiment, one or more amino acids selected from amino acid residues can be replaced with a different amino acid residue such that the antibody has altered Clq binding and/or reduced or abolished complement dependent cytotoxicity (CDC). This approach is described in further detail in U.S. Pat. No. 6,194,551, the contents of which are herein incorporated by reference for this purpose.

(205) In another embodiment, one or more amino acid residues are altered to thereby alter the ability of the antibody to fix complement. This approach is described further in WO94/29351, the contents of which are herein incorporated by reference for this purpose.

(206) In yet another embodiment, the Fc region is modified to increase the ability of the antibody to mediate antibody dependent cellular cytotoxicity (ADCC) and/or to increase the affinity of the antibody for an Fcγ receptor by modifying one or more amino acids. This approach is described further in WO00/42072, the contents of which are herein incorporated by reference for this purpose. Moreover, the binding sites on human IgG.sub.1 for FcγRI, FcγRII, FcγRIII and FcRn have been mapped and variants with improved binding have been described (see Shields, R. L. et al., 2001 J. Biol. Chem. 276:6591-6604, the contents of which are herein incorporated by reference for this purpose).

(207) In still another embodiment, the glycosylation of an antibody is modified. For example, an aglycosylated antibody can be made (i.e., the antibody lacks glycosylation). Glycosylation can be altered to, for example, increase the affinity of the antibody for “antigen”. Such carbohydrate modifications can be accomplished by, for example, altering one or more sites of glycosylation within the antibody sequence. For example, one or more amino acid substitutions can be made that result in elimination of one or more variable region framework glycosylation sites to thereby eliminate glycosylation at that site. Such aglycosylation may increase the affinity of the antibody for antigen. Such an approach is described in further detail in U.S. Pat. Nos. 5,714,350 and 6,350,861, the contents of each of which are herein incorporated by reference for this purpose.

(208) Additionally or alternatively, an antibody can be made that has an altered type of glycosylation, such as a hypofucosylated antibody having reduced amounts of fucosyl residues or an antibody having increased bisecting GlcNAc structures. Such altered glycosylation patterns have

been demonstrated to increase the ADCC ability of antibodies. Such carbohydrate modifications can be accomplished by, for example, expressing the antibody in a host cell with altered glycosylation machinery. Cells with altered glycosylation machinery have been described in the art and can be used as host cells in which to express recombinant antibodies of the disclosure to thereby produce an antibody with altered glycosylation. For example, EP 1,176,195 (the contents of which are herein incorporated by reference for this purpose) describes a cell line with a functionally disrupted FUT8 gene, which encodes a fucosyl transferase, such that antibodies expressed in such a cell line exhibit hypofucosylation. WO03/035835 describes a variant CHO cell line, Lec13 cells, with reduced ability to attach fucose to Asn (297)-linked carbohydrates, also resulting in hypofucosylation of antibodies expressed in that host cell (see also Shields, R. L. et al., 2002 J. Biol. Chem. 277:26733-26740). WO99/54342 describes cell lines engineered to express glycoprotein-modifying glycosyl transferases (e.g., beta(1,4)-N acetylglucosaminyltransferase III (GnTIII)) such that antibodies expressed in the engineered cell lines exhibit increased bisecting GlcNac structures which results in increased ADCC activity of the antibodies (see also Umana et al., 1999 Nat. Biotech. 17:176-180). The contents of each of the foregoing applications and references are herein incorporated by reference for this purpose.

(209) Another modification of the antibodies herein that is contemplated by the disclosure is pegylation. An antibody can be pegylated, for example, to increase the biological (e.g., serum) half-life of the antibody. To pegylate an antibody, the antibody, or fragment thereof, typically is reacted with polyethylene glycol (PEG), such as a reactive ester or aldehyde derivative of PEG, under conditions in which one or more PEG groups become attached to the antibody or antibody fragment. The pegylation can be carried out by an acylation reaction or an alkylation reaction with a reactive PEG molecule (or an analogous reactive water-soluble polymer). As used herein, the term “polyethylene glycol” is intended to encompass any of the forms of PEG that have been used to derivatize other proteins, such as mono (C1-C10) alkoxy- or aryloxy-polyethylene glycol or polyethylene glycol-maleimide. In certain embodiments, the antibody to be pegylated is an aglycosylated antibody. Methods for pegylating proteins are known in the art and can be applied to the antibodies of the disclosure. See, for example, EP0154316 and EP0401384, the contents of each of which are herein incorporated by reference for this purpose.

(210) Another modification of the antibodies that is contemplated by the disclosure is a conjugate or a protein fusion of at least the antigen-binding region of the antibody of the disclosure to serum protein, such as human serum albumin or a fragment thereof to increase half-life of the resulting molecule. Such an approach is described, for example, in EP0322094, the contents of which are herein incorporated by reference for this purpose.

(211) Another possibility is a fusion of at least the antigen-binding region of the antibody of the disclosure to proteins capable of binding to serum proteins, such as human serum albumin to increase half-life of the resulting molecule. Such approach is described, for example, in EP0486525, the contents of which are herein incorporated by reference for this purpose.

(212) Nucleic Acid Molecules Encoding Antibodies of the Disclosure

(213) Another aspect of the disclosure pertains to nucleic acid molecules that encode the antibodies of the disclosure. The term “nucleic acid” is used herein interchangeably with the term “polynucleotide,” and refers to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. The term encompasses nucleic acids containing known nucleotide analogs or modified backbone residues or linkages, which are synthetic, naturally-occurring, and non-naturally-occurring, which have similar binding properties as the reference nucleic acid, and which are metabolized in a manner similar to the reference nucleotides. Examples of such analogs include, without limitation, phosphorothioates, phosphoramidates, methyl phosphonates, chiral-methyl phosphonates, 2-O-methyl ribonucleotides, peptide-nucleic acids (PNAs). In some embodiments, the nucleic acid may be an mRNA.

(214) Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses

conservatively modified variants thereof (e.g., degenerate codon substitutions) and complementary sequences, as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (See: Batzer et al., *Nucleic Acids Res* 1991; 25(19):5081; Ohtsuka et al., *J Biol Chem* 1985; 260(5):2605-8; Rossolini et al., *Mol Cell Probes* 1994; 8(2):91-8; the contents of each of which are herein incorporated by reference for this purpose).

(215) Provided herein are exemplary full length heavy and light chain nucleotide sequences of anti-NPR1 antibodies. In some embodiments, the nucleic acid molecules are one or more of those identified in Table 2, e.g., those encoding an anti-NPR1 antibody or antigen binding fragment thereof. In some other embodiments, the nucleic acid molecules described herein comprise nucleotide sequences that are substantially identical (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to the nucleotide sequences of those identified in Table 2. When expressed from appropriate expression vectors, polypeptides encoded by these polynucleotides are capable of binding to a NPR1 protein (e.g., human NPR1).

(216) Also provided herein are polynucleotides which encode at least one CDR region, and usually all three CDR regions, from the heavy and/or light chain of an anti-NPR1 antibody or antigen binding fragment of the disclosure. Further provided herein are polynucleotides which encode all or substantially all of the variable region sequence of the heavy chain and/or the light chain of an exemplary anti-NPR1 antibody or antigen binding fragment of the disclosure. Because of the degeneracy of the genetic code, a variety of nucleic acid sequences will encode each of the immunoglobulin amino acid sequences.

(217) In some embodiments, the nucleic acid molecules disclosed herein encode both a variable region and a constant region of an antibody. In some embodiments, the nucleic acid molecules disclosed herein comprise nucleotides encoding a full-length heavy chain sequence that is substantially identical (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to the heavy chain sequence of one of the antibodies described herein including those in Table 2. In some embodiments, the nucleic acid molecules disclosed herein comprise nucleotides encoding a full-length light chain sequence that is substantially identical (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to the light chain sequence of one of the antibodies described herein including those in Table 2.

(218) The nucleic acids may be present in whole cells, in a cell lysate, or may be nucleic acids in a partially purified or substantially pure form. A nucleic acid is "isolated" or "rendered substantially pure" when purified away from other cellular components or other contaminants, e.g., other cellular nucleic acids or proteins, by standard techniques, including alkaline/SDS treatment, CsCl banding, column chromatography, agarose gel electrophoresis and others well known in the art. See, F. Ausubel, et al., ed. 1987 *Current Protocols in Molecular Biology*, Greene Publishing and Wiley Interscience, New York, the contents of which are herein incorporated by reference for this purpose. A nucleic acid of the disclosure can be, for example, DNA or RNA and may or may not contain intronic sequences. In an embodiment, the nucleic acid is a cDNA molecule. The nucleic acid may be present in a vector such as a phage display vector, or in a recombinant plasmid vector.

(219) Nucleic acids of the disclosure can be obtained using standard molecular biology techniques. For antibodies expressed by hybridomas (e.g., hybridomas prepared from transgenic mice carrying human immunoglobulin genes as described further herein), cDNAs encoding the light and heavy chains of the antibody made by the hybridoma can be obtained by standard PCR amplification or cDNA cloning techniques. For antibodies obtained from an immunoglobulin gene library (e.g., using phage display techniques), nucleic acid encoding the antibody can be recovered from various phage clones that are members of the library.

(220) The polynucleotide sequences can be produced by de novo solid-phase DNA synthesis or by PCR mutagenesis of an existing sequence (e.g., sequences as described herein in, for example,

Table 2). Direct chemical synthesis of nucleic acids can be accomplished by methods known in the art, such as the phosphotriester method of Narang et al., 1979, *Meth. Enzymol.* 68:90; the phosphodiester method of Brown et al., *Meth. Enzymol.* 68:109, 1979; the diethylphosphoramidite method of Beaucage et al., *Tetra. Lett.*, 22:1859, 1981; and the solid support method of U.S. Pat. No. 4,458,066 (the contents of each of which are herein incorporated by reference for this purpose). Introducing mutations to a polynucleotide sequence by PCR can be performed as described in, e.g., *PCR Technology: Principles and Applications for DNA Amplification*, H. A. Erlich (Ed.), Freeman Press, NY, N.Y., 1992; *PCR Protocols: A Guide to Methods and Applications*, Innis et al. (Ed.), Academic Press, San Diego, Calif, 1990; Mattila et al., *Nucleic Acids Res.* 19:967, 1991; and Eckert et al., *PCR Methods and Applications* 1:17, 1991.

(221) Once DNA fragments encoding VH and VL segments are obtained, these DNA fragments can be further manipulated by standard recombinant DNA techniques, for example to convert the variable region genes to full-length antibody chain genes, to Fab fragment genes or to an scFv gene. In these manipulations, a VL- or VH-encoding DNA fragment is operatively linked to another DNA molecule, or to a fragment encoding another protein, such as an antibody constant region or a flexible linker. The term “operatively linked”, as used in this context, is intended to mean that the two DNA fragments are joined in a functional manner, for example, such that the amino acid sequences encoded by the two DNA fragments remain in-frame, or such that the protein is expressed under control of a desired promoter.

(222) The isolated DNA encoding the VH region can be converted to a full-length heavy chain gene by operatively linking the VH-encoding DNA to another DNA molecule encoding heavy chain constant regions (CH1, CH2 and CH3). The sequences of human heavy chain constant region genes are known in the art (see e.g., Kabat, E. A., et al., 1991 *Sequences of Proteins of Immunological Interest*, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, the contents of which are herein incorporated by reference for this purpose) and DNA fragments encompassing these regions can be obtained by standard PCR amplification. The heavy chain constant region can be an IgG1, IgG2, IgG3, IgG4, IgA, IgE, IgM, or IgD constant region. In some embodiments, the heavy chain constant region is an IgG1 isotype. For a Fab fragment heavy chain gene, the VH-encoding DNA can be operatively linked to another DNA molecule encoding only the heavy chain CH1 constant region.

(223) The isolated DNA encoding the VL region can be converted to a full-length light chain gene (as well as to a Fab light chain gene) by operatively linking the VL-encoding DNA to another DNA molecule encoding the light chain constant region, CL. The sequences of human light chain constant region genes are known in the art (see, e.g., Kabat, E. A., et al., 1991 *Sequences of Proteins of Immunological Interest*, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, the contents of which are herein incorporated by reference for this purpose) and DNA fragments encompassing these regions can be obtained by standard PCR amplification. The light chain constant region can be a kappa or a lambda constant region.

(224) To create an scFv gene, the VH- and VL-encoding DNA fragments are operatively linked to another fragment encoding a flexible linker, such that the VH and VL sequences can be expressed as a contiguous single-chain protein, with the VL and VH regions joined by the flexible linker (see e.g., Bird et al., 1988 *Science* 242:423-426; Huston et al., 1988 *Proc. Natl. Acad. Sci. USA* 85:5879-5883; McCafferty et al., 1990 *Nature* 348:552-554; the contents of each of which are herein incorporated by reference for this purpose).

(225) Vectors

(226) Various expression vectors can be employed to express the polynucleotides encoding the antibody of the disclosure or antigen-binding fragment thereof. Both viral-based and nonviral expression vectors can be used to produce the antibodies in a mammalian host cell. Nonviral vectors and systems include plasmids, episomal vectors, typically with an expression cassette for expressing a protein or RNA, and human artificial chromosomes (see, e.g., Harrington et al., *Nat*

Genet. 15:345, 1997, the contents of which are herein incorporated by reference for this purpose). For example, nonviral vectors useful for expression of the polynucleotides and polypeptides of the multispecific antibody of the disclosure or domains thereof in mammalian (e.g., human) cells include pThioHis A, B and C, pcDNA3.1/His, pEBVHis A, B and C, (Invitrogen, San Diego, Calif.), MPS V vectors, and numerous other vectors known in the art for expressing other proteins. Useful viral vectors include vectors based on retroviruses, adenoviruses, adenoassociated viruses, herpes viruses, vectors based on SV40, papilloma virus, HBP Epstein Barr virus, vaccinia virus vectors and Semliki Forest virus (SFV). See, Brent et al., supra; Smith, Annu. Rev. Microbiol. 49:807, 1995; and Rosenfeld et al., Cell 68:143, 1992, the contents of each of which are herein incorporated by reference for this purpose.

(227) The choice of expression vector depends on the intended host cells in which the vector is to be expressed. Expression vectors for mammalian host cells can include expression control sequences, such as an origin of replication, a promoter, and an enhancer (see, e.g., Queen, et al., Immunol. Rev. 89:49-68, 1986, the contents of which are herein incorporated by reference for this purpose), and necessary processing information sites, such as ribosome binding sites, RNA splice sites, polyadenylation sites, and transcriptional terminator sequences. These expression vectors usually contain promoters derived from mammalian genes or from mammalian viruses. Suitable promoters may be constitutive, cell type-specific, stage-specific, and/or modulatable or regulatable. Useful promoters include, but are not limited to, the metallothionein promoter, the constitutive adenovirus major late promoter, the dexamethasone-inducible MMTV promoter, the SV40 promoter, the MRP polIII promoter, the constitutive MPS V promoter, the tetracycline-inducible CMV promoter (such as the human immediate-early CMV promoter), the constitutive CMV promoter, and known promoter-enhancer combinations.

(228) Cultures of transformed organisms can be expanded under non-inducing conditions without biasing the population for coding sequences whose expression products are better tolerated by the host cells. In addition to promoters, other regulatory elements may also be required or desired for efficient expression of the antibody of the disclosure or fragments thereof. These elements typically include an ATG initiation codon and adjacent ribosome binding site or other sequences. In addition, the efficiency of expression may be enhanced by the inclusion of enhancers appropriate to the cell system in use (see, e.g., Scharf et al., Results Probl. Cell Differ. 20:125, 1994; and Bittner et al., Meth. Enzymol., 153:516, 1987; the contents of each of which are herein incorporated by reference for this purpose). For example, the SV40 enhancer or CMV enhancer may be used to increase expression in mammalian host cells.

(229) Accordingly, the disclosure provides a cloning or expression vector comprising one or more of the nucleic acid sequences of the antibodies shown in Table 2. Furthermore, the disclosure provides a cloning or expression vector comprising a nucleic acid encoding one or more of the nucleotide sequences shown in Table 2.

(230) Host Cells

(231) For expression of the light and heavy chains, the expression vector or expression vectors encoding the heavy and light chains may be transferred into a host cell by standard techniques.

(232) Methods for introducing expression vectors containing the polynucleotide sequences of interest vary depending on the type of cellular host. For example, calcium chloride transfection is commonly utilized for prokaryotic cells, whereas calcium phosphate treatment or electroporation may be used for other cellular hosts. (See generally Sambrook, et al., supra, the contents of which are herein incorporated by reference for this purpose). Other methods include, e.g., electroporation, calcium phosphate treatment, liposome-mediated transformation, injection and microinjection, ballistic methods, virosomes, immunoliposomes, polycationic nucleic acid conjugates, naked DNA, artificial virions, fusion to the herpes virus structural protein VP22 (Elliot and O'Hare, Cell 88:223, 1997, the contents of which are herein incorporated by reference for this purpose), agent-enhanced uptake of DNA, and ex vivo transduction.

(233) It is theoretically possible to express the antibodies of the disclosure in either prokaryotic or eukaryotic host cells. Expression of antibodies in eukaryotic cells, in particular mammalian host cells, is discussed because such eukaryotic cells, and in particular mammalian cells, are more likely than prokaryotic cells to assemble and secrete a properly folded and immunologically active antibody. Prokaryotic expression of antibody genes has been reported to be ineffective for production of high yields of active antibody (Boss, M. A. and Wood, C. R., 1985 Immunology Today 6:12-13, the contents of which are herein incorporated by reference for this purpose).

(234) For long-term, high-yield production of recombinant proteins, stable expression will often be desired. For example, cell lines which stably express the antibodies or antigen-binding fragments thereof of the disclosure can be prepared using expression vectors of the disclosure which contain viral origins of replication or endogenous expression elements and a selectable marker gene. Following the introduction of the vector, cells may be allowed to grow for 1-2 days in an enriched media before they are switched to selective media. The purpose of the selectable marker is to confer resistance to selection, and its presence allows growth of cells which successfully express the introduced sequences in selective media. Resistant, stably transfected cells can be proliferated using tissue culture techniques appropriate to the cell type. The present disclosure thus provides a method of producing the antibodies or antigen-binding fragments of the disclosure, wherein said method comprises the step of culturing a host cell comprising a nucleic acid encoding the antibodies or antigen-binding fragments.

(235) In some embodiments, mammalian host cells are used to express and produce the anti-NPR1 antibodies or antigen binding fragments of the present disclosure. For example, they can be either a hybridoma cell line expressing endogenous immunoglobulin genes or a mammalian cell line harboring an exogenous expression vector. These include any normal mortal or normal or abnormal immortal animal or human cell. For example, a number of suitable host cell lines capable of secreting intact immunoglobulins have been developed including the CHO cell lines, various COS cell lines, HeLa cells, myeloma cell lines, transformed B-cells, and hybridomas. Exemplary host cells include but are not limited to Chinese hamster ovary (CHO) cells, human embryonic kidney (HEK) cells (e.g., HEK293, HEK293T, HEK293F), monkey kidney (COS) cells (e.g., COS-1, COS-7), baby hamster kidney (BHK) cells (e.g., BHK-21), African green monkey kidney cells (e.g., BSC-1), HeLa cells, human hepatocellular carcinoma cells (e.g., Hep G2), myeloma cells (e.g., NS0, 653, SP2/0), lymphoma cells, oocyte cells, and cells from a transgenic animal (e.g., mammary epithelial cells), or any derivative, immortalized, or transformed cell thereof. In particular, for use with NS0 myeloma cells, another expression system is the GS gene expression system shown in WO87/04462, WO89/01036 and EP0338841, the contents of each of which are herein incorporated by reference for this purpose. When recombinant expression vectors encoding antibody nucleic acid are introduced into mammalian host cells, the antibodies are produced by culturing the host cells for a period of time sufficient to allow for expression of the antibody in the host cells or secretion of the antibody into the culture medium in which the host cells are grown. Antibodies can be recovered from the culture medium using standard protein purification methods. Such purified antibodies of the disclosure may be used for any purpose including, but not limited to, the methods and uses described herein, and/or as part of a pharmaceutical composition as described herein.

(236) In a further alternative, the host cell may be a yeast or a filamentous fungi engineered for mammalian-like glycosylation pattern, and capable for producing antibodies lacking fucose as glycosylation pattern (see, for example, EP1297172, the contents of which are herein incorporated by reference for this purpose).

(237) Accordingly, the disclosure provides a host cell comprising one or more of the vectors, or nucleic acid sequences of the disclosure described above.

(238) Generation of Monoclonal Antibodies of the Disclosure

(239) Monoclonal antibodies (mAbs) can be produced by a variety of techniques, including

conventional monoclonal antibody methodology e.g., the somatic cell hybridization technique of Kohler and Milstein, 1975 *Nature* 256:495, the contents of which are herein incorporated by reference for this purpose. Many techniques for producing monoclonal antibody can be employed e.g., viral or oncogenic transformation of B lymphocytes.

(240) Hybridomas may be prepared using, for example, the murine system. Immunization protocols and isolation of immunized splenocytes for fusion may be performed according to any appropriate procedure. Chimeric or humanized antibodies can be prepared based on the sequence of a murine monoclonal antibody prepared as described herein. DNA encoding the heavy and light chain immunoglobulins can be obtained from the murine hybridoma of interest and engineered to contain non-murine (e.g., human) immunoglobulin sequences using standard molecular biology techniques. For example, to create a chimeric antibody, the murine variable regions can be linked to human constant regions using any known methods (see e.g., U.S. Pat. No. 4,816,567, the contents of which are herein incorporated by reference for this purpose). To create a humanized antibody, the murine CDR regions can be inserted into a human framework using any known methods. See e.g., U.S. Pat. Nos. 5,225,539, and 5,530,101; 5,585,089; 5,693,762 and 6,180,370, the contents of each of which are herein incorporated by reference for this purpose.

(241) Human monoclonal antibodies can also be generated using transgenic or transchromosomic mice carrying parts of the human immune system rather than the mouse system. These transgenic and transchromosomic mice include mice referred to herein as HuMAb Mice® and KM mice, respectively, and are collectively referred to herein as “human Ig mice.”

(242) The HuMAb Mouse® (Medarex, Inc.) contains human immunoglobulin gene miniloci that encode un-rearranged human heavy (u and y) and K light chain immunoglobulin sequences, together with targeted mutations that inactivate the endogenous u and K chain loci (see e.g., Lonberg, et al., 1994 *Nature* 368(6474):856-859, the contents of which are herein incorporated by reference for this purpose). Accordingly, the mice exhibit reduced expression of mouse IgM or K, and in response to immunization, the introduced human heavy and light chain transgenes undergo class switching and somatic mutation to generate high affinity human IgGK monoclonal (Lonberg, N. et al., 1994 *supra*; reviewed in Lonberg, N., 1994 *Handbook of Experimental Pharmacology* 113:49-101; Lonberg, N. and Huszar, D., 1995 *Intern. Rev. Immunol.* 13:65-93, and Harding, F. and Lonberg, N., 1995 *Ann. N. Y. Acad. Sci.* 764:536-546; the contents of each of which are herein incorporated by reference for this purpose). The preparation and use of HuMAb Mice®, and the genomic modifications carried by such mice, is further described in Taylor, L. et al., 1992 *Nucleic Acids Research* 20:6287-6295; Chen, J. et al., 1993 *International Immunology* 5:647-656; Tuaillon et al., 1993 *Proc. Natl. Acad. Sci. USA* 94:3720-3724; Choi et al., 1993 *Nature Genetics* 4:117-123; Chen, J. et al., 1993 *EMBO J.* 12:821-830; Tuaillon et al., 1994 *J. Immunol.* 152:2912-2920; Taylor, L. et al., 1994 *International Immunology* 579-591; and Fishwild, D. et al., 1996 *Nature Biotechnology* 14:845-851, the contents of each of which are hereby incorporated by reference for this purpose. See further, U.S. Pat. Nos. 5,545,806; 5,569,825; 5,625,126; 5,633,425; 5,789,650; 5,877,397; 5,661,016; 5,814,318; 5,874,299; and 5,770,429; U.S. Pat. No. 5,545,807; WO92/103918, WO93/12227, WO94/25585, WO97/113852, WO98/24884 and WO99/45962; and WO01/14424; the contents of each of which are hereby incorporated by reference for this purpose.

(243) In another embodiment, human antibodies can be raised using a mouse that carries human immunoglobulin sequences on transgenes and transchromosomes such as a mouse that carries a human heavy chain transgene and a human light chain transchromosome. Such mice, referred to herein as “KM mice”, are described in detail in WO02/43478, the contents of which are hereby incorporated by reference for this purpose.

(244) Still further, alternative transgenic animal systems expressing human immunoglobulin genes are available in the art and can be used to raise antibodies of the disclosure. For example, an alternative transgenic system referred to as the Xenomouse (Abgenix, Inc.) can be used. Such mice are described in, e.g., U.S. Pat. Nos. 5,939,598; 6,075,181; 6,114,598; 6,150,584 and 6,162,963, the

contents of each of which are hereby incorporated by reference for this purpose.

(245) Moreover, alternative transchromosomal animal systems expressing human immunoglobulin genes are available in the art and can be used to raise antibodies of the disclosure. For example, mice carrying both a human heavy chain transchromosome and a human light chain transchromosome, referred to as “TC mice” can be used; such mice are described in Tomizuka et al., 2000 Proc. Natl. Acad. Sci. USA 97:722-727, the contents of which are hereby incorporated by reference for this purpose. Furthermore, cows carrying human heavy and light chain transchromosomes have been described in the art (Kuroiwa et al., 2002 Nature Biotechnology 20:889-894, the contents of which are hereby incorporated by reference for this purpose) and can be used to raise antibodies of the disclosure.

(246) Human monoclonal antibodies of the disclosure can also be prepared using phage display methods for screening libraries of human immunoglobulin genes. Such phage display methods for isolating human antibodies are established in the art or described in the examples below. See for example: U.S. Pat. Nos. 5,223,409; 5,403,484; and 5,571,698; 5,427,908 and 5,580,717; 5,969,108 and 6,172,197; and 5,885,793; 6,521,404; 6,544,731; 6,555,313; 6,582,915 and 6,593,081, the contents of each of which are hereby incorporated by reference for this purpose.

(247) Human antibodies or antigen binding fragments of the disclosure can also be prepared using SCID mice into which human immune cells have been reconstituted such that a human antibody response can be generated upon immunization. Such mice are described in, for example, U.S. Pat. Nos. 5,476,996 and 5,698,767, the contents of each of which are hereby incorporated by reference for this purpose.

(248) Antibodies of the disclosure may be prepared by any of the methods described herein.

(249) Generation of Hybridomas Producing Antibodies or Antigen Binding Fragments of the Disclosure

(250) To generate hybridomas producing the antibodies or antigen binding fragments of the disclosure, splenocytes and/or lymph node cells from immunized mice can be isolated and fused to an appropriate immortalized cell line, such as a mouse myeloma cell line. The resulting hybridomas can be screened for the production of antigen-specific antibodies. For example, single cell suspensions of splenic lymphocytes from immunized mice can be fused to one-sixth the number of P3X63-Ag8.653 nonsecreting mouse myeloma cells (ATCC, CRL 1580) with 50% PEG. Cells are plated at approximately 2×10^5 in flat bottom microtiter plates, followed by a two week incubation in selective medium containing 20% fetal Clone Serum, 18% “653” conditioned media, 5% origen (IGEN), 4 mM L-glutamine, 1 mM sodium pyruvate, 5 mM HEPES, 0.055 mM 2-mercaptoethanol, 50 units/ml penicillin, 50 mg/ml streptomycin, 50 mg/ml gentamycin and 1X HAT (Sigma; the HAT is added 24 hours after the fusion). After approximately two weeks, cells can be cultured in medium in which the HAT is replaced with HT. Individual wells can then be screened by ELISA for human monoclonal IgM and IgG antibodies. Once extensive hybridoma growth occurs, medium can be observed usually after 10-14 days. The antibody secreting hybridomas can be replated, screened again, and if still positive for human IgG, the monoclonal antibodies can be subcloned at least twice by limiting dilution. The stable subclones can then be cultured in vitro to generate small amounts of antibody in tissue culture medium for characterization.

(251) To purify antibodies or antigen binding fragments thereof, selected hybridomas can be grown in two-liter spinner-flasks for antibody purification. Supernatants can be filtered and concentrated before affinity chromatography with protein A-sepharose (Pharmacia, Piscataway, N.J.). Eluted IgG can be checked by gel electrophoresis and high performance liquid chromatography to ensure purity. The buffer solution can be exchanged into PBS, and the concentration can be determined by OD280 using 1.43 extinction coefficient. The antibodies or antigen binding fragments can be aliquoted and stored at -80°C .

(252) Hybridomas producing the antibodies or antigen binding fragments of the disclosure may be produced, for example, using the methods described herein.

(253) Generation of Transfectomas Producing Antibodies or Antigen Binding Fragments of the Disclosure

(254) Antibodies or antigen binding fragments of the disclosure can also be produced in a host cell transfectoma using, for example, a combination of suitable recombinant DNA techniques and gene transfection methods (e.g., Morrison, S. (1985) Science 229:1202, the contents of which are incorporated herein by reference for this purpose).

(255) For example, to express the antibodies, or antigen-binding fragments thereof, DNAs encoding partial or full-length light and heavy chains, can be obtained by standard molecular biology techniques (e.g., PCR amplification or cDNA cloning using a hybridoma that expresses the antibody of interest) and the DNAs can be inserted into expression vectors such that the genes are operatively linked to transcriptional and translational control sequences. In this context, the term “operatively linked” is intended to mean that an antibody gene is ligated into a vector such that transcriptional and translational control sequences within the vector serve their intended function of regulating the transcription and translation of the antibody gene. The expression vector and expression control sequences are chosen to be compatible with the expression host cell used. The antibody light chain gene and the antibody heavy chain gene can be inserted into separate vector or both genes may be inserted into the same expression vector. The antibody genes are inserted into the expression vector by standard methods (e.g., ligation of complementary restriction sites on the antibody gene fragment and vector, or blunt end ligation if no restriction sites are present). The light and heavy chain variable regions of the antibodies described herein can be used to create full-length antibody genes of any antibody isotype by inserting them into expression vectors already encoding heavy chain constant and light chain constant regions of the desired isotype such that the VH segment is operatively linked to the CH segment(s) within the vector and the VL segment is operatively linked to the CL segment within the vector. Additionally or alternatively, the recombinant expression vector can encode a signal peptide that facilitates secretion of the antibody chain from a host cell. The antibody chain gene can be cloned into the vector such that the signal peptide is linked in frame to the amino terminus of the antibody chain gene. The signal peptide can be an immunoglobulin signal peptide or a heterologous signal peptide (i.e., a signal peptide from a non-immunoglobulin protein).

(256) In addition to the antibody chain genes, the recombinant expression vectors of the disclosure carry regulatory sequences that control the expression of the antibody chain genes in a host cell. The term “regulatory sequence” is intended to include promoters, enhancers and other expression control elements (e.g., polyadenylation signals) that control the transcription or translation of the antibody chain genes. Such regulatory sequences are described, for example, in Goeddel (Gene Expression Technology; Methods in Enzymology 185, Academic Press, San Diego, CA 1990, the contents of which are incorporated herein by reference for this purpose). It will be appreciated by those skilled in the art that the design of the expression vector, including the selection of regulatory sequences, may depend on such factors as the choice of the host cell to be transformed, the level of expression of protein desired, etc. Regulatory sequences for mammalian host cell expression include viral elements that direct high levels of protein expression in mammalian cells, such as promoters and/or enhancers derived from cytomegalovirus (CMV), Simian Virus 40 (SV40), adenovirus (e.g., the adenovirus major late promoter (AdMLP)), and polyoma. Alternatively, nonviral regulatory sequences may be used, such as the ubiquitin promoter or P-globin promoter. Still further, regulatory elements composed of sequences from different sources, such as the SRa promoter system, which contains sequences from the SV40 early promoter and the long terminal repeat of human T cell leukemia virus type 1 (Takebe, Y. et al., 1988 Mol. Cell. Biol. 8:466-472, the contents of which are incorporated herein by reference for this purpose).

(257) In addition to the antibody chain genes and regulatory sequences, the recombinant expression vectors of the disclosure may carry additional sequences, such as sequences that regulate replication of the vector in host cells (e.g., origins of replication) and selectable marker genes. The

selectable marker gene facilitates selection of host cells into which the vector has been introduced (see, e.g., U.S. Pat. Nos. 4,399,216, 4,634,665 and 5,179,017, the contents of each of which are incorporated herein by reference for this purpose). For example, typically the selectable marker gene confers resistance to drugs, such as G418, hygromycin or methotrexate, on a host cell into which the vector has been introduced. Selectable marker genes include the dihydrofolate reductase (DHFR) gene (for use in dhfr-host cells with methotrexate selection/amplification) and the neo gene (for G418 selection).

(258) For expression of the light and heavy chains, the expression vector(s) encoding the heavy and light chains is or are transfected into a host cell by standard techniques. The various forms of the term “transfection” are intended to encompass a wide variety of techniques commonly used for the introduction of exogenous DNA into a prokaryotic or eukaryotic host cell, e.g., electroporation, calcium-phosphate precipitation, DEAE-dextran transfection and the like. It is theoretically possible to express the antibodies of the disclosure in either prokaryotic or eukaryotic host cells. Expression of antibodies in eukaryotic cells, in particular mammalian host cells, is discussed because such eukaryotic cells, and in particular mammalian cells, are more likely than prokaryotic cells to assemble and secrete a properly folded and immunologically active antibody. Prokaryotic expression of antibody genes has been reported to be ineffective for production of high yields of active antibody (Boss, M. A. and Wood, C. R., 1985 Immunology Today 6:12-13, the contents of which are incorporated herein by reference for this purpose).

(259) Mammalian host cells for expressing the antibodies of the disclosure are described elsewhere herein. When recombinant expression vectors encoding antibody genes are introduced into mammalian host cells, the antibodies are produced by culturing the host cells for a period of time sufficient to allow for expression of the antibody in the host cells or secretion of the antibody into the culture medium in which the host cells are grown. Antibodies can be recovered from the culture medium using standard protein purification methods. Accordingly, the disclosure provides a process for the production of an anti-NPR1 antibody of the disclosure, or antigen-binding fragment thereof, comprising culturing a host cell of the disclosure and isolating the antibody or antigen-binding fragment thereof.

(260) Uses and Methods of Treatment

(261) Methods of Treatment

(262) Provided herein are methods of treating a disease associated with NPR1 loss of function by using the anti-NPR1 antibodies or antigen binding fragments thereof disclosed herein (e.g., an antibody or group of antibodies as defined in Table 2, Table 3, or Table 4). In some embodiments, the antibody or antigen binding fragment thereof may be selected from WW01_LALA, WW03_LALA, WW05_LALA, WW06_LALA, XX01_LALA, XX01_DAPA, XX01_N30S_DAPA, XX03_LALA, XX04_LALA, XX06_LALA, XX06_DAPA, XX07_LALA, XX08_LALA, XX08_DAPA, XX08_N30S_DAPA, XX08_N30Q_DAPA, XX09_LALA, XX11_LALA, XX12_LALA, XX13_LALA, XX14_LALA, XX15_LALA, XX15_DAPA, XX16_LALA, XX16_DAPA, XX17_LALA, XX17_DAPA, XX18_LALA, XX18_DAPA, XX19_LALA, XX19_DAPA, XX20_LALA, XX20_DAPA, YY01_LALA, YY02_LALA, YY03_LALA, YY04_LALA, YY05_LALA, YY06_LALA, YY07_LALA, ZZ05_LALA, ZZ12_LALA, ZZ13_LALA, ZZ14_LALA, ZZ15_LALA, ZZ16_LALA, and ZZ17_LALA.

(263) In some embodiments, the antibody or antigen binding fragment thereof may be selected from WW01_LALA, WW03_LALA, XX01_LALA, XX01_DAPA, XX01_N30S_DAPA, XX03_LALA, XX04_LALA, XX06_LALA, XX06_DAPA, XX07_LALA, XX08_LALA, XX08_DAPA, XX08_N30S_DAPA, XX08_N30Q_DAPA, XX09_LALA, XX11_LALA, XX12_LALA, XX13_LALA, XX14_LALA, XX15_LALA, XX15_DAPA, XX16_LALA, XX16_DAPA, XX17_LALA, XX17_DAPA, XX18_LALA, XX18_DAPA, XX19_LALA, XX19_DAPA, XX20_LALA, XX20_DAPA, YY01_LALA, YY03_LALA, YY04_LALA, ZZ12_LALA, and ZZ13_LALA. In some embodiments, the antibody or antigen binding fragment

thereof may be selected from WW05_LALA, WW06_LALA, YY05_LALA, YY06_LALA, YY07_LALA, ZZ05_LALA, ZZ14_LALA, and ZZ16_LALA. In some embodiments, the antibody or antigen binding fragment thereof may be XX16_DAPA. In some embodiments, the antibody or antigen binding fragment thereof may be XX16_LALA.

(264) In some embodiments, the disease associated with NPR1 loss of function is a cardiovascular disorder. In some embodiments, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI). In some embodiments, the disease associated with NPR1 loss of function is heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, or cytokine release syndrome. In some embodiments, the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments, the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments, the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments, the hypertension is selected from resistant hypertension and hypertensive heart disease.

(265) In some embodiments, the disease associated with NPR1 loss of function is a kidney disorder. In some embodiments, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

(266) NPR1-related disorders also include any other disorders which are directly or indirectly associated with aberrant NPR1 activity and/or expression. Provided herein are also methods of treating a NPR1 related disorder directly or indirectly associated with aberrant NPR1 activity and/or expression by using the anti-NPR1 antibodies or antigen binding fragments disclosed herein (e.g., from Table 2, Table 3, or Table 4, such as XX16_DAPA or XX16_LALA).

(267) In some embodiments, the present disclosure provides methods of treating an undesirable condition, disease, or disorder associated with natriuretic peptide receptor activity in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of an antibody or antigen binding fragment disclosed herein. In some embodiments, the present disclosure provides a use of an antibody or antigen binding fragment disclosed herein for treatment of an undesirable condition, disease or disorder associated with natriuretic peptide receptor activity in a subject in need thereof. In some embodiments, the present disclosure provides an antibody or antigen binding fragment disclosed herein for use in a method for treating an undesirable condition, disease or disorder associated with natriuretic peptide receptor activity. In some embodiments, the present disclosure provides an antibody or antigen binding fragment disclosed herein for use in manufacturing a medicament for treating an undesirable condition, disease or disorder associated with natriuretic peptide receptor activity. Such conditions, diseases and disorders include, but are not limited to, cardiovascular disorders (e.g., hypertension, peripheral vascular disease, heart failure (including but not limited to heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure), coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy (e.g., ventricular

hypertrophy), diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, or myocardial infarction (MI)), hypertension (e.g., resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, or pulmonary arterial hypertension), preeclampsia, asthma, glaucoma, cytokine release syndrome, and/or a kidney disorder (e.g., diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD)).

(268) In some embodiments, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as one of the antibodies described herein. For example, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as XX16. In another embodiment, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as WW03. In another embodiment, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as WW06.

(269) All the aforementioned embodiments for the methods of protection and treatment according to the present invention are equally applicable to the use of any one of the antibodies or antigen binding fragments as described herein for the manufacture of a medicament for use according to the present invention, the use of any one of the antibodies or antigen binding fragments described herein according to the present invention, any one of the antibodies or antigen binding fragments described herein for use according to the present invention, the pharmaceutical compositions comprising any one of the antibodies or antigen binding fragments described herein for the use according to the present invention, the use of the pharmaceutical compositions comprising any one of the antibodies or antigen binding fragments described herein according to the present invention, and the use of the pharmaceutical compositions comprising any one of the antibodies or antigen binding fragments described herein for the manufacture of a medicament for use according to the present invention.

Combination Therapies

(270) The various treatments described above can be combined with other treatment partners or therapeutic agents such as the current standard of care for a disease associated with NPR1 loss of function, e.g., the current standard of care for one or more of the diseases or disorders discussed herein. For example, the NPR1 antibodies or an antigen-binding fragment thereof described herein can be combined with one or more of an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), or combinations thereof. As a non-limiting set of examples, the NPR1 antibody or antigen binding from may be combined with an additional therapeutic agent selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propanolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nifedipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, and combinations thereof. Exemplary diuretics and

digitalis glycosides include, but are not limited to, chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, and combinations thereof. In some embodiments, the NPR1 antibodies or an antigen-binding fragment thereof described herein may be combined with an angiotensin receptor-neprilysin inhibitor (ARNi) such as a combination of sacubitril and valsartan (e.g., Entresto®). In some embodiments, the NPR1 antibodies or an antigen-binding fragment thereof described herein can be combined with one or more of a corticosteroid (e.g., an inhaled corticosteroid such as fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, or fluticasone furoate; or an oral or intravenous corticosteroid such as prednisone or methylprednisolone), a leukotriene modifier (e.g., montelukast, zafirlukast, or zileuton), a bronchodilator (e.g., a long-acting beta agonist (e.g., salmeterol or formoterol), a short-acting beta agonist (e.g., albuterol or levalbuterol), theophylline or ipratropium), or combinations thereof (e.g., a combination of fluticasone and salmeterol, a combination of budesonide and formoterol, or a combination of formoterol and mometasone). In some embodiments, the NPR1 antibodies or an antigen-binding fragment thereof described herein can be combined with one or more of a beta-adrenoceptor antagonist (e.g., timolol, levobunolol, metipranolol, carteolol, or betaxolol), a carbonic anhydrase inhibitor (e.g., acetazolamide, dorzolamide, brinzolamide, or methazolamide), an alpha 2-adrenoceptor agonist (e.g., brimonidine or apraclonidine), a parasympathomimetic (e.g., cholinomimetics like pilocarpine), a prostaglandin analog (e.g., latanoprost, latanoprostene bunod, travoprost, bimatoprost, or tafluprost), a rho kinase inhibitor (e.g., netarsudil or ripasudil), or combinations thereof (e.g., a combination of rho kinase inhibitor and latanoprost).

(271) Accordingly, the methods of treating a disease associated with NPR1 loss of function described herein can further include administering a second agent to the subject in need of treatment.

(272) The term “combination” refers to either a fixed combination in one dosage unit form, or a combined administration where an anti-NPR1 antibody or antigen-binding fragment thereof described herein and a combination partner (e.g., another drug as explained below, also referred to as “therapeutic agent” or “co-agent”) may be administered independently at the same time or separately within time intervals, especially where these time intervals allow that the combination partners show a cooperative, e.g., synergistic effect. The single components may be packaged in a kit or separately. One or both of the components (e.g., powders or liquids) may be reconstituted or diluted to a desired dose prior to administration. The terms “co-administration” or “combined administration” or the like as utilized herein are meant to encompass administration of the selected combination partner to a single subject in need thereof (e.g., a patient), and are intended to include treatment regimens in which the agents are not necessarily administered by the same route of administration and/or at the same time. The term “pharmaceutical combination” as used herein means a product that results from the mixing or combining of more than one therapeutic agent and includes both fixed and non-fixed combinations of the therapeutic agents. The term “fixed combination” means that the therapeutic agents, e.g., a compound of the present invention and a combination partner, are both administered to a patient simultaneously in the form of a single entity or dosage. The term “non-fixed combination” means that the therapeutic agents, e.g., a compound of the present invention and a combination partner, are both administered to a patient as separate entities either simultaneously, concurrently or sequentially with no specific time limits, wherein such administration provides therapeutically effective levels of the two compounds in the body of the patient. The latter also applies to cocktail therapy, e.g., the administration of three or more therapeutic agent.

(273) The term “pharmaceutical combination” as used herein refers to either a fixed combination in one dosage unit form, or non-fixed combination or a kit of parts for the combined administration where two or more therapeutic agents may be administered independently at the same time or

separately within time intervals, especially where these time intervals allow that the combination partners show a cooperative, e.g., synergistic effect.

(274) The term “combination therapy” refers to the administration of two or more therapeutic agents to treat a therapeutic condition or disorder described in the present disclosure. Such administration encompasses co-administration of these therapeutic agents in a substantially simultaneous manner, such as in a single capsule having a fixed ratio of active ingredients. Alternatively, such administration encompasses co-administration in multiple, or in separate containers (e.g., tablets, capsules, powders, and liquids) for each active ingredient. Powders and/or liquids may be reconstituted or diluted to a desired dose prior to administration. In addition, such administration also encompasses use of each type of therapeutic agent in a sequential manner, either at approximately the same time or at different times. In either case, the treatment regimen will provide beneficial effects of the drug combination in treating the conditions or disorders described herein.

(275) Pharmaceutical Compositions, Dosages, and Methods of Administration

(276) Also provided herein are compositions, e.g., pharmaceutical compositions, for use in treatment of an NPR1-associated disease. Such compositions include one or more anti-NPR1 antibodies or an antigen-binding fragment thereof as described herein and may include a pharmaceutically acceptable carrier. Such compositions can further include another agent, e.g., a current standard of care for the disease to be treated.

(277) Pharmaceutical compositions typically include a pharmaceutically acceptable carrier. As used herein the term “pharmaceutically acceptable carrier” refers to a carrier or a diluent that does not cause significant irritation to a subject and does not abrogate the biological activity and properties of the administered anti-NPR1 antibody or antigen binding fragment and/or any additional therapeutic agent in the composition. Pharmaceutically acceptable carriers may enhance or stabilize the composition or can be used to facilitate preparation of the composition. Pharmaceutically acceptable carriers may include saline, solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration. An adjuvant may also be included in any of these formulations. Pharmaceutical compositions are typically formulated to be compatible with its intended route of administration. Examples of routes of administration include parenteral (e.g., intravenous, intraarterial, intraperitoneal), oral, intracranial, intrathecal, or intranasal (e.g., inhalation), intradermal, subcutaneous, or transmucosal administration. In some embodiments, the pharmaceutical compositions are formulated to deliver anti-NPR1 antibodies or antigen-binding fragments thereof to cross the blood-brain barrier. The phrases “physiologically acceptable carrier” and “pharmaceutically acceptable carrier” may be used interchangeably.

(278) As used herein, the term “excipient” refers to an inert substance added to a pharmaceutical composition to further facilitate administration of an active ingredient. Formulations for parenteral administration can, for example, contain excipients such as sterile water or saline, polyalkylene glycols such as polyethylene glycol, vegetable oils, or hydrogenated naphthalenes. Other exemplary excipients include, but are not limited to, calcium bicarbonate, calcium phosphate, various sugars and types of starch, cellulose derivatives, gelatin, ethylene-vinyl acetate co-polymer particles, and surfactants, including, for example, polysorbate 20.

(279) A pharmaceutical composition of the present disclosure can be administered by a variety of methods known in the art. The route and/or mode of administration may vary depending upon the desired results. In some embodiments, the administration is intravitreal, intravenous, intramuscular, intraperitoneal, or subcutaneous. The pharmaceutically acceptable carrier should be suitable for intravitreal, intravenous, intramuscular, subcutaneous, parenteral, spinal, or epidermal administration (e.g., by injection or infusion). Depending on the route of administration, the active compound(s), i.e., the anti-NPR1 antibody or antigen binding fragment and optionally the additional therapeutic agent, may be coated in a material to protect the compound(s) from the

action of acids and other natural conditions that may inactivate the compound(s).

(280) Typically, a therapeutically effective dose or efficacious dose of the anti-NPR1 antibodies or antigen binding fragments is employed in the pharmaceutical compositions of the present disclosure. The anti-NPR1 antibodies or antigen binding fragments may be formulated into pharmaceutically acceptable dosage forms by conventional methods known to those of skill in the art.

(281) Methods of formulating suitable pharmaceutical compositions are known in the art, see, e.g., Remington: The Science and Practice of Pharmacy. 21st ed., 2005; and the books in the series Drugs and the Pharmaceutical Sciences: a Series of Textbooks and Monographs (Dekker, NY), the contents of each of which are incorporated by reference herein for this purpose. For example, solutions or suspensions used for parenteral, intradermal, or subcutaneous application can include the following components: a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerin, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide. The parenteral preparation can be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

(282) Dosage regimens for anti-NPR1 antibodies and antigen binding fragments with or without an additional therapeutic agent may be adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus of one or both agents may be administered at one time, several divided doses may be administered over a predetermined period of time, or the dose of one or both agents may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. For any particular subject, specific dosage regimens may be adjusted over time according to the individual's need, and the professional judgment of the treating clinician. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

(283) An "effective amount" is an amount sufficient to effect beneficial or desired results. For example, a therapeutic amount is one that achieves the desired therapeutic effect. This amount can be the same or different from a prophylactically effective amount, which is an amount necessary to prevent onset of disease or disease symptoms. An effective amount can be administered in one or more administrations, applications or dosages. A therapeutically effective amount of a therapeutic compound (i.e., an effective dosage) depends on the therapeutic compounds selected. The skilled artisan (such as a medical doctor) will appreciate that certain factors may influence the dosage and timing required to effectively treat a subject, including but not limited to the severity of the disease or disorder, previous treatments, the general health and/or age of the subject, and other diseases present. Moreover, treatment of a subject with a therapeutically effective amount of the therapeutic compounds described herein can include a single treatment or a series of treatments.

(284) Dosage, toxicity and therapeutic efficacy of the therapeutic compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ (the dose lethal to 50% of the population) and the ED₅₀ (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD₅₀/ED₅₀. Compounds which exhibit high therapeutic indices are preferred. While compounds that exhibit toxic side effects may be used, care should be taken to design a delivery system that targets such compounds to the site of affected tissue in order to minimize potential damage to uninfected cells and, thereby, reduce side effects.

(285) Dosage regimens for anti-NPR1 antibodies and antigen binding fragments alone or in combination with an additional therapeutic agent may be adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus of one or both agents may be administered at one time, several divided doses may be administered over a predetermined period of time, or the dose of one or both agents may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. For any particular subject, specific dosage regimens may be adjusted over time according to the individual's need, and the professional judgment of the treating clinician. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

(286) Dosage values for compositions comprising an anti-NPR1 antibody or antigen binding fragment, and/or any additional therapeutic agent(s), may be selected based on the unique characteristics of the active compound(s), and the particular therapeutic effect to be achieved. A physician or veterinarian can start doses of the antibodies of the disclosure employed in the pharmaceutical composition at levels lower than that required to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved. In general, effective doses of the compositions of the present disclosure, for the treatment of obesity or another disorder described herein may vary depending upon many different factors, including means of administration, target site, physiological state of the patient, whether the patient is human or an animal, other medications administered, and whether treatment is prophylactic or therapeutic. The selected dosage level may also depend upon a variety of pharmacokinetic factors including the activity of the particular compositions of the present disclosure employed, or the ester, salt or amide thereof, the route of administration, the time of administration, the rate of excretion of the particular compound being employed, the duration of the treatment, other drugs, compounds and/or materials used in combination with the particular compositions employed, the age, sex, weight, condition, general health and prior medical history of the patient being treated, and like factors. Treatment dosages may be titrated to optimize safety and efficacy.

(287) Kits

(288) Also provided herein are kits including one or more of the compositions provided herein (e.g., an antibody or antigen binding fragment thereof described in Table 2, Table 3, or Table 4) and instructions for use. Instructions for use can include instructions for diagnosis or treatment of an NPR1-associated disease. Kits as provided herein may be used in accordance with any of the methods described herein. Those skilled in the art will be aware of other suitable uses for kits provided herein, and will be able to employ the kits for such uses. Kits as provided herein can also include a mailer (e.g., a postage paid envelope or mailing pack) that can be used to return the sample for analysis, e.g., to a laboratory. The kit can include one or more containers for the sample, or the sample can be in a standard blood collection vial. The kit can also include one or more of an informed consent form, a test requisition form, and instructions on how to use the kit in a method described herein. Methods for using such kits are also included herein. One or more of the forms (e.g., the test requisition form) and the container holding the sample can be coded, for example, with a bar code for identifying the subject who provided the sample.

(289) The disclosure is further illustrated by the following examples and claims, which are illustrative and are not meant to be further limiting. One skilled in the art will recognize many methods and materials similar or equivalent to those described herein, which could be used in the practice of the described compositions and methods. Such equivalents are within the scope of the present disclosure and claims. The contents of all references, including issued patents and published patent applications, cited throughout this application are hereby incorporated by reference.

EMBODIMENTS

(290) In more detail, the disclosure provides the following embodiments:

(291) Embodiment 1. An isolated antibody or antigen binding fragment that (i) binds to natriuretic peptide receptor 1 (NPR1); and (ii) is capable of activating NPR1 in the absence of atrial natriuretic peptide (ANP). Embodiment 2. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen-binding fragment thereof of embodiment 1, which does not bind to and/or does not activate natriuretic peptide receptor 2 (NPR2) and/or natriuretic peptide receptor 3 (NPR3).

(292) Embodiment 3. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of embodiment 1 or embodiment 2 which binds to (a) human NPR1; and (b) mouse NPR1 and/or rat NPR1.

(293) Embodiment 4. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of embodiment 1 or embodiment 2 which binds to (a) human NPR1; and (b) cyno NPR1.

(294) Embodiment 5. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen-binding fragment thereof of any one of embodiments 1-4, which is ANP non-competitive.

(295) Embodiment 6. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen-binding fragment thereof of any one of embodiments 1, 2, or 4, which is ANP competitive.

(296) Embodiment 7. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 which is capable of stabilizing the ANP-NPR1 complex.

(297) Embodiment 8. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 99-133 of SEQ ID NO: 1.

(298) Embodiment 9. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5, 7, or 8, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least two amino acid residues within amino acids 99-133 of SEQ ID NO: 1.

(299) Embodiment 10. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-9, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least 3, 4, 5, 6, 7, or 8 amino acid residues within amino acids 99-133 of SEQ ID NO: 1.

(300) Embodiment 11. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-10, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 99-111 of SEQ ID NO: 1.

(301) Embodiment 12. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-11, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 99-103 of SEQ ID NO: 1.

(302) Embodiment 13. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-12, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 105-111 of SEQ ID NO: 1.

(303) Embodiment 14. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-13, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least 2, 3, or 4 amino acid residues within amino acids 105-111 of SEQ ID NO: 1.

(304) Embodiment 15. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-14, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope of human NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 99-103 of SEQ ID NO: 1, (ii) 105-111 of SEQ ID NO: 1, (iii) 131-134 of SEQ ID NO: 1, and additionally binds to amino acid 375 and/or 378 of SEQ ID NO: 1.

(305) Embodiment 16. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 8-14, wherein the epitope is a conformational epitope, and wherein the conformational epitope additionally comprises at least one amino acid residue selected from the group consisting of amino acids 33, 34, 76, 82, and 104 of SEQ ID NO: 1.

(306) Embodiment 17. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of embodiment 15, wherein the conformational epitope additionally comprises at least one amino acid residue selected from the group consisting of amino acids 33, 34, 76, 82, 104, 374, and 375 of SEQ ID NO: 1.

(307) Embodiment 18. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-17, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 82, 102, 103, 105, 106, 109, 132, and 375 of SEQ ID NO: 1.

(308) Embodiment 19. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-18, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 34, 82, 102, 103, 105, 106, 107, 109, 132, 133, 375, and 378 of SEQ ID NO: 1.

(309) Embodiment 20. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-18, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 79, 82, 99, 102, 103, 105, 106, 109, 131, 132, and 375 of SEQ ID NO: 1.

(310) Embodiment 21. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, or 6, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 188-219 of SEQ ID NO: 1.

(311) Embodiment 22. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least 2, 3, 4, 5, 6, or 7 amino acids within amino acids 188-219 of SEQ ID NO: 1.

(312) Embodiment 23. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21, or 22, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope within NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) 201-208 of SEQ ID NO: 1, (iii) 215-238 of SEQ ID NO: 1, and (iv) 294-297 of SEQ ID NO: 1.

(313) Embodiment 24. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21-23, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 188, 192, 194, 197, 201, 208, and 219 of SEQ ID NO: 1.

(314) Embodiment 25. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21-24, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 188, 192, 194, 197, 201, 208, 219, and 295 of SEQ ID NO: 1.

(315) Embodiment 26. An isolated anti-NPR1 antibody or antigen binding fragment; or the

antibody or antigen binding fragment of any one of embodiments 1-5 or 7-20, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein the antibody or antigen binding fragment comprises the CDRs of one of the ANP non-competitive groups described in Table 3 or Table 4.

(316) Embodiment 27. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21-25, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein the antibody or antigen binding fragment comprises the CDRs of one of the ANP competitive groups described in Table 3 or Table 4.

(317) Embodiment 28. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, or 26, wherein the antibody or antigen binding fragment is WW01_LALA, WW03_LALA, XX01_LALA, XX01_DAPA, XX01_N30S_DAPA, XX03_LALA, XX04_LALA, XX06_LALA, XX06_DAPA, XX07_LALA, XX08_LALA, XX08_DAPA, XX08_N30S_DAPA, XX08_N30Q_DAPA, XX09_LALA, XX11_LALA, XX12_LALA, XX13_LALA, XX14_LALA, XX15_LALA, XX15_DAPA, XX16_LALA, XX16_DAPA, XX17_LALA, XX17_DAPA, XX18_LALA, XX18_DAPA, XX19_LALA, XX19_DAPA, XX20_LALA, XX20_DAPA, YY01_LALA, YY03_LALA, YY04_LALA, ZZ12_LALA, and ZZ13_LALA.

(318) Embodiment 29. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, or 27, wherein the antibody or antigen binding fragment is WW05_LALA, WW06_LALA, YY05_LALA, YY06_LALA, YY07_LALA, ZZ05_LALA, ZZ14_LALA, and ZZ16_LALA.

(319) Embodiment 30. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, or 28, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y 5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y 5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1SX.sub.2GX.sub.3Y (SEQ

ID NO: 431), wherein X.sub.1 is S or E, X.sub.2 is D or K, or X.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1Y.sub.2Y.sub.3Y.sub.4PR (SEQ ID NO: 432); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is Y or W, Y.sub.3 is E, V, R, A, T, or M, and Y.sub.4 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in IX.sub.1SX.sub.2GX.sub.3YX.sub.4 (SEQ ID NO: 433), wherein X.sub.1 is S or E, X.sub.2 is D or K, X.sub.3 is S or N, and X.sub.4 is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4Y 5PRT (SEQ ID NO: 430); wherein Y.sub.1 is M or Q, Y.sub.2 is S, E, T, or I, Y.sub.3 is Y or W, Y.sub.4 is E, V, R, A, T, or M, and Y.sub.5 is K, V, R, or A; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1IX.sub.2SX.sub.3GX.sub.4YX.sub.5X.sub.6YADSVKG (SEQ ID NO: 429), wherein X.sub.1 is A or V, X.sub.2 is S or E, X.sub.3 is D or K, X.sub.4 is S or N, X.sub.5 is I or T, and X.sub.6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in X.sub.1SX.sub.2GX.sub.3Y (SEQ ID NO: 431), wherein X.sub.1 is S or E, X.sub.2 is D or K, or X.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1WY.sub.2Y.sub.3PR (SEQ ID NO: 435); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in IX.sub.1SX.sub.2GX.sub.3YX.sub.4 (SEQ ID NO: 433), wherein X.sub.1 is S or E, X.sub.2 is D or K, X.sub.3 is S or N, and X.sub.4 is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (c) (I) HCDR1 comprises or

consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 119; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 119; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 120, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in Y.sub.1WY.sub.2Y.sub.3PR (SEQ ID NO: 435); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 121, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY.sub.1WY.sub.2Y.sub.3PRT (SEQ ID NO: 434); wherein Y.sub.1 is S, E, T, or I, Y.sub.2 is V, R, A, T, or M, and Y.sub.3 is K, V, R, or A; (d) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THYIH (SEQ ID NO: 436), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5TY.sub.6YADSVKG (SEQ ID NO: 437), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, Y.sub.5 is S, N, or M, and Y.sub.6 is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5TY.sub.6YADSVKG (SEQ ID NO: 437), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, Y.sub.5 is S, N, or M, and Y.sub.6 is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1TH (SEQ ID NO: 438), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5 (SEQ ID NO: 439), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, and Y.sub.5 is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists

of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THY (SEQ ID NO: 440), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in IY.sub.1Y.sub.2Y.sub.3GY.sub.4Y.sub.5T (SEQ ID NO: 441), wherein Y.sub.1 is S or G, Y.sub.2 is S or G, Y.sub.3 is S or Q, Y.sub.4 is S, Q, or G, and Y.sub.5 is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (e) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THYIH (SEQ ID NO: 436), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1SGY.sub.2Y.sub.3TYYADSVKG (SEQ ID NO: 442), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1SGY.sub.2Y.sub.3TYYADSVKG (SEQ ID NO: 442), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1TH (SEQ ID NO: 438), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SY.sub.1SGY.sub.2Y.sub.3 (SEQ ID NO: 443), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1THY (SEQ ID NO: 440), wherein X.sub.1 is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY.sub.1SGY.sub.2Y.sub.3T (SEQ ID NO: 444), wherein Y.sub.1 is S or G, Y.sub.2 is S or Q, and Y.sub.3 is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (f) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3X.sub.4X.sub.5 (SEQ ID NO: 445), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, X.sub.3 is W or Y, X.sub.4 is I or L, and X.sub.5 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2X.sub.3X.sub.4 (SEQ ID NO: 447), wherein X.sub.1 is S, K, or R, X.sub.2 is W

or Y, X.sub.3 is I or L, and X.sub.4 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2Y (SEQ ID NO: 448), wherein X.sub.1 is S or T, and X.sub.2 is S, K, or R, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4E (SEQ ID NO: 449), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, and Y.sub.4 is S, H, or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3 (SEQ ID NO: 450), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, and X.sub.3 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IY.sub.1QY.sub.2Y.sub.3Y.sub.4EY.sub.5 (SEQ ID NO: 451), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, Y.sub.4 is S, H, or L, and Y.sub.5 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (g) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3X.sub.4X.sub.5 (SEQ ID NO: 445), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, X.sub.3 is W or Y, X.sub.4 is I or L, and X.sub.5 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2X.sub.3X.sub.4 (SEQ ID NO: 447), wherein X.sub.1 is S, K, or R, X.sub.2 is W or Y, X.sub.3 is I or L, and X.sub.4 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFX.sub.1FSX.sub.2Y (SEQ ID NO: 448), wherein X.sub.1 is S or T, and X.sub.2 is S, K, or R, HCDR2 comprises or consists of an amino acid sequence as set forth in HQY.sub.1Y.sub.2Y.sub.3E (SEQ ID NO: 456), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, and Y.sub.3 is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid

sequence as set forth in GFX.sub.1FSX.sub.2YX.sub.3 (SEQ ID NO: 450), wherein X.sub.1 is S or T, X.sub.2 is S, K, or R, and X.sub.3 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IHQY.sub.1Y.sub.2Y.sub.3EY.sub.4 (SEQ ID NO: 458), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, and Y.sub.4 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (h) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 452), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 454), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1IY.sub.2QY.sub.3Y.sub.4Y.sub.5EY.sub.6Y.sub.7YVESVKG (SEQ ID NO: 446), wherein Y.sub.1 is S or N, Y.sub.2 is K or H, Y.sub.3 is S, Q, or H, Y.sub.4 is G or A, Y.sub.5 is S, H, or L, Y.sub.6 is T or K, and Y.sub.7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1Y (SEQ ID NO: 455), wherein X.sub.1 is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1QY.sub.2Y.sub.3Y.sub.4E (SEQ ID NO: 449), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, and Y.sub.4 is S, H, or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2 (SEQ ID NO: 457), wherein X.sub.1 is S or R, and X.sub.2 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IY.sub.1QY.sub.2Y.sub.3Y.sub.4EY.sub.5 (SEQ ID NO: 451), wherein Y.sub.1 is K or H, Y.sub.2 is S, Q, or H, Y.sub.3 is G or A, Y.sub.4 is S, H, or L, and Y.sub.5 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; or (i) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 452), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1

comprises or consists of an amino acid sequence as set forth in X.sub.1YX.sub.2IX.sub.3 (SEQ ID NO: 454), wherein X.sub.1 is S or R, X.sub.2 is W or Y, and X.sub.3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY.sub.1Y.sub.2Y.sub.3EY.sub.4Y.sub.5YVESVKG (SEQ ID NO: 453), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, Y.sub.4 is T or K, and Y.sub.5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1Y (SEQ ID NO: 455), wherein X.sub.1 is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in HQY.sub.1Y.sub.2Y.sub.3E (SEQ ID NO: 456), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, and Y.sub.3 is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX.sub.1YX.sub.2 (SEQ ID NO: 457), wherein X.sub.1 is S or R, and X.sub.2 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IHQY.sub.1Y.sub.2Y.sub.3EY.sub.4 (SEQ ID NO: 458), wherein Y.sub.1 is Q or H, Y.sub.2 is G or A, Y.sub.3 is H or L, and Y.sub.4 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239.

(320) Embodiment 31. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, or 29, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in GX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 459), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in GNSNRPY.sub.1 (SEQ ID NO: 460), wherein Y.sub.1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7V (SEQ ID NO: 461), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 229, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in GX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 459), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in GNSNRPY.sub.1 (SEQ ID NO: 460), wherein Y.sub.1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7V (SEQ ID NO: 461), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; (III) HCDR1 comprises or consists of an amino acid

sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313, HCDR3 comprises or consists of an amino acid sequence as set forth in GX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 459), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in YZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7 (SEQ ID NO: 462), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in ARGX.sub.1X.sub.2X.sub.3GX.sub.4LGFDH (SEQ ID NO: 463), wherein X.sub.1 is A or S, X.sub.2 is V or L, X.sub.3 is A or P, and X.sub.4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ.sub.1Z.sub.2Z.sub.3Z.sub.4Z.sub.5Z.sub.6Z.sub.7V (SEQ ID NO: 461), wherein Z.sub.1 is Y, D, or G, Z.sub.2 is T, S, or A, Z.sub.3 is S, P, or F, Z.sub.4 is S, T, or P, Z.sub.5 is H, S, or R, Z.sub.6 is G, S, or F, and Z.sub.7 is P, S, or V; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YAX.sub.3X.sub.4 (SEQ ID NO: 464), wherein X.sub.1 is S or G, X.sub.2 is S or T, X.sub.3 is I or M, and X.sub.4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1ISY.sub.2Y.sub.3GY.sub.4Y.sub.5Y.sub.6Y.sub.7YAY.sub.8SVKG (SEQ ID NO: 465), wherein Y.sub.1 is A or S, Y.sub.2 is A, S, or G, Y.sub.3 is S or H, Y.sub.4 is G or Y, Y.sub.5 is S or Y, Y.sub.6 is T or A, Y.sub.7 is Y, R, or N, and Y.sub.8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YAX.sub.2X.sub.3 (SEQ ID NO: 466), wherein X.sub.1 is S or T, X.sub.2 is I or M, and X.sub.3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in Y.sub.1ISY.sub.2Y.sub.3GY.sub.4Y.sub.5Y.sub.6Y.sub.7YAY.sub.8SVKG (SEQ ID NO: 465), wherein Y.sub.1 is A or S, Y.sub.2 is A, S, or G, Y.sub.3 is S or H, Y.sub.4 is G or Y, Y.sub.5 is S or Y, Y.sub.6 is T or A, Y.sub.7 is Y, R, or N, and Y.sub.8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2Y (SEQ ID NO: 467), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SY.sub.1Y.sub.2GY.sub.3Y.sub.4 (SEQ ID NO: 468), wherein Y.sub.1 is A, S, or G, Y.sub.2 is S or H, Y.sub.3 is G or Y, and Y.sub.4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YA (SEQ ID NO: 469), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY.sub.1Y.sub.2GY.sub.3Y.sub.4T (SEQ ID NO: 470), wherein Y.sub.1 is S or G, Y.sub.2 is S or H, Y.sub.3 is G or Y, and Y.sub.4 is S or Y, HCDR3 comprises or consists of an amino acid

sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; or (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YAX.sub.3X.sub.4 (SEQ ID NO: 464), wherein X.sub.1 is S or G, X.sub.2 is S or T, X.sub.3 is I or M, and X.sub.4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1Y.sub.2GYYY.sub.3Y.sub.4YAY.sub.5SVKG (SEQ ID NO: 471), wherein Y.sub.1 is A or S, Y.sub.2 is S or H, Y.sub.3 is T or A, Y.sub.4 is R or N, and Y.sub.5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X.sub.1YAX.sub.2X.sub.3 (SEQ ID NO: 466), wherein X.sub.1 is S or T, X.sub.2 is I or M, and X.sub.3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY.sub.1Y.sub.2GYYY.sub.3Y.sub.4YAY 5SVKG (SEQ ID NO: 471), wherein Y.sub.1 is A or S, Y.sub.2 is S or H, Y.sub.3 is T or A, Y.sub.4 is R or N, and Y.sub.5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2Y (SEQ ID NO: 467), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SY.sub.1Y.sub.2GY (SEQ ID NO: 472), wherein Y.sub.1 is A or S, and Y.sub.2 is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX.sub.1X.sub.2YA (SEQ ID NO: 469), wherein X.sub.1 is S or G, and X.sub.2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY.sub.1Y.sub.2G (SEQ ID NO: 473), wherein Y.sub.1 is A, S, or G, and Y.sub.2 is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

(321) Embodiment 32. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, or 30, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30,

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of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 32, 370, and 381, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 230, 371, and 382, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 34, 372, and 383, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 231, 373, and 384, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239.

(322) Embodiment 33. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, 29, or 31, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 321 and 354, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 229, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 321 and 354, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 325, 356, and 362; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 315 and 349, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; or (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 270 and 407, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 271, 389, and 408, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and

LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 273 and 409, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 271, 389, and 408, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 32 and 410, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 274, 390, and 411, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 275 and 412, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 276, 391, and 413, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

(323) Embodiment 34. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, 30, or 32, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (a) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3); (b) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3); (c) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3); (d) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42

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(HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (y) (I) SEQ ID NO: 226 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 230 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 231 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (z) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 284 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 285 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 277 (HCDR3), SEQ ID NO: 286 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 283 (LCDR3); or (aa) (I) SEQ ID NO: 291 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (II) SEQ ID NO: 294 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (III) SEQ ID NO: 295 (HCDR1), SEQ ID NO: 296 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 305 (LCDR3); or (IV) SEQ ID NO: 297 (HCDR1), SEQ ID NO: 298 (HCDR2), SEQ ID NO: 299 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 304 (LCDR3).

(324) Embodiment 35. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, 29, 31, or 33, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (a) (I) SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (II) SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (III) SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3); or (IV) SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3); (b) (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3); (c) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348

(HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 362 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 361 (LCDR3); (d) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 390 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 391 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); (e) (I) SEQ ID NO: 407 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 409 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 410 (HCDR1), SEQ ID NO: 411 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 412 (HCDR1), SEQ ID NO: 413 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); (f) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 325 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 315 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 322 (LCDR3); (g) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); or (h) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 356 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 355 (LCDR3).

(325) Embodiment 36. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, 30, 32, or 34, wherein the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region

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comprising an amino acid sequence of SEQ ID NO: 24; (bb) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 48; (cc) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 374, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 244; (dd) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 385, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 244; (ee) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 233, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 244; (ff) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 278, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 287; or (gg) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 300, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 306.

(326) Embodiment 37. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, 29, 31, 33, or 35, wherein the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 61, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 72; (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 85, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 96; (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 363; (d) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 392, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344; (e) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 414, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344; (f) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 316, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 327; (g) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 333, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344; or (h) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 357.

(327) Embodiment 38. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, 30, 32, 34, or 36, wherein the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 130; (d) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 130; (e) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 149; (f) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 149; (g) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (h) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (i) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (j) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (k)

a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (l) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (m) a heavy chain comprising an amino acid sequence of SEQ ID NO: 105, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (n) a heavy chain comprising an amino acid sequence of SEQ ID NO: 108, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (o) a heavy chain comprising an amino acid sequence of SEQ ID NO: 117, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (p) a heavy chain comprising an amino acid sequence of SEQ ID NO: 124, and a light chain comprising an amino acid sequence of SEQ ID NO: 50; (q) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 130; (r) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (s) a heavy chain comprising an amino acid sequence of SEQ ID NO: 141, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (t) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 149; (u) a heavy chain comprising an amino acid sequence of SEQ ID NO: 156, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (v) a heavy chain comprising an amino acid sequence of SEQ ID NO: 159, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (w) a heavy chain comprising an amino acid sequence of SEQ ID NO: 163, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (x) a heavy chain comprising an amino acid sequence of SEQ ID NO: 170, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (y) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (z) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (aa) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (bb) a heavy chain comprising an amino acid sequence of SEQ ID NO: 195, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (cc) a heavy chain comprising an amino acid sequence of SEQ ID NO: 195, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (dd) a heavy chain comprising an amino acid sequence of SEQ ID NO: 15, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (ee) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 50; (ff) a heavy chain comprising an amino acid sequence of SEQ ID NO: 376, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (gg) a heavy chain comprising an amino acid sequence of SEQ ID NO: 387, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (hh) a heavy chain comprising an amino acid sequence of SEQ ID NO: 235, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (ii) a heavy chain comprising an amino acid sequence of SEQ ID NO: 280, and a light chain comprising an amino acid sequence of SEQ ID NO: 289; or (jj) a heavy chain comprising an amino acid sequence of SEQ ID NO: 302, and a light chain comprising an amino acid sequence of SEQ ID NO: 308.

(328) Embodiment 39. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, 29, 31, 33, 35, or 37, wherein the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 63, and a light chain comprising an amino acid sequence of SEQ ID NO: 74; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 87, and a light chain comprising an amino acid sequence of SEQ ID NO: 98; (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 365; (d) a heavy chain comprising an amino acid sequence of SEQ ID NO: 394, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; (e) a heavy chain comprising an amino acid sequence of SEQ ID NO: 416, and a light chain comprising an

amino acid sequence of SEQ ID NO: 346; (f) a heavy chain comprising an amino acid sequence of SEQ ID NO: 318, and a light chain comprising an amino acid sequence of SEQ ID NO: 329; (g) a heavy chain comprising an amino acid sequence of SEQ ID NO: 335, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; or (h) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 359.

(329) Embodiment 40. The antibody or antigen binding fragment of any one of embodiments 1-39, which is an antigen binding fragment selected from the group consisting of a Fab, Fab', F(ab')₂, Fv, single domain antibody (dAb), and a single chain variable fragment (scFv), optionally wherein the antigen binding fragment is selected from the group consisting of a Fab, Fab', Fv, single domain antibody (dAb), and a single chain variable fragment (scFv).

(330) Embodiment 41. The antibody or antigen binding fragment of any one of embodiments 1-40, which is monoclonal.

(331) Embodiment 42. The antibody or antigen binding fragment of any one of embodiments 1-41, which is fully human.

(332) Embodiment 43. The antibody or antigen binding fragment of any one of embodiments 1-42, which is an IgG antibody, optionally which is an IgG_{sub.1} antibody.

(333) Embodiment 44. The antibody or antigen binding fragment of any one of embodiments 1-43, which is an IgG_{sub.1} antibody having a kappa light chain.

(334) Embodiment 45. The antibody or antigen binding fragment of any one of embodiments 1-44, which is a fully human antibody of the IgG1 isotype and has a kappa light chain.

(335) Embodiment 46. The antibody or antigen binding fragment of any one of embodiments 1-45, wherein the antibody or antigen binding fragment has further modifications as described herein, e.g., wherein the antibody or antigen binding fragment additionally has mutations in the Fc region according to the EU index of Kabat, wherein the mutations comprise at least D265A and P329A; or wherein the mutations comprise at least L234A and L235A.

(336) Embodiment 47. The antibody or antigen binding fragment of any one of embodiments 1-40, wherein the antibody or antigen binding fragment is: a) monoclonal; and/or b) fully human; and/or c) an IgG antibody, optionally an IgG_{sub.1} antibody; and/or d) has a kappa light chain; and/or e) has mutations in the Fc region according to the EU index of Kabat, optionally wherein the mutations comprise at least D265A and P329A; and/or f) has mutations in the Fc region according to the EU index of Kabat, optionally wherein the mutations comprise at least L234A and L235A.

(337) Embodiment 48. The antibody or antigen binding fragment of any one of embodiments 1-47, wherein the antibody or antigen binding fragment is therapeutic.

(338) Embodiment 49. An isolated antibody or antigen binding fragment that binds to the same epitope on human NPR1 as the antibody or antigen binding fragment of any one of embodiments 1 to 48.

(339) Embodiment 50. An isolated antibody or antigen binding fragment that competes for binding to human NPR1 with the antibody or antigen binding fragment of any one of embodiments 1 to 49.

(340) Embodiment 51. An isolated nucleic acid or nucleic acids encoding the amino acid sequence of the antibody or antigen binding fragment of any one of embodiments 1 to 50.

(341) Embodiment 52. A vector comprising the isolated nucleic acid(s) of embodiment 51.

(342) Embodiment 53. A host cell comprising the isolated nucleic acid(s) of embodiment 51 or the vector of embodiment 52.

(343) Embodiment 54. A method of producing the antibody or antigen binding fragment of any one of embodiments 1 to 50, comprising culturing the host cell of embodiment 53 under conditions suitable to produce the antibody or antigen binding fragment.

(344) Embodiment 55. The method of embodiment 54, wherein the method additionally comprises purification of the antibody or antigen binding fragment.

(345) Embodiment 56. A pharmaceutical composition comprising a purified antibody or antigen

binding fragment produced by the method of embodiment 55 and a pharmaceutically acceptable carrier.

(346) Embodiment 57. A pharmaceutical composition comprising an antibody or antigen binding fragment of any one of embodiments 1 to 50 and a pharmaceutically acceptable carrier.

(347) Embodiment 58. The pharmaceutical composition of embodiment 56 or 57 or a combination comprising an antibody or antigen binding fragment of any one of embodiments 1 to 50, wherein the composition further comprises an additional therapeutic agent.

(348) Embodiment 59. The pharmaceutical composition or combination of embodiment 58, wherein the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), and combinations thereof.

(349) Embodiment 60. The pharmaceutical composition or combination of embodiment 58 or embodiment 59, wherein the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propanolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nifedipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, and combinations thereof.

(350) Embodiment 61. The pharmaceutical composition or combination of embodiment 58 or embodiment 59, wherein the additional therapeutic agent is selected from chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, and combinations thereof.

(351) Embodiment 62. The pharmaceutical composition or combination of any one of embodiments 58-61, wherein the additional therapeutic agent is an angiotensin receptor-neprilysin inhibitor (ARNi).

(352) Embodiment 63. The pharmaceutical composition or combination of embodiment 58, wherein the additional therapeutic agent is selected from a corticosteroid, a leukotriene modifier, a bronchodilator, and combinations thereof.

(353) Embodiment 64. The pharmaceutical composition or combination of embodiment 63, wherein the additional therapeutic agent is selected from fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline and ipratropium, and combinations thereof.

(354) Embodiment 65. The pharmaceutical composition or combination of embodiment 63 or embodiment 64, wherein the additional therapeutic agent is selected from salmeterol, formoterol, albuterol, and levalbuterol, and combinations thereof.

(355) Embodiment 66. The pharmaceutical composition or combination of embodiment 58, wherein the additional therapeutic agent is selected from a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof, and combinations thereof.

(356) Embodiment 67. The pharmaceutical composition or combination of embodiment 66, wherein the additional therapeutic agent is selected from timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil and ripasudil, and combinations thereof.

(357) Embodiment 68. The pharmaceutical composition of embodiment 58, wherein the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an

angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), an angiotensin receptor-neprilysin inhibitor (ARNi), a corticosteroid, a leukotriene modifier, a bronchodilator, a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof.

(358) Embodiment 69. The pharmaceutical composition of embodiment 68, wherein the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nicardipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline, ipratropium, salmeterol, formoterol, albuterol, and levalbuterol, timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil and ripasudil, and combinations thereof.

(359) Embodiment 70. The antibody or antigen binding fragment thereof of any of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69 for use (i) in therapy, (ii) as a medicament or (iii) in the manufacture of a medicament for the treatment of a disease.

(360) Embodiment 71. Use of the antibody or antigen binding fragment of any of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69 for the manufacture of a medicament for the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

(361) Embodiment 72. Use of the antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69 for the manufacture of a medicament for the treatment of a cardiovascular disorder in a subject in need of such treatment.

(362) Embodiment 73. The use of embodiment 72, wherein the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(363) Embodiment 74. Use of the antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69, for the manufacture of a medicament for the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

(364) Embodiment 75. The use of embodiment 73 or embodiment 74, wherein the subject has heart failure, and wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure.

(365) Embodiment 76. The use of embodiment 73 or embodiment 74, wherein the subject has hypertrophic cardiomyopathy, and wherein the hypertrophic cardiomyopathy is ventricular hypertrophy.

(366) Embodiment 77. The use of embodiment 73 or embodiment 74, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension.

(367) Embodiment 78. The use of any one of embodiments 73, 74, or 77, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

(368) Embodiment 79. Use of the antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69, for the manufacture of a medicament for the treatment of a kidney disorder in a subject in need of such treatment.

(369) Embodiment 80. The use of embodiment 79, wherein the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

(370) Embodiment 81. The antibody or antigen binding fragment of any of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69 for use in the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

(371) Embodiment 82. The antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69 for use in the treatment of a cardiovascular disorder in a subject in need of such treatment.

(372) Embodiment 83. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 82, wherein the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

(373) Embodiment 84. The antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69, for use in the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

(374) Embodiment 85. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 83 or

embodiment 84, wherein the subject has heart failure, and wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure.

(375) Embodiment 86. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 83 or embodiment 84, wherein the subject has hypertrophic cardiomyopathy, and wherein the hypertrophic cardiomyopathy is ventricular hypertrophy.

(376) Embodiment 87. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 83 or embodiment 84, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension.

(377) Embodiment 88. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiments 83, 84, or 87, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

(378) Embodiment 89. The antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69, for use in the the treatment of a kidney disorder in a subject in need of such treatment.

(379) Embodiment 90. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 89, wherein the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

EXAMPLES

(380) The following examples provide illustrative embodiments of the disclosure. One of ordinary skill in the art will recognize the numerous modifications and variations that may be performed without altering the spirit or scope of the disclosure. Such modifications and variations are encompassed within the scope of the disclosure. The examples provided do not in any way limit the disclosure.

(381) This disclosure provides anti-NPR1 antibodies that specifically bind and activate NPR1, e.g., antibodies and antigen binding fragments that (i) bind to NPR1; and (ii) activate NPR1 in the absence of ANP. Antibodies that specifically bind and activate NPR1 could have different possible modes of action: (1) the antibody induces a conformational change within the NPR1 monomers to activate the receptor; (2) the antibody directly mimics the structure and function of the natural ligand ANP and activates the receptor by binding in the ANP binding pocket of NPR1; or (3) the antibody stabilizes the preformed functionally active complex of hNPR1 and ANP (NPR1-ANP-complex).

Example 1: HuCAL® Phage Library Panning and Screening

(382) For the selection of NPR1-specific antibodies covering the described different methods of action, 13 different panning strategies were applied (see Table 5). Ten strategies were performed exclusively on protein (strategies 1-6 and 10-13). In addition, three differential cell pannings were performed (strategies 7-9). In total, four panning strategies (strategies 3 and 11-13) aimed for the enrichment of ANP competing antibodies (elution with ANP, pre-adsorption of phage on NPR1-ANP-complexes, and anti-idiotypic pannings on murine anti-ANP antibodies).

(383) TABLE-US-00005 TABLE 5 Overview of HuCAL ® panning strategies Strategy 1.sup.st round 2.sup.nd round 3.sup.rd round comments 1 hNPR1 solution hNPR1 solution hNPR1 solution Solution panning with human antigen only 2 hNPR1 capture hNPR1 capture hNPR1 capture Fc capture panning with human antigen only 3 hNPR1 solution hNPR1 solution + hNPR1 solution + Solution panning aiming for ANP elution ANP elution enrichment of ANP competitors 4 hNPR1 solution hNPR1 solution + hNPR1 solution + Solution panning aiming for pH 5.8 elution pH 5.8 elution enrichment of pH dependent binders 5 hNPR1 solution rNPR1 capture hNPR1 solution Solution/Fc capture panning aiming for rat crossreactivity 6 rNPR1 capture hNPR1 capture rNPR1 capture Fc capture panning aiming for rat crossreactivity 7 hNPR1 capture CHO-K1 NPR1 rNPR1 capture Fc capture/cell panning aiming cells for rat crossreactivity 8 hNPR1 solution CHO-K1 NPR1 hNPR1 solution Solution/cell panning with cells human antigen only 9 CHO-K1 NPR1 hNPR1 solution CHO-K1 NPR1 Solution/cell panning with cells cells human antigen only 10 hNPR1-ANP- rNPR1-ANP- hNPR1-ANP- Solution panning aiming for complex complex complex enrichment of NPR-ANP solution capture solution complex stabilizers and rat crossreactivity 11 pre-adsorption pre-adsorption pre-adsorption Solution panning aiming for on NPR1-ANP on NPR1-ANP on NPR1-ANP enrichment of ANP competitors complex, save complex, save complex, save the unbound the unbound the unbound phage still in phage still in phage still in solution, bind to solution, bind to solution, bind to hNPR1 in hNPR1 in hNPR1 in solution without solution without solution without ANP and ANP and ANP and capture capture capture NPR1/phage NPR1/phage NPR1/phage complexes complexes complexes 12 mouse anti- hNPR1 solution mouse anti- Fc capture/cell panning with ANP-mAb ANP-mAb anti-ANP mAb capture capture 13 mouse anti- mouse anti- mouse anti- Fc capture panning with anti- ANP mix of ANP mix of ANP mix of ANP mAb aiming for enrichment two mAbs two mAbs two mAbs of ANP competitors capture capture + capture + ANP elution ANP elution

(384) For Fc capture panning, NPR1-hFc was immobilized on a 96-well plate via an appropriate capture antibody (a goat or mouse anti-human Fc antibody). The antigen was immobilized in an appropriate number of wells of a 96-well plate and wells were subsequently blocked prior to the addition of phage-antibodies. In parallel to well preparation, phage-antibodies were blocked. During blocking of phage, additional blocking reagents were added to the blocking buffer to avoid selection of antibodies against the hFc-tag or the capture antibody (goat or mouse y globulin). Following the blocking procedure, two pre-adsorption steps on human y globulin and on the counter-target hNPR3-hFc were performed to avoid selection of antibodies against the Fc-tag or the counter-target. The pre-blocked and pre-adsorbed phage mix was added to each well with immobilized NPR1-hFc and the phage-antibodies were allowed to bind to the antigen. Intensive washing ensured removal of non-specifically bound phage, followed by elution of specifically bound phage. The second and third round of solid phase panning was performed according to the protocol of the first panning round. Amounts of antigen were decreased and washing conditions with increased stringency were applied.

(385) For solution panning, NPR1 was biotinylated and the retained activity of biotinylated NPR1 for ANP binding was confirmed. During solution panning, the Fab displaying phage and the biotinylated NPR1-hFc were incubated in solution, which facilitated the accessibility of the antigen by the phage. An appropriate amount of Streptavidin beads was blocked and, in parallel, an appropriate amount of phage-antibodies was blocked. During blocking of phage, human y globulin, the counter-target hNPR3-hFc and the Flag-TEV linker peptide were added to the blocking buffer to avoid selection of antibodies against the hFc-tag, the counter-target, or the linker peptide. For removal of Streptavidin-, Biotin-, or bead-binding phage, pre-adsorption steps of blocked phage particles were performed using blocked Streptavidin beads with and without coupled biotinylated irrelevant antigen. Subsequently, biotinylated NPR1-hFc/NPR1-hFc-ANP-complex was added to the pre-adsorbed and blocked phage particles and the phage-antibodies were allowed to bind to the antigen in solution. For enrichment of antibody phage binding to the ANP-binding site of NPR1

(ANP competitive antibodies) the pre-formed NPR1-ANP-complex was added to the phage blocking solution or the ANP peptide was used for elution of the bound phage. Thereby, the ANP peptide was used at least in 250-fold molar excess to the NPR1 antigen or the NPR1 expressing cells. The phage-antigen complexes were captured using blocked Streptavidin beads and phage particles bound to the Streptavidin beads were collected with a magnetic separator. Phage bound nonspecifically were washed off by several washing steps. Specifically bound phage were eluted from Streptavidin beads. The eluate was transferred to an *E. coli* culture for phage infection. The second and third round of bead-based solution panning was performed according to the protocol of the first panning round. Amounts of antigen were decreased and washing conditions with increased stringency were applied.

(386) For whole cell panning, an appropriate amount of phage-antibodies was blocked. During blocking of phage, counter-target hNPR3-hFc was added to the blocking buffer to avoid selection of antibodies against the counter-target. In parallel, an appropriate amount of target cells expressing NPR1 and an appropriate amount of adsorption cells without expression of antigen (parental cells) per phage pool were blocked. The blocked target cells were spun down, resuspended in the pre-blocked phage particles and the phage-antibodies were allowed to bind to the NPR1 presented on the cell. The phage-cell complexes were washed several times. For enrichment of antibody phage binding to the ANP-binding site of NPR1 (ANP competitive antibodies) the pre-formed NPR1-ANP-complex was added to the phage blocking solution or the ANP peptide was used for elution of the bound phage. Thereby, the ANP peptide was used at least in 250-fold molar excess to the NPR1 antigen or the NPR1 expressing cells. Specifically bound phage were eluted from target cells. After centrifugation, the supernatant (eluate) was applied to adsorption cells for removal of phage binding to cell surface molecules other than the target antigen (post-adsorption). The final supernatant was transferred to an *E. coli* culture for phage infection. The second and third round of the whole cell panning were performed according to the protocol of the first panning round. Washing conditions with increased stringency were applied.

(387) The outputs of the panning rounds were subsequently subcloned into bacterial expression vectors and bacterial lysates (BEL) were used for primary and secondary screening. The outputs were analyzed for binding to human and rat NPR1 during the primary screening (ELISA-based). Clones binding to human NPR3 were deselected. Secondary screening was performed on hNPR1 expressing CHO-K1 cells. Further screenings regarding ANP competition and binding solely in presence of ANP were performed. Approximately 1700 clones fulfilled the screening selection criteria and 760 clones were selected for sequencing. The sequencing of 760 clones resulted in 210 HCDR3 unique hits, whose binding properties are summarized in Table 6. Of these clones, 72 demonstrated significant ANP competition, while 7 clones bound only in presence of ANP.

(388) TABLE-US-00006 TABLE 6 Binding properties of 210 HCDR3 unique hits (HuCAL®)

Binding to hNPR1	Number of HCDR3 unique candidates	hNPR1	rNPR1	expr. cells	210 of which	7
47 human/	28 human/rat cross reactive	cell binders	Yes	Yes	Yes	clones only bind rat cross
19 human/rat cross reactive	Yes	Yes	No	in presence of reactive (not binding to cells)	ANP	156 human
53 human specific cell binders	Yes	No	Yes	specific	103 human specific (not binding to cells)	Yes
No	No					No

Example 2: Antibody Reformatting, Expression, and Purification

(389) After confirmation of binding, the VH and VL domains of the 210 HCDR3 unique clones were subcloned into a vector with a human IgG constant region. 180 of the 210 clones were selected for expression and 166 of the 180 passed the production quality control. They were characterized in regard to binding to relevant cell lines and functional activity. 40 of the 166 candidates were then selected for exploratory scale production, and 31 of these candidates were characterized in detail as shown below with respect to binding to relevant antigens and cell lines, ANP competition, and functionality in a cell based cGMP production assay.

(390) For production of the IgG candidates, eukaryotic HKB11 cells were transfected with

mammalian expression vector DNA encoding both heavy and light chains of IgG. Cell culture supernatants were harvested at appropriate times and subjected to Protein A affinity chromatography. If needed, a second purification step was performed to remove aggregates. Buffer exchange was performed to 1x Dulbecco's PBS (pH 7.2) and samples were sterile filtered (0.2 μ m pore size).

Example 3: Antigen Binding, ANP Competition and Cellular cGMP Production-HuCAL® Candidates

(391) The 31 IgGs which passed the exploratory scale production quality control were tested via ELISA for binding to the following antigens: human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target). The clones were also tested by flow cytometry for binding to human NPR1 expressing CHO K1 cells in the absence and presence of ANP and on parental CHO K1 cells. The binding properties of the five functional candidates are shown in FIG. 1 and FIG. 2.

(392) The same 31 IgGs were tested for ANP competition using a Fluorescence Resonance Energy Transfer (FRET)-based assay in which the NPR1-specific antibodies competed with ANP for binding to NPR1. In this FRET based assay (see FIG. 3), Eu-labeled Streptavidin (for measurement of IgGs) or Eu-labeled anti-hFc antibody (for measurement of FabCys) was used as an energy donor, while Cy5-labeled ANP was used as an acceptor (for all measurements). The resulting fluorescent signal was decreased by ANP-competitive antibodies. The assay was performed as follows: antibodies were mixed with NPR1 and incubated for 5 minutes at room temperature. After addition of the Eu-labeled donor and incubation for 30 minutes at room temperature, the Cy5-ANP solution was added. After a further 60-minute incubation, readout was performed using a TECAN Infinite M1000 Pro using an excitation wavelength of 317 nm and an emission wavelength of 665 nm. Percentage of ANP competition was calculated according to the following formulae:

$$\text{Ratio}^* = [(A_{\text{sub.665 nm}} / A_{\text{sub.620 nm}}) * 10^4]$$

$$\text{Ratio} = (\text{Ratio}^* - \text{Ratio}_{\text{sub.neg}})$$

Competition % = $[100 - (\text{Ratio} / (\text{Ratio}_{\text{sub.pos}} / 100))]$ Ratio.sub.neg: mean Ratio*data values of control without NPR1 Ratio.sub.pos: mean Ratio data values of control without agonist (reaction buffer)

(393) 15 of the 31 IgGs were ANP competitive, but only two of these candidates showed functionality in the cGMP assay (WW04 and WW06). The other three functional candidates WW01, WW02 and WW03 demonstrated a “negative” ANP competition in this assay indicating the stabilization of the NPR1-ANP-complex. The FRET assay results for the five functional candidates are depicted in FIG. 4.

(394) Additionally, as discussed above, the 31 IgGs were tested for their functional activity in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. For the functional characterization of the selected antibodies the production of cyclic guanosine 3',5'-cyclic monophosphate (cGMP) upon binding to and stimulation of NPR1 expressed on the cell surface of CHO-K1 cells was monitored. Cellular cGMP is a major second messenger that mediates cell activities and is synthesized by activated NPR1 triggered by ANP or NPR1-specific antibodies. Therefore, a commercial assay kit was used (Cisbio Bioassays CisBio HTRF Assay Kit CisBio (Cat. #62GM2PEB)). The assay was performed according to manufacturer's instructions with minor deviations. In brief, cells were adjusted to 1×10^5 cells/mL, 20 μ L/well were seeded in 96 well microtiter plates and were incubated overnight. After addition of 10 μ L/well of the antibodies in different concentrations, the plate was incubated for 30 min at 37° C. to allow for cGMP production. In parallel, a standard curve using a calibrator (contained in the kit) was generated. The cells were lysed and a mix of cGMP-d2 and anti-cGMP-Cryptate was added and incubated for 1 h at room temperature. The readout was performed using a Tecan M1000 Pro using an excitation wavelength of 317 nm and an emission wavelength of 665 nm. cGMP concentration (Delta F [%]) was calculated according to the following formulae:

Ratio=[(A.sub.665 nm/B.sub.620 nm)*10.sup.4]

Mean Ratio=(Σratios/2)

CV=[(Std deviation/Mean ratio)*100]

Delta F=[(((Calibrator or sample Ratio–Ratio.sub.neg)/Ratio.sub.neg)*100] Ratio.sub.neg: negative control

(395) Five candidates with significant functional activity were identified using the cellular cGMP assay: WW01, WW02, WW03, WW04, and WW06. These five candidates were functionally active and could be assigned to different methods of action. WW01, WW02, and WW03 were able to stabilize the NPR1-ANP-complex, while WW06 was determined to be ANP competitive. These candidates were all derived from initial panning codes 10 and 11 (aiming for method of action 2 or 3). The results of the assay for the cellular production of cGMP in the absence or presence of 0.075 nM ANP induced by the five functional candidates (IgG format) are shown in FIG. 5.

Example 4: Production and Characterization of HuCAL® Candidates in FabCys Format

(396) The functional clones were also tested for functionality in FabCys format. Eukaryotic HKB11 cells were transfected with mammalian expression vector DNA encoding both heavy and light chains of disulfide-bridged FabCys. Cell culture supernatants were harvested at appropriate times and subjected to metal ion affinity chromatography using a liquid handling station. Buffer exchange was performed to 1x Dulbecco's PBS (pH 7.2) and samples were sterile filtered (0.2 μm pore size).

(397) The five functional candidates WW01, WW02, WW03, WW04 and WW06 were additionally analyzed with regard to their monovalent affinities for human and rat NPR1 and the counter-target human NPR3 in absence and presence of ANP in monovalent FabCys format. The results of the affinity determination, epitope binning, and cGMP assay are summarized in Table 7.

(398) TABLE-US-00007 TABLE 7 Summary of Affinity, Epitope, and cGMP data for WW01, WW02, WW03, WW04, and WW06 in FabCys Format

Characterization	monovalent FabCys	cGMP assay	Epitope	cGMP conc [nM]	Bin	on at 2 μM FabCys	Affinity	FabCys K.sub.D [nM]	hNPR1 +0.075 hNPR1 + rNPR1 + (-ANP- Without nM Antibody hNPR1 ANP rNPR1 ANP complex)	ANP ANP WW01	— 1.5 — 2.0 B 30 121 WW02	weak weak weak weak n.a. 4 20 WW03	1000 0.1 2600 1.0 B 32 108 WW04	0.8 0.4 66 73 D 4 0 WW06	5.3 12 weak weak D 281 251
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(399) For candidates WW01 and WW03 no or very weak binding to human and rat NPR1 was observed in the absence of ANP, while the affinities in the presence of ANP were in the low nanomolar to subnanomolar range. Both shared the same epitope bin “B”. The affinity of candidates WW02 was too weak for adequate determination of K.sub.D values and the epitope bin. WW04 and WW06 had affinities in the double-digit nanomolar to subnanomolar range, which were independent from the presence or absence of ANP. Both shared the same epitope bin “A”. WW06 was the only candidate which did not exhibit rat cross-reactivity.

(400) While for WW02, WW03 and WW04 no binding to the counter-target hNPR3 in the absence or presence of ANP was observed, additional binding to the counter-target was detected for WW01 as well as for WW06 at higher concentrations.

Example 5: Reformatting HuCAL® Candidates into IgG Format

(401) Subcloning from the FabCys vector into an IgG.sub.1_LALA vector for expression in mammalian cells was performed via amplification of the Fab-encoding insert using one biotinylated primer and one non-biotinylated primer. The amplified product was bound on streptavidin beads, digested using restriction enzymes, and washed, resulting in the release of the purified insert into the supernatant. The insert was cloned into the acceptor vector, the DNA was transformed and single clones were quality controlled via colony PCR and sequencing.

(402) The five functional candidates WW01, WW02, WW03, WW04, and WW06 in IgG format were characterized as described above. Binding data (ELISA, flow cytometry, ANP competition) and functional data (cGMP assay) as well as affinities, and epitope bins are shown in Table 8.

Interestingly, WW06 had a significantly increased functional activity in FabCys format compared to IgG format as shown in FIG. 6.

(403) TABLE-US-00008 TABLE 8 Summary of Affinity, Epitope, and cGMP data for WW01, WW02, WW03, WW04, and WW06 in IgG Format Characterization in bivalent IgG format Flow cGMP Cytometry assay [S/BG] at cGMP Elisa +/- 100 nM ANP IgG binding/EC.sub.50 [nM] 2 μM IgG ANF conc hNPR1 W74R rNPR1 hNPR3 hNPR1 competition [nM] at + + + + CHO assay 2 μM IgG 100 100 100 100 hNPR1 KI + ANP +0.075 nM nM nM nM- CHO 100 nM competition Without nM Antibody hNPR1 ANP W74R ANP rNPR1 ANP hNPR3 ANP K1 ANP by IgG ANP ANP WW01 no 1.0 no no no no no no 19.9 222.2 No 36 197 specific specific specific specific specific specific specific binding binding binding binding binding binding binding binding WW02 no 531.6 no no no no no no 2.0 17.4 No 6 85 specific specific specific specific specific specific specific binding binding binding binding binding binding binding binding WW03 42.7 1.4 no no no no no no 2.2 214.3 No 31 192 specific specific specific specific specific specific specific binding binding binding binding binding binding binding binding WW04 2.5 4.2 3.5 1.8 no no no no 1.9 46.6 Yes 8 107 specific specific specific specific binding binding binding binding WW06 2.5 1.1 0.9 0.8 no no no no 4.6 1.1 Yes 13 38 specific specific specific specific binding binding binding binding

Example 6: Generation of HuCAL® Maturation Libraries

(404) To increase affinity and biological activity of the selected antibody fragments (WW01, WW02, WW03, WW04, and WW06), LCDR3 and HCDR2 regions were exchanged in parallel by diversified cassettes/modules (Prassler et al. (2009): In vitro affinity maturation of HuCAL® antibodies: complementarity determining region exchange and RapMAT technology; Immunotherapy 1 (4), pp. 571-583, the contents of which are hereby incorporated by reference for this purpose), while the framework regions were kept constant. Parental Fab fragments were transferred from the corresponding expression vector into a library cloning vector for affinity maturation.

(405) The generation of HuCAL® maturation libraries was performed for each maturation candidate individually. For LCDR3 optimization, an approximately 400 bp DNA fragment encoding for the LCDR3, framework 4 as well as the constant region of the light chain was removed from the sequence encoding the parental antibody by restriction digest. In order to reduce the background of the parental undiversified sequence the excised fragment was replaced by an approximately 520 bp dummy sequence via ligation, before a repertoire of DNA fragments encoding for diversified LCDR3 regions together with framework 4 and the constant domain (diversified LCDR3 cassette) was inserted via restriction digest and ligation.

(406) In a second library set the HCDR2-encoding sequence was diversified, while the connecting framework regions were kept constant. In order to reduce the background of the parental undiversified sequence an approximately 150 bp DNA fragment containing the parental HCDR2 and the framework 3 sequences was replaced by an approximately 590 bp dummy sequence via restriction digest and ligation, before the diversified HCDR2 cassette (including framework 3) was inserted also via restriction digest and ligation.

(407) The ten maturation libraries were successfully cloned and had library sizes between $9.2 \times 10^{8.8}$ and $2.2 \times 10^{9.9}$ cfu. Ligation mixtures were electroporated into *E. coli* cells yielding $>10^{8.8}$ independent colonies. Amplification of the library was performed as described previously (Rauchenberger et al. (2003): Human combinatorial Fab library yielding specific and functional antibodies against the human fibroblast growth factor receptor 3; *J Biol Chem* 278 (40), pp. 38194-38205, the contents of which are hereby incorporated by reference for this purpose). For quality control, approx. 10-20 single clones per library were picked randomly and sequenced.

(408) For the selection of affinity improved candidates, phage derived from maturation libraries were subjected to three rounds of maturation panning as described further below. Panning stringency was increased by prolonged washing steps. In addition, off-rate selection was performed (Hawkins et al. (1992): Selection of phage antibodies by binding affinity. Mimicking affinity

maturation. In *J. Mol. Biol.* 226 (3), pp. 889-896, the contents of which are hereby incorporated by reference for this purpose).

Example 7: Pannings and Screenings-HuCAL®

(409) The maturation libraries were used for four different maturation panning strategies. Strategies #3 and #4 aimed for the enrichment of progenies with improved affinities compared to the parental clones. In addition, strategies #1 and #2 aimed for the enrichment of clones with improved affinities for NPR1 instead of NPR1-ANP-complex. The rationale behind that was the idea to generate candidates which are able to directly active NPR1 by a conformational change. During the panning process, all maturation libraries were kept separately. The panning strategies are summarized in Table 9 in detail. The outputs of the third panning rounds were subsequently sub-cloned into a bacterial expression vector and bacterial lysates (BEL) were used for SET screening.

(410) TABLE-US-00009 TABLE 9 Overview of HuCAL® Maturation Panning Strategies

Parental Strategy	antibodies	1.sup.st round	2.sup.nd round	3.sup.rd round	comments
1	WW01 CHO-K1 NPR1 hNPR1 CHO-K1 NPR1	Cell/solution panning aiming improved candidates WW03 WW04 WW06	2	WW01 CHO-K1 NPR1 hNPR1-ANP- CHO-K1 NPR1	Cell/solution panning aiming WW02 cells complex cells WW03 solution for improved candidates WW04
3	WW01 hNPR1-ANP- CHO-K1 hNPR1-ANP-	Solution/cell aiming for WW02 complex NPR1-ANP- complex improved NPRI-ANP- WW03 solution complex solution complex WW04	Cells	4	WW04 Preadsorption Preadsorption Preadsorption Cell/solution panning aiming WW06 on NPR1-ANP on NPR1- on NPR1-ANP for improved ANP complex ANP complex complex competitors CHO-K1 NPR1 hNPR1 CHO-K1 NPR1 cells solution cells

(411) The outputs of the 3rd panning rounds were used for Solution Equilibrium Titration (SET) screening. 88 clones per subcode (2640 clones in total) were analyzed in SET screening for improved affinity for hNPR1 and/or hNPR1-ANP-complex compared to the parental clones.

(412) During SET screening, 82 HCDR2 or LCDR3 unique improved derivatives were identified. Compared to their parental clones, the affinities of WW01 and WW03 derivatives were improved up to 20-fold both for hNPR1 and hNPR1-ANP-complex. The affinities of the WW04 derivatives were not improved significantly, while the WW06 derivatives had up to 3-fold improved affinities compared to the parental clone. See FIG. 7 (which shows affinities for hNPR1) and FIG. 8 (which shows affinities for hNPR1-ANP complex). Affinities (K.sub.D [pM]) are indicated at the x-axis below the parental clone name for both figures.

Example 8: Characterization of Matured Candidates-HuCAL®

(413) 74 of the 82 improved candidates were successfully subcloned into FabCys format and 61 of the 74 clones passed the production quality control and were characterized in regard to binding to relevant antigens, binding to relevant cell lines, ANP competition, and functional activity in the cGMP production assay in comparison to their parental clones. All 16 derivatives of WW01 and all 27 derivatives of WW03 had up to 20-fold improved binding and functional activity. The majority of the derivatives also showed improved binding and functionality in the presence of ANP. Some derivatives showed binding to W74R (constitutively active hNPR1 mutant), which was not true for the parental FabCys. One of the four derivatives of WW04 had two-fold improved binding and functional activity, while the rest behaved like the parental FabCys. All 14 derivatives of WW06 had improved binding for NPR1 and remained competitive with ANP. The functional activities of 10 of the 14 progenies were improved up to three-fold compared to the parental FabCys. Some derivatives displayed rat cross-reactivity, which was not true for the parental FabCys.

(414) After FabCys characterization, a further 40 of the 61 derivatives were selected for IgG conversion and further characterization. Ten potential candidates shown in Table 10 were then assayed with respect to binding to relevant antigens, binding to relevant cell lines, ANP competition, and functional activity in the cGMP production assay in comparison to their parental clones. They were further analyzed via 3P assay and their affinities for human and rat NPR1 in absence and presence of ANP were determined via SET K.sub.D measurement. WW01 and WW03

derivative antibodies were analyzed in IgG format and the WW06 derivatives in FabCys format. (415) For protein panel profiling (Frese et al. (2013): An automated immunoassay for early specificity profiling of antibodies; mAbs 5 (2), pp. 279-287, the contents of which are herein incorporated by reference for this purpose), 32 different proteins and controls were coated on two 384-well MSD standard plates at a concentration of 1.0 µg/mL at 4° C. overnight. The coating solution was discarded and plates were blocked with 50 µL 3% (w/v) BSA in PBS for one hour at RT on a microtiter plate shaker (~500 rpm) followed by three washing steps with 50 µL washing buffer (PBS with 0.05% (v/v) Tween 20). Antibody samples were diluted to 100 nM and 10 nM in assay buffer (PBS with 0.5% (w/v) BSA, 0.05% (v/v) Tween 20). As controls, a reference antibody (Fab or IgG, depending on the sample format) and assay buffer were used. Samples and controls were added at 30 µL/well and incubated for three hours at RT on a microtiter plate shaker. The plates were washed three times and 30 µL detection antibody (ECL-labeled anti-human Fab) were added per well and incubated for one hour on a microtiter plate shaker (~500 rpm). After washing the MSD plate and adding 35 µL/well MSD Read Buffer T with surfactant, electrochemiluminescence signals were detected using a Sector Imager 6000 (Meso Scale Discovery; Gaithersburg, MD, USA). For evaluation, signals of the antibody sample on a certain protein were divided by the respective signals of the reference mAb resulting in a binding ratio (BR). The cumulative binding ratio (CBR) of all proteins except the controls (25 in total) was then calculated: CBR up to 150 represented an antibody or fragment thereof without detectable non-specific binding. Values above represented an antibody or fragment thereof with increased non-specific binding compared to a reference mAb.

(416) TABLE-US-00010 TABLE 10 Overview of Matured HuCAL® Candidates Matured Parental Matured antibody antibody CDR XX01 WW01 HCDR2 XX02 WW06 HCDR2 XX03 WW03 HCDR2 XX04 WW03 LCDR3 XX05 WW06 HCDR2 XX06 WW03 LCDR3 XX07 WW03 LCDR3 XX08 WW01 HCDR2 XX10 WW06 HCDR2 XX12 WW03 LCDR3

(417) Furthermore, the clones were tested via ELISA for binding to the following antigens: human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, human NPR3 (counter target), each in the absence and presence of ANP, and BSA. The clones were also analyzed by flow cytometry for binding to human NPR1 expressing CHO K1 cells in absence and presence of ANP (100 nM) and to parental CHO K1 cells. The binding properties of the 10 candidates are shown in FIGS. 9 and 10. The ANP competition results for nine of the candidates are shown in FIG. 11.

(418) Negative values for XX01, XX03, XX04, XX06, XX07, and XX12 suggest enhancement of ANP binding by these antibodies. The functional activity of the 10 candidates was analyzed using the cellular cGMP production assay and results are shown in FIG. 12. As expected from the functional data of the parental clone WW06, its derivatives XX02, XX05, and XX10 showed weak to no functional activity in the IgG format, whereas they had very high functionality in monovalent FabCys format.

(419) The affinities for XX01-XX08, XX10, and XX12 in monovalent FabCys format were determined via SET K.sub.D measurement. The results are summarized in Table 11 in comparison to the affinities of the parental clones (determined in another experiment via Biacore®). The affinities for human and rat NPR1 of WW01 and WW03 derivatives were improved up to 2,300-fold, while the affinities for the NPR1-ANP-complexes were only slightly improved (maximal 5-fold). They had affinities between 10 and 46 nM for hNPR1 and between 100 and 300 pM for hNPR1-ANP-complex. All WW01 and WW03 progenies displayed rat cross-reactivity with rat/human K.sub.D ratios <5. The affinities of the WW06 derivatives for human NPR1 and hNPR1-ANP-complex were improved maximally 8-fold and had K.sub.D values between 1 and 5 nM, while no binding to rat NPR1 or rat NPR1-ANP-complex could be observed.

(420) TABLE-US-00011 TABLE 11 Affinities of Parental and Matured HuCAL® Candidates Affinity (SET measurement) FabCys K.sub.D [nM] Antibody hNPR1 hNPR1 + ANP rNPR1 rNPR1 + ANP WW01 weak 1.5 weak 2.0 XX01 43 0.3 1.4 0.5 XX08 46 0.3 1.8 0.2 WW03 1000

0.1 2600 1.0 XX03 22 0.2 1.1 0.1 XX04 32 0.1 3.3 0.3 XX06 10 0.1 2.3 0.3 XX07 21 0.2 2.9 0.5
 XX12 16 0.2 3.0 0.5 WW06 5.3 12 — — XX02 1.0 1.4 — — XX05 3.2 5.1 — — XX10 2.2 4.0 —
 —

Example 9: Cross-Cloning and PTM Removal

(421) Parental clone WW03 had a 'DG' site in HCDR2. The majority of the WW03 derivatives (26 out of 27) were diversified in LCDR3. Only one candidate (XX03) was diversified in HCDR2 including the mutation of 'DG' into 'DK' at amino acid position 54 in the heavy chain variable region (see, e.g., position 54 of SEQ ID NO: 122). The light chains of the functional LCDR3 diversified clones were cross-cloned with the heavy chain of XX03 to engineer these clones without loss of functionality. Furthermore, the 'DG' to 'DK' mutation was inserted in the original heavy chains of several LCDR3 diversified derivatives. An overview of exemplary cross-cloned and D54K engineered candidates is shown in Table 12.

(422) TABLE-US-00012 TABLE 12 Overview of VL-VH Cross-cloned and D54K Engineered Clones (WW03 derivatives) Light chain Heavy chain Antibody origin origin XX13 XX06 WW03 D54K XX14 XX09 XX15 XX04 XX03 XX16 XX06 XX17 XX07 XX18 XX09 XX19 XX11 XX20 XX12

(423) Exemplary functional data of a cross-clone (XX16) compared to the original clone (XX06) and ANP are shown in FIG. 13. All cross-clones had similar or even better functional activity compared to their original clones. XX16 even had a maximal cGMP concentration comparable to the natural ligand ANP in an in vitro study analyzing their activity in hNPR1 transformed CHO cells (see FIG. 14).

(424) Cross-cloned and PTM removed clones were tested for their specificity via 3P assay. Both D54K engineered clones XX13 and XX14 showed non-specific binding to several antigens and were deselected, while no cross-clone showed non-specific binding. Results are shown in Table 13.

(425) TABLE-US-00013 TABLE 13 In vitro functional data for XX15 and XX16 cGMP Generation for human, rat, cyno NPR1 and human NPR2 expressing CHO cells hNPR1 Y max rNPR1 cNPR1 hNPR2 hNPR1 (% of rNPR1 Y max cNPR1 Y max hNPR2 Y max EC50 (nM) ANP) EC50 (nM) (% of ANP) EC50 (nM) (% of ANP) EC50 (% of Antibody +/-STDEV +/-STDEV +/-STDEV +/-STDEV +/-STDEV +/-STDEV (nM) ANP) XX16 3.2 +/- 0.4 97 +/- 1 8 +/- 2 94 +/- 8 12 +/- 2 86 +/- 2 >500 <1 XX15 9.4 +/- 0.4 98 +/- 7 23 +/- 6 100 +/- 10 30 +/- 2 95 +/- 3 >500 <1 XX18 40 +/- 10 110 +/- 10 26 +/- 9 99 +/- 8 50 +/- 10 90 +/- 5 >500 <1

Example 10: Crystal Structure of Anti-NPR1 Antibodies

(426) Crystal structures for several molecules in complex with hNPR1 were created as described below.

(427) For Fab03-WW03, the Fab construct of WW03 was complexed to the extracellular domain of hNPR1 (C264T) with a molar ratio of 2 Fab molecules for every 1 NPR1 molecule. The complex was incubated for 1 hour in the cold room rocking and then loaded onto a Superdex 200 16/60 column in the buffer 20 mM HEPES pH7.4, 100 mM NaCl. The complexed protein was separated from a small aggregate peak and the excess Fab and then concentrated to 19.8 mg/mL. The complex crystallized in space group P212121 and diffracted to a resolution of 2.89 Å. The model was built using molecular replacement with the hNPR1 structure and a Fab molecule, iteratively built in Coot and refined with Buster to an Rfree of 21.7%.

(428) For Fab06-WW06, the Fab construct of WW06 was complexed to the extracellular domain of hNPR1 (C264T) with a molar ratio of 2 Fab molecules for every 1 NPR1 molecule. The complex was incubated for 1 hour in the cold room rocking and then loaded onto a Superdex 200 16/60 column in the buffer 20 mM HEPES pH7.4, 100 mM NaCl. The complexed protein was separated from a small aggregate peak and the excess Fab and then concentrated to approximately 20.0 mg/mL. The complex crystallized in space group P212121 and diffracted to a resolution of 2.17 Å. The model was built using molecular replacement with the hNPR1 structure and a Fab molecule, iteratively built in Coot and refined with Buster to an Rfree of 20.9%.

(429) For Fab16-XX16, the Fab construct of XX16 was complexed to the extracellular domain of hNPR1 (C264T) with a molar ratio of 2 Fab molecules for every 1 NPR1 molecule. The complex was incubated for 1 hour in the cold room rocking and then loaded onto a Superdex 200 16/60 column in the buffer 20 mM HEPES pH7.4, 100 mM NaCl. The complexed protein was separated from a small aggregate peak and the excess Fab and then concentrated to approximately 20.0 mg/mL. The complex crystallized in space group P212121 and diffracted to a resolution of 3.02 Å. The model was built using molecular replacement with the hNPR1 structure and a Fab molecule, iteratively built in Coot and refined with Buster to an Rfree of 24.4%.

(430) The crystal structure of Fab06 in complex with hNPR1 is shown in FIG. 15. FIG. 16 shows the conformation of the hNPR1 ECD as it would be in complex with Fab06; the Fab06 were removed from the image to more clearly reveal the conformation of hNPR1 induced by Fab binding. The structures shown in FIGS. 15 and 16 explain the discrepancy between Fab and IgG functional data shown in FIG. 6. The Fab C-termini are 180 degrees apart precluding a single IgG spanning a receptor dimer. The structure of hNPR1-ECD in complex with ANP (which was also determined as a part of this work) is shown for comparison in FIG. 17 (left), and the crystal structure of Fab16 (the XX16 antibody in Fab format) in complex with hNPR1 extracellular domain (right). The WW06 Fab was significantly more potent (cGMP production) when tested on CHO cells expressing hNPR1 W74R/C232T (constitutively active mutant) compared to WT hNPR1 (FIG. 18, panels A and B) suggesting that the mutation destabilizes the head-to-head conformation of NPR1 and facilitates antibody binding/receptor activation.

Example 11: Ylanthia® Phage Library Panning and Screening

(431) Six panning strategies were performed, which reflected the most successful strategies from the initial pannings (HuCAL®) or modifications of these strategies aiming for specific methods of action. The panning strategies are summarized in Table 14 in detail. Strategies #2 and #5 were identical to strategies performed in the initial HuCAL® campaign and were selected because all five initial functional candidates were derived from these panning strategies. Strategies #3 and #4 were variations from initial strategies with focus on hNPR1 alternating with hNPR1 expressing cells. In addition, a constitutively active mutant of NPR1 (W74R) was used as an antigen in strategies #1 and #6. Bacterial lysates (BEL) of the outputs of the 3rd panning rounds in phage display vector pYPDis were directly used for primary and secondary screenings.

(432) TABLE-US-00014 TABLE 14 Overview of panning strategies-Ylanthia® Strategy 1.sup.st round 2.sup.nd round 3.sup.rd round Comments 1 W74R hNPR1 capture W74R hNPR1 capture W74R hNPR1 capture Fc capture panning with active W74R mutant only 2 hNPR1-ANP-complex rNPR1-ANP-complex hNPR1-ANP-complex Solution/Fc capture panning solution capture solution aiming for enrichment of NPR1- ANP-complex 3 CHO-K1 NPR-ANP- hNPR1-ANP-complex CHO-K1 NPR-ANP- Cell/solution panning aiming for complex cell solution complex cell enrichment of NPR1-ANP-complex stabilizers 4 hNPR1 solution CHO-K1 NPR1 cell + hNPR1 solution + Solution/cell aiming for ANP elution ANP elution enrichment of ANP competitors 5 pre-adsorption on pre-adsorption on pre-adsorption on Solution panning aiming for NPR1-ANP complex, NPR1-ANP complex, NPR1-ANP complex, enrichment of ANP competitors save the unbound save the unbound save the unbound phage still in solution, phage still in solution, phage still in solution, bind to bind to bind to hNPR1 in solution hNPR1 in solution hNPR1 in solution without ANP and without ANP and without ANP and capture NPR1/phage capture NPR1/phage capture NPR1/phage complexes complexes complexes 6 CHO-K1 NPR1 cell W74R hNPR1 capture CHO-K1 NPR1 cell Cell/Fc capture panning with human antigen and active W74R mutant

(433) The outputs of the 3.sup.rd panning rounds were analyzed for binding to relevant antigens and cell lines. 368 clones per subcode (in total 4416 clones) were screened in ELISA-based primary screening on human NPR1 in absence and presence of ANP, constitutively active hNPR1 mutant (W74R) and counter-target human hNPR3. The primary screening yielded 810 hits, which were analyzed with respect to binding of relevant cell lines (human NPR1 expressing CHO-K1

cells in absence and presence of ANP, parental CHO-K1 cells) and rat NPR1 in secondary screening. In total, 380 clones from primary and secondary screening were selected for sequencing with priority for exclusive binding to NPR1-ANP-complex, good cell binding, and rat cross-reactivity. The VL and VH sequencing resulted in 138 HCDR3 unique clones with different binding properties (Table 15). Of these clones six bound only in presence of ANP.

(434) TABLE-US-00015 TABLE 15 Binding properties of HCDR3 unique hits-Ylanthia® Binding to: hNPR1 expressing Number of unique candidates hNPR1 rNPR1 cells 140 (of which 6 53 human/rat 2 human/rat cross-reactive cell binders Yes Yes Yes clones bound only cross-reactive 51 human/rat cross-reactive Yes Yes No in presence of (not binding to cells) ANP) 79 human 14 human specific cell binders Yes No Yes specific 65 human specific Yes No No (not binding to cells)

(435) Since candidate WW06 derived from the initial HuCAL® pannings was significantly more active in FabCys format compared to IgG format, the functional screening was performed in FabCys format rather than IgG format.

(436) The sub-cloning of 138 HCDR3 unique clones into the FabCys format was performed via YClone®. 111 out of 138 clones were successfully converted into FabCys format and 95 clones were selected for further analysis. 92 of the 95 FabCys passed the production quality control and were analyzed in detail. Afterwards, 30 of the 92 clones with the most promising properties were selected for IgG conversion via AmplyFly®, exploratory scale expression and S-DAS. 24 of the 30 IgGs passed the production quality control and were analyzed in detail.

(437) All 92 FabCys and 24 IgGs were tested for binding to relevant antigens via ELISA and relevant cell lines by flow cytometry. Furthermore, the clones were tested for ANP competition and functional activity in the cellular cGMP production assay. Eight functional candidates were identified and analyzed for specificity in the Protein Panel Profiling assay (3P assay) in IgG format. For comparison, one of the functional candidates from the initial campaign (WW03) was analyzed. YY02 and YY03 showed low non-specific binding; and YY01, YY04, YY05, YY06, YY07 and WW03 did not show non-specific binding in this assay. All 92 FabCys and 24 IgGs were tested via ELISA for binding on the following antigens: human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, human NPR3 (counter target) in the absence or presence of ANP (100 nM) and irrelevant antigens. The clones were also tested by flow cytometry for binding on human NPR1 expressing CHO K1 cells in the absence and presence of ANP and on parental CHO K1 cells. The binding properties of the seven functional candidates in IgG format are shown in FIGS. 19 and 20. All 92 FabCys and 24 IgGs were tested for ANP competition. 22 out of 92 FabCys and 12 out of 24 IgGs showed a significant ANP competition >70% at a concentration of 1 µM FabCys/IgG. The results of the seven functional candidates in IgG format are shown in FIG. 21. YY02, YY05, YY06, and YY07 showed clear ANP competition. YY03 and YY04 showed a “negative” ANP competition in this assay indicating the stabilization of the NPR1-ANP-complex.

(438) All 92 FabCys and 24 IgGs were tested for their functional activity in the cellular cGMP production assay using human NPR1 expressing CHO-K1 cells in presence and absence of ANP. 8 of the 92 FabCys and the same 8 out of 24 IgGs were functionally active and could be assigned to the different methods of action. Five out of eight clones showed much higher functional activity in presence of ANP, including YY01, YY02, YY03, and YY04. Three other clones behaved ANP independent in the functional assay, namely YY05, YY06, and YY07. All eight clones were derived from the panning strategies #2, #3 and #5, whereby #2 and #5 were exact repetitions of the initial HuCAL® panning strategies #10 and #11, which led to the identification of the five functional clones from the initial HuCAL® campaign. The results of the cGMP assay for seven functional candidates in IgG format are shown in FIG. 22. As seen before for candidate WW06 derived from the initial HuCAL® campaign, the functional activities of candidates YY05 and YY07 significantly increased in monovalent FabCys format compared to the bivalent IgG format (FIG. 23).

(439) The seven functional candidates YY01-YY07 were analyzed with regard to their monovalent

affinities for human and rat NPR1 in absence and presence of ANP in monovalent FabCys format. The results of the affinity determination and the epitope binning are summarized in Table 16.

(440) TABLE-US-00016 TABLE 16 Characterization in monovalent FabCys format-Ylanthia® Characterization in monovalent FabCys format cGMP assay cGMP conc Affinity [nM] FabCys K.sub.D [nM] Epitope at 1 µM Bin on Binning FabCys hNPR1 rNPR1 hNPR1 FabCys +0.075 + + (-ANP- Without nM Antibody hNPR1 ANP rNPR1 ANP complex) ANP ANP YY01 — 16 — 14 B 15 46 YY02 45 1.1 280 4.4 B — — YY03 — 0.1 0.1 0.1 B — — YY04 4.9 1.1 — 1 B 18 67 YY05 0.9 350 700 16000 A 207 227 YY06 0.5 1 1600 1400 C 20 75 YY07 1.8 590 — — A 77 88

(441) The affinities were in the low nM to low µM range and strongly depended on the presence or absence of ANP. Four of the seven functional candidates showed significantly improved binding in presence of ANP (YY01, YY02, YY03, and YY04). YY05 and YY07 competed with ANP for binding to NPR1 and showed much higher affinities in absence of ANP. The affinities of YY06 were independent from ANP. Some candidates exhibited non-specific binding to the reference flow cell, while others had such high affinities that their K.sub.D values approach the assay limit. YY01, YY02, YY03, and YY04 share one epitope bin, which is the same as for WW03 from the initial HuCAL® campaign. YY05 and YY07 share another epitope bin, which is the same as for WW06 from the initial HuCAL® campaign. YY06 binds to a single epitope bin.

Example 12: Reformatting into IgG-Ylanthia®

(442) Subcloning of the Ylanthia® candidates from the FabCys vector into the IgG1_LALA vector for expression in mammalian cells was performed via amplification of the Fab-encoding insert using two biotinylated primers. The amplified product was bound on streptavidin beads, digested using restriction enzymes, and washed, resulting in the release of the purified insert into the supernatant. The insert was cloned into the acceptor vector, the DNA was transformed and single clones were quality controlled via colony PCR and sequencing.

(443) Five Ylanthia® candidates were selected for affinity maturation. YY01 and YY04 stabilize the NPR1-ANP-complex, YY06 behaves in an ANP-independent manner, and YY05 and YY07 are ANP-competitive. Binding data (ELISA, flow cytometry, ANP competition), functional data (cGMP assay), affinities, and epitope bins are shown in Table 17.

(444) TABLE-US-00017 TABLE 17 Summary of Affinity, Epitope, and cGMP data for YY01, YY04, YY08, YY06, and YY07 in IgG Format Characterization in bivalent IgG format Flow Cytometry [S/BG] at 1 µM IgG hNPR1 cGMP Elisa +/- 100 nM ANP IgG binding at 1 µM IgG CHO ANP assay cGMP hNPR1 W74R rNPR1 hNPR3 KI competition conc [nM] + + + + + assay at 1µM IgG 100 100 100 100 100 ANP +0.075 nM nM nM nM hNPR1 nM competition Without nM Antibody hNPR1 ANP W74R ANP rNPR1 ANP hNPR3 ANP CHOK1 ANP by IgG ANP ANP YY01 54 76 33 53 26 50 18 6 35 21 yes 5 14 YY04 31 60 11 13 8 54 69 18 2 33 no 10 15 YY05 81 32 69 23 11 16 7 4 15 5 yes 13 15 YY06 91 102 83 86 3 4 6 6 26 18 yes 12 15 YY07 68 24 53 20 3 3 6 5 00 44 yes 4 6

Example 13: Generation of Ylanthia® Maturation Libraries

(445) To increase affinity and biological activity and to reduce non-specificity of selected antibody candidates, LCDR3 and HCDR1/HCDR2 regions were optimized in parallel using diversified Ylanthia® maturation modules (YMM) previously generated with Slonomics® technology (van den Brulle et al. (2008): A novel solid phase technology for high-throughput gene synthesis; *Biotechniques* 45 (3), pp. 340-343, the contents of which are herein incorporated by reference for this purpose).

(446) Cloning of the maturation libraries was performed in vectors encoding the parental Fab fragments. The generation of the maturation libraries was performed for five parental antibodies (YY01, YY04, YY05, YY06, and YY07). For the library generation, all maturation candidates were treated individually. The maturation libraries were successfully cloned and had library sizes between 6.2×10^{sup.8} and 4.5×10^{sup.9} cfu.

(447) In order to monitor the cloning efficiency, the parental HCDR1/2 and LCDR3 were replaced

by MBP-stuffers prior to insertion of the diversified YMM. Digested vector fragments were ligated with a 2-fold molar excess of the insert fragments carrying the diversified HCDR1/2 or LCDR3s. Ligation mixtures were electroporated in *E. coli* cells yielding in >10⁸ independent colonies. Amplification of the library was performed as described in the literature (Tiller et al. (2013): A fully synthetic human Fab antibody library based on fixed VH/VL framework pairings with favorable biophysical properties; mAbs 5 (3), pp. 445-470, the contents of which are herein incorporated by reference for this purpose). For quality control, approx. 10-20 single clones per library were randomly picked and sequenced.

Example 14: Pannings and Screenings—Ylanthia®

(448) The nine maturation libraries were used for four different maturation panning strategies, which aimed for the enrichment of progenies with improved affinities compared to the parental clones. Furthermore, rat material was included where appropriate and the pannings were performed with high stringencies concerning antigen concentration and washing conditions. During the panning process, the libraries of YY01 and YY04 (only LCDR3) as well as YY05 and YY07 were pooled, while the libraries of YY06 were kept separately. The panning strategies are summarized in Table 18 in detail. The bacterial lysates (BEL) of the outputs after the third panning rounds were directly used for an ELISA-based pre-screening and for SET screening.

(449) TABLE-US-00018 TABLE 18 Overview of Ylanthia® Maturation Panning Strategies

Parental Strategy	antibodies	1.sup.st round	2.sup.nd round	3.sup.rd round	comments
1 YY01/YY04	hNPR1-ANP- rNPR1-ANP- hNPR1-ANP- Solution	panning	aiming	complex	solution complex
2 YY06	hNPR1-ANP- hNPR1-ANP- hNPR1-ANP- Solution	panning	aiming	complex	solution complex
3 YY05/YY07	NPR1-ANP-complex	Preadsorption on	Preadsorption on	Solution panning	aiming
4 YY05/YY07	NPR1-ANP-complex	Preadsorption on	Preadsorption on	Solution panning	aiming

(450) The outputs of the third panning rounds were used for an ELISA-based pre-screening to ensure that only clones binding to NPR1 and/or NPR1-ANP-complex were selected for further Solution Equilibrium Titration (SET) screening. 880 clones in total were analyzed in SET screening for improved affinity for hNPR1 and/or hNPR1-ANP-complex compared to the parental clones.

(451) During SET screening, 263 HCDR1//2 or LCDR3 unique improved derivatives were identified, which resulted in 112 unique clones after sequencing and conversion to IgG.sub.1_LALA format. Compared to their parental clones, the affinities of the YY05 and YY07 derivatives were not improved for NPR1 but were improved up to 200-fold for NPR1-ANP-complex. The derivatives of YY01 had similar affinities to the parental clone YY04, whose derivatives had only slightly improved affinities. The affinities of the YY06 derivatives were improved 4- to 40-fold for NPR1 and 7- to 70-fold for NPR1-ANP-complex. See FIG. 24 (which shows affinities for hNPR1) and FIG. 25 (which shows affinities for hNPR1-ANP complex). Affinities (K_{sub.D} [pM]) are indicated at the x-axis below the parental clone name for both figures.

Example 15: Characterization of Matured Candidates—Ylanthia®

(452) 95 of 112 improved candidates were selected for advanced production. 77 of the 95 clones passed the production quality control and were characterized in regard to binding to relevant antigens, binding to relevant cell lines, and functional activity in the cGMP production assay in comparison to their parental clones. After detailed IgG characterization, 17 candidates (detailed in Table 19) were selected, produced in exploratory scale IgG production, and further analyzed via 3P assay. Furthermore, they were converted to FabCys format for their affinity determination on human and rat NPR1 in absence and presence of ANP via SET K_{sub.D} measurement.

(453) TABLE-US-00019 TABLE 19 Overview of Matured Ylanthia® Candidates Matured Parental Matured antibody antibody CDR ZZ01 YY07 LCDR3 ZZ02 YY05 LCDR3 ZZ03 YY07 LCDR3 ZZ04 YY07 LCDR3 ZZ05 YY07 LCDR3 ZZ06 YY07 LCDR3 ZZ07 YY07 LCDR3 ZZ08 YY07 LCDR3 ZZ09 YY05 HCDR1 + 2 ZZ10 YY05 HCDR1 + 2 ZZ11 YY04 LCDR3 ZZ12 YY01 HCDR1 + 2 ZZ13 YY01 HCDR1 + 2 ZZ14 YY06 HCDR1 + 2 ZZ15 YY06 HCDR1 + 2 ZZ16 YY06 HCDR1 + 2 ZZ17 YY06 HCDR1 + 2

(454) Affinities for 16 of the matured Ylanthia® candidates (in monovalent FabCys format) were determined via SET measurement and/or via Octet. The results are summarized in Tables 20 and 21 below in comparison to the affinities of the parental clones.

(455) TABLE-US-00020 TABLE 20 Summary of Affinity data (SET) for YY01, YY06, ZZ12, ZZ13, ZZ15, ZZ16, and ZZ17 in FabCys Format Affinity (SET measurement) FabCys K.sub.D [nM] hNPR1 + rNPR1 + Antibody hNPR1 ANP rNPR1 ANP comment YY01 — 16.00 — 14.00 K.sub.D determination via Octet ZZ12 6.70 0.32 2.10 1.40 ZZ13 9.90 1.30 2.30 1.40 YY04 4.90 1.10 — 1.00 K.sub.D determination via Octet ZZ11 46.00 2.40 21.00 18.00 YY05 0.90 350.00 700.00 16000.00 K.sub.D determination via Octet; assay limit reached ZZ02 0.37 0.58 — — ZZ09 0.49 0.57 — — ZZ10 0.52 1.40 — — YY06 0.50 1.00 1600.00 1400.00 K.sub.D determination via Octet ZZ15 0.053 0.074 — — ZZ16 0.040 0.034 62.00 51.00 ZZ17 0.046 0.050 52.00 36.00 YY07 1.80 590.0 — — ZZ01 0.65 0.85 — — ZZ03 0.75 1.30 — — ZZ04 0.30 0.38 — — ZZ05 0.74 1.20 — — ZZ06 0.35 0.50 — — ZZ07 0.62 1.50 — — ZZ08 0.57 1.30 — —

(456) TABLE-US-00021 TABLE 21 Summary of Affinity data (Octet) for YY01, YY06, ZZ12, ZZ13, ZZ15, ZZ16, and ZZ17 in FabCys Format Affinity (Octet measurement) FabCys K.sub.D [nM] hNPR1 + rNPR1 + Antibody hNPR1 ANP rNPR1 ANP comment YY01 — 26.00 — 8.80 Heterogenous ZZ12 — 0.04 — 0.10 binding ZZ13 — 0.14 66.00 0.10 YY06 0.62 0.80 100.00 38.00 ZZ15 0.26 0.19 — — ZZ16 0.14 0.11 — — ZZ17 0.16 0.09 — —

Example 16: Epitope Determination for ANP Competitive and ANP Non-Competitive Agonist Antibodies

(457) The crystal structure of the hNPR1 and XX16 Fab complex was used to identify the XX16 Fab epitope on hNPR1. The interaction surface on hNPR1 by XX16 Fab was formed by several continuous and discontinuous (i.e., noncontiguous) sequences as detailed in Table 22. These residues form the three-dimensional conformational epitope recognized by XX16 Fab.

(458) Results of the epitope mapping of XX16 Fab (ANP non-competitive) are shown in Table 22. hNPR1 residues are numbered based upon SEQ ID NO: 1 (P16066). Fab residues are numbered based upon their linear amino acid sequence. hNPR1 residues shown have at least one atom within 5 Å of an atom in the XX16 Fab, to account for potential water mediated interactions.

(459) TABLE-US-00022 TABLE 22 Epitope Mapping of XX16 Fab Antibody XX16 Fab Antigen (NPR1) Amino Amino acid Number Chain acid Number Chain SER 31 H TRP 106 A TYR 32 H TRP 106 A TRP 106 B TRP 33 H ASN 109 A GLU 52 H HIS 108 A ASN 109 A SER 53 H GLU 107 A ASN 109 A LYS 54 H GLU 107 A ASN 56 H GLY 33 A ASN 34 A THR 36 A TRP 76 A TYR 57 H ASN 34 A TRP 76 A ASN 109 A PRO 110 A ALA 111 A VAL 374 A THR 375 A PHE 59 H THR 375 A ARG 100 H TRP 106 A ASP 103 B GLU 107 B TYR 101 H VAL 102 A TRP 106 A VAL 102 B ASP 103 B TRP 106 B GLU 107 B SER 102 H LYS 105 A TRP 106 A MET 103 H VAL 102 A LYS 105 A TRP 106 A TRP 132 A LEU 99 B VAL 102 B ASP 103 B ILE 104 H LYS 105 A HIS 131 A TRP 132 A LEU 99 B ALA 100 B ASP 103 B TYR 105 H LYS 105 A TRP 132 A SER 106 H LYS 105 A HIS 131 A TRP 132 A ARG 133 A TYR 107 H LYS 105 A ASN 109 A PRO 110 A ALA 111 A VAL 134 A THR 375 A ASP 113 H GLU 107 B ARG 18 L GLN 336 B SER 30 L GLU 86 B ASN 87 B ALA 88 B SER 31 L SER 84 B GLU 86 B TYR 32 L HIS 131 A TYR 49 L ASP 103 B LEU 104 B GLU 107 B HIS 108 B SER 52 L LEU 43 B LEU 82 B THR 53 L VAL 81 B LEU 82 B GLY 83 B LEU 104 B HIS 108 B LEU 54 L ARG 79 B VAL 81 B HIS 108 B GLN 55 L GLU 107 B HIS 108 B SER 56 L GLU 107 B HIS 108 B VAL 58 L ARG 79 B PRO 59 L ARG 79 B SER 60 L ARG 79 B SER 65 L THR 44 B GLY 66 L THR 44 B SER 67 L THR 44 B GLU 86

B ASN 87 B SER 76 L GLN 336 B TRP 92 L HIS 131 A ARG 133 A ALA 88 B ARG 93 L ARG 133 A GLU 378 A LYS 94 L ASP 376 A

(460) Critical epitope residues for the binding of XX16 Fab and NPR1, which were determined using structural analysis and affinity maturation data, include (first tier) 99-103, 105-111, 131-133, 378; and (second tier): 33-34, 76, 82, 104, 374-375. Amino acids 99-103 of NPR1 (SEQ ID NO: 1) comprise both a E106 backbone that enabled the affinity maturation by D54K in HCDR2, and W106 from both NPR1 protomers that clamp Y101 of HCDR3. Certain critical epitope residues are shown in Table 23 below. Regions of NPR1 encompassing these critical residues include N34, W76, L82, V102-A111, H131-V134, and V374-E378.

(461) TABLE-US-00023 TABLE 23 Critical Epitope Residues of XX16 Fab (ANP non-competitive) Critical Critical epitope epitope residue residue (amino acid) (number) Explanation N 34 H-bond with N56 (backbone) of VH L 82 H-bond (chain B) with T53 (side chain) of VL V 102 Packing against M103 (side chain) of VH D 103 Salt bridge (chain B) with R100 (side chain) of VH K 105 H-bond (backbone) with M103 (backbone) and Y105 (backbone) of VH W 106 Packing between Y32 (side chain) and Y101 (side chain) of VH E 107 Salt bridge (chain B) with R100 (side chain) of VH E 107 H-bond (backbone) with K54 (side chain) of VH N 109 H-bond with E52 (side chain), S53 (side chain) and W33 (side chain) of VH W 132 Packing against M103 (side chain) of VH R 133 H-bond with S106 (side chain) of VH T 375 Packing against Y57 (side chain) of VH E 378 Salt bridge with R93 (side chain) of VL

(462) The crystal structure of the hNPR1 and WW03 Fab complex was used to identify the WW03 Fab epitope on hNPR1. The interaction surface on hNPR1 by WW03 Fab was formed by several continuous and discontinuous (i.e., noncontiguous) sequences as detailed in Table 24. These residues form the three-dimensional conformational epitope recognized by WW03 Fab.

(463) Results of the epitope mapping of WW03 Fab (ANP non-competitive) are shown in Table 24. hNPR1 residues are numbered based upon SEQ ID NO: 1 (P16066). Fab residues are numbered based upon their linear amino acid sequence. hNPR1 residues shown have at least one atom within 5 Å of an atom in the WW03 Fab, to account for potential water mediated interactions.

(464) TABLE-US-00024 TABLE 24 Epitope Mapping of WW03 Fab Antibody WW03 Fab Antigen (NPR1) Amino Amino acid Number Chain acid Number Chain SER 31 H TRP 106 A TYR 32 H TRP 106 A TRP 106 B TRP 33 H ASN 109 A SER 52 H HIS 108 A ASN 109 A SER 53 H TRP 106 A GLU 107 A ASN 109 A ASP 54 H THR 36 A ARG 79 A SER 56 H ASN 34 A TYR 57 H ASN 109 A PRO 110 A ALA 111 A VAL 374 A THR 375 A TYR 59 H THR 375 A ARG 100 H ASP 103 B TRP 106 B GLU 107 B TYR 101 H VAL 102 A TRP 106 A VAL 102 B ASP 103 B TRP 106 B GLU 107 B SER 102 H LYS 105 A TRP 106 A ASN 109 A MET 103 H VAL 102 A LYS 105 A TRP 106 A TRP 132 A LEU 99 B VAL 102 B ASP 103 B ILE 104 H HIS 131 A TRP 132 A LEU 99 B ASP 103 B TYR 105 H LYS 105 A TRP 132 A SER 106 H LYS 105 A TRP 132 A ARG 133 A TYR 107 H LYS 105 A ASN 109 A PRO 110 A ALA 111 A VAL 134 A THR 375 A ASP 113 H GLU 107 B ARG 18 L GLN 336 B SER 30 L ASN 87 B ALA 88 B SER 31 L SER 84 B TYR 32 L HIS 131 A TYR 49 L ASP 103 B GLU 107 B HIS 108 B SER 52 L LEU 43 B LEU 82 B THR 53 L VAL 81 B LEU 82 B GLY 83 B LEU 104 B HIS 108 B LEU 54 L ARG 79 B VAL 81 B HIS 108 B GLN 55 L GLU 107 B HIS 108 B SER 56 L GLU 107 B HIS 108 B VAL 58 L ARG 79 B PRO 59 L ARG 79 B SER 60 L ARG 79 B GLY 66 L THR 44 B SER 67 L THR 44 B GLU 86 B ASN 87 B TYR 92 L ARG 133 A GLU 93 L ARG 133 A LYS 94 L ASP 376 A

(465) Critical epitope residues for the binding of WW03 Fab and NPR1, which were determined using structural analysis and affinity maturation data, include the residues shown in Table 25. Regions of NPR1 encompassing these critical residues include R79, L82, L99-A111, H131-V134, and V374-T375.

(466) TABLE-US-00025 TABLE 25 Critical Epitope Residues of WW03 Fab (ANP non-competitive) Critical Critical epitope epitope residue residue (amino acid) (number) Explanation R 79 Salt bridge with D54 (side chain) of VH; H-bond (chain B) with S60 (side chain) of VL L 82

H-bond (backbone) with T53 (side chain) of VL L 99 Packing against M103 (side chain) and 1104 (side chain) of VH V 102 Packing against M103 (side chain) of VH D 103 Salt bridge with R100 (side chain) of VH K 105 H-bond with M103 (backbone) and Y105 (backbone) of VH W 106 Packing between Y32 (side chain) and Y101 (side chain) of VH; stacking (chain B) against Y101 (side chain) of VH N 109 H-bond with S53 (side chain) of VH H 131 H-bond with 1104 (backbone) of VH W 132 H-bond with M103 (side chain) of VH T 375 H-bond with Y59 (side chain) of VH; pack against Y57 (side chain) of VH

(467) The crystal structure of the hNPR1 and WW06 Fab complex was used to identify the WW06 Fab epitope on hNPR1. The interaction surface on hNPR1 by WW06 Fab was formed by several continuous and discontinuous (i.e., noncontiguous) sequences as detailed in Table 26. These residues form the three-dimensional conformational epitope recognized by WW06 Fab.

(468) Results of the epitope mapping of the WW06 Fab (ANP competitive) are shown in Table 26. hNPR1 residues are numbered based upon SEQ ID NO: 1 (P16066). WW06 Fab residues are numbered based upon their linear amino acid sequence. hNPR1 residues shown have at least one atom within 5 Å of an atom in the WW06 Fab, to account for potential water mediated interactions.

(469) TABLE-US-00026 TABLE 26 Epitope Mapping of WW06 Fab (ANP competitive) Antibody WW06 Fab Antigen (NPR1) Amino Amino acid Number Chain acid Number Chain Y 27 H R 294 B H G 295 B S 28 H R 294 B H G 295 B H D 296 B S 31 H R 230 A H D 296 B Y 32 H G 295 B H D 296 B H G 297 B W 33 H Y 188 A H E 219 A R 50 H F 197 A H E 219 A D 52 H Y 188 A H A 221 A H H 227 A D 54 H D 224 A N 55 H Y 188 A H P 190 A H A 221 A H D 223 A Y 57 H Y 188 A H R 189 A H P 190 A H G 191 A H D 192 A H E 193 A H E 194 A H F 197 A T 58 H E 194 A R 59 H E 194 A H F 197 A H E 201 A Q 65 H D 94 A H E 194 A R 98 H G 295 B W 99 H E 219 A L 100 H R 230 A S 101 H H 217 A P 102 H K 238 A G 103 H D 216 A H H 217 A H T 234 A H K 238 A Y 104 H L 218 A H E 219 A H F 220 A H H 227 A H R 230 A H L 231 A H T 234 A H K 238 A A 105 H K 238 A L 106 H R 233 A H T 234 A H R 237 A H K 238 A G 107 H R 237 A Q 109 H G 297 B Q H V 300 B I 29 L R 208 A G 30 L R 208 A A 31 L F 204 A L R 208 A G 32 L R 208 A Y 33 L Y 186 A L E 201 A L F 204 A L H 217 A Y 93 L F 197 A L E 201 A L 95 L L 144 A L E 201 A L F 204 A L M 205 A Q 96 L L 144 A L G 145 A L V 148 A L M 205 A S 98 L F 197 A L F 198 A L E 201 A R 100 L E 219 A

(470) Critical epitope residues for the binding of WW06 and NPR1, which were determined using structural analysis and affinity maturation data, include the residues shown in Table 27. Regions of NPR1 encompassing these critical residues include Y188-F198, E201-R208, V215-K238, and R294-G297.

(471) TABLE-US-00027 TABLE 27 Critical Epitope Residues of WW06 (ANP competitive) Critical Critical epitope epitope residue residue (amino acid) (number) Explanation Y 188 Packing against W33 (side chain) of VH D 192 H-bond with Y57 (side chain) of VH E 194 Salt bridge with R59 (side chain) of VH F 197 Stacking against R59 (side chain) of VH E 201 H-bond with S98 (side chain) of VL R 208 H-bond with G30 (backbone) of VL E 219 Salt bridge with R50 (side chain) of VH G 295 H-bond (backbone, chain B) with S28 (backbone) of VH

Example 17: Mouse In Vivo Characterization of Effect of WW06 on Plasma cGMP

(472) WW06 FabCys was used in an in vivo study in hNPR1 transgenic mice to determine the effect of this antibody on plasma cGMP levels in vivo.

(473) For analysis of plasma cGMP samples, the LC-MS/MS detection method using .sup.15N.sub.2, .sup.13C cGMP as an internal standard was adopted from Oeckl and Ferger, Journal of Neuroscience Methods 203 (2012) 338-343; and Zhang et al., J. Chromatogr B: Analyt Technol Biomed Life Sci 2009; 877:513-20; the contents of each of which are hereby incorporated by reference for this purpose).

(474) Plasma cGMP concentration in hNPR1 Tg mice which were intravenously administered the WW06 Fab increased at the 5 minute time point and the signal returned to baseline by 3 h. As expected and consistent with the data shown in FIGS. 26 and 27, the T1/2 of the FabCys antibody

was <30 min. Each value shown is the average of three points collected from three individual animals. Dose response data are shown in FIG. 26, and PK data are shown in FIG. 27.

Example 18: Effect of XX16 on Heart Weight and NT-proBNP in ANP KO and WT Mice

(475) XX16 was used in an in vivo study to determine its effect on heart weight and NT-proBNP levels in ANP knockout (KO) and wild-type (WT) mice.

(476) ANP knockout mice are hypertensive and have cardiac hypertrophy (increased HW/BW ratio). NT-proBNP is a biomarker of cardiac dysfunction. XX16 was administered at 0.3 or 3 mg/kg subcutaneously once every two weeks for four weeks in ANP knockout and wild type mice.

(477) Results are shown in FIG. 28. At two weeks after the second treatment, XX16 dose-dependently reduced heart weight/body weight ratio and NT-proBNP in both wild type and ANP KO mice in comparison to vehicle treated animals.

Example 19: Acute Effect of XX16 on Blood Pressure and Urinary Flow Rate in Hypertensive Rats

(478) XX16 was used to determine its effect on blood pressure and urinary flow rate in hypertensive rats (spontaneous hypertensive rat stroke prone, SHRsp). Animals were administered 0.3 mg/kg XX16, 1 mg/kg XX16, or a vehicle control intravenously (one time). Blood pressure was measured using a femoral artery catheter. Measurements were taken three hours after the intravenous administration and results are shown in FIG. 29.

(479) Intravenous XX16 treatment normalized mean arterial pressure and increased urinary flow rate acutely in hypertensive rats (SHRsp) in comparison to vehicle treated animals.

Example 20: Chronic Effects of XX16 on Blood Pressure and Plasma cGMP in Wistar-Han Rats

(480) Chronic hemodynamic effects of XX16 in telemetry implanted normal rats were evaluated.

XX16 (at doses of 0.1, 0.3, 1, 3, 10, and 30 mg/kg) was administered subcutaneously one time.

Results are shown in FIG. 30. Subcutaneous administration of XX16 decreased mean arterial pressure (MAP) in all groups except for the 0.1 mg/kg group. There was a floor (around 95 mmHg) for the MAP reduction. A dose-dependent effect for the time to return baseline MAP was observed; this effect was greater than 70 days for the 30 mg/kg treatment group. Plasma cGMP concentration also had a ceiling around 90 nmol/L, after which the plasma cGMP gradually reduced toward baseline. This effect is very similar to the blood pressure response.

Example 21: Comparison of Extant Anti-NPR1 Antibodies with Antibodies of the Application

(481) The WW03 antibody was compared with antibody 5591-IgG of PCT Application No. WO2010/065293A1 in terms of the ability of both antibodies to produce cGMP in human cells expressing hNPR1 or rat cells expressing rNPR1 (both in the presence or absence of 0.075 nM human or rat ANP, respectively). The WW03 antibody displayed superior potentiation in the absence of ANP on both cell lines. Additionally, the WW03 antibody demonstrated superior potentiation on rat cells expressing rNPR1 (both in the presence and absence of ANP).

(482) In summary, previous antibodies (e.g., those of WO2010/065293, including 5591-IgG) demonstrate no activity in vivo (e.g., through analysis of activation using cGMP assays). In contrast, the antibodies of the instant application demonstrated in vivo activity in both mouse and rat. Further, the epitope binding of the antibodies described herein has been demonstrated to be dissimilar to the antibodies of WO2010/065293 using crystal structure data. The activity, cross-reactivity, and crystallographic data described herein demonstrate the differing and superior effects of the antibodies described in this application as compared to previous antibodies.

Claims

1. An isolated nucleic acid or nucleic acids encoding the amino acid sequence of an anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: TABLE-US-00028 (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119

(HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3).

2. A vector comprising the isolated nucleic acid(s) of claim 1.

3. A host cell comprising the isolated nucleic acid(s) of claim 1.

4. A method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing the host cell of claim 3 under conditions suitable for producing the antibody or antigen binding fragment.

5. The isolated nucleic acid(s) of claim 1, wherein the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; or (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136.

6. A vector comprising the isolated nucleic acid(s) of claim 5.

7. A host cell comprising the isolated nucleic acid(s) of claim 5.

8. A method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing the host cell of claim 7 under conditions suitable for producing the antibody or antigen binding fragment.

9. The isolated nucleic acid(s) of claim 1, wherein the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; or (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 138.

10. A vector comprising the isolated nucleic acid(s) of claim 9.

11. A host cell comprising the isolated nucleic acid(s) of claim 9.

12. A method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing the host cell of claim 11 under conditions suitable for producing the antibody or antigen binding fragment.
